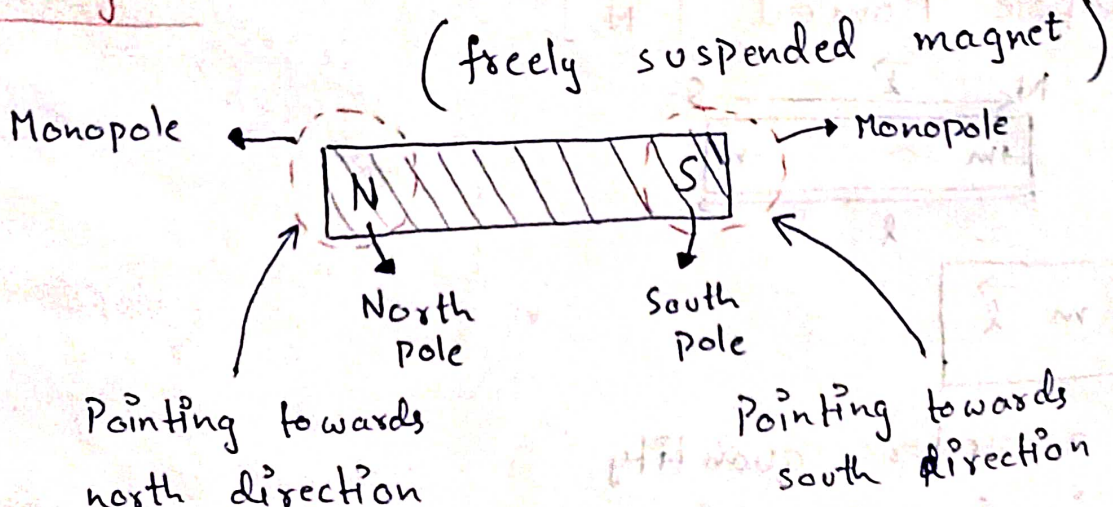
MAGNETIC EFFECT OF CURRENT (EMF)

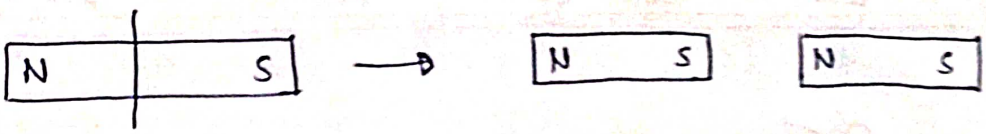
If two bodies (neutral bodies) are attracting / repelling each other, then this force should be magnetic force and this property of the bodies is known as magnetism.

* Cause of magnetism is motion of charged particles.
* Bodies which possess property of magnetism are known as magnets.

Bar Magnet :



* A bar magnet is a magnetic dipole.
* Monopole can't exist independently.



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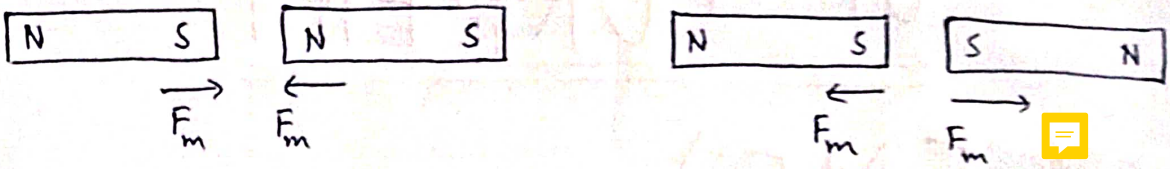
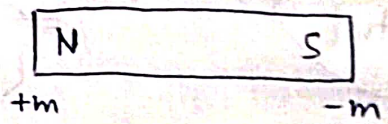


Like poles repel, while unlike poles attract.

Pole Strength

North pole : $+m$

South pole : $-m$



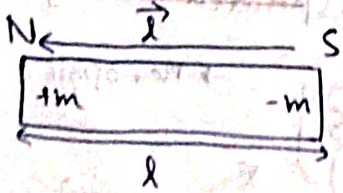
→ $m \uparrow \Rightarrow$ Magnetic effect ↑

→ $m \downarrow \Rightarrow$ Magnetic effect ↓

→ $m = 0 \Rightarrow$ Magnetic effect = 0

* It is a scaler quantity.
* It is of two types — +ve (N)
  -ve (S)
* Unit → Ampere-Metre
* Dimension → $[IL] = [AL]$

Magnetic Dipole Moment $[\vec{M}]$



$$\vec{M} = m\vec{l}$$

* It is a vector quantity.
* Direction : S to N
  $-m$ to $+m$

Unit → Amp-m^2

* Dimension → $[I L^2]$

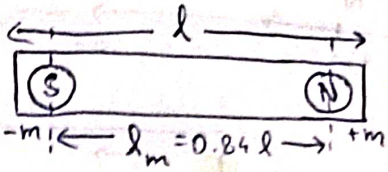
Practically, (Avoid)

Magnetic length

$M = m(l_m)$

$= m(0.84)l$

$= 0.84\, ml$



Magnetic Field

The region around a moving charged particle or magnet, in which there is some magnetic effects is known as magnetic field.

Strength of Magnetic Field / Magnetic Induction / Magnetic Flux Density [$\vec{B}$]

→ Magnetic effect ↑ ⇒ B ↑

→ Magnetic effect ↓ ⇒ B ↓

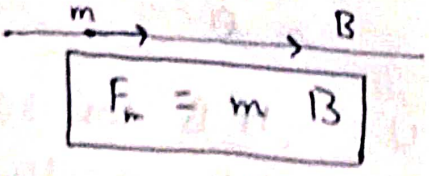
→ Magnetic effect = 0 ⇒ B = 0

* Unit → $\frac{\text{Newton}}{\text{Amp-m}} = \frac{\text{kg-m/sec}^2}{\text{Amp-m}} = \frac{\text{kg}}{\text{Amp-sec}^2} = \text{Tesla [T]}$

$= \frac{\text{Weber}}{m^2}$

* Dimension → $[M I^{-1} T^{-2}]$

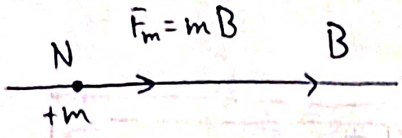
* It is a vector quantity

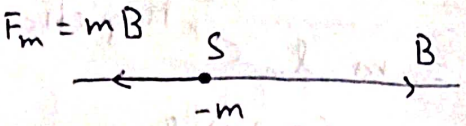


however, magnetic monopoles do not exist

hence, there must be another '-m' pole somewhere, which doesn't affect the force calculation

Magnetic Force on a Monopole placed in Magnetic Field

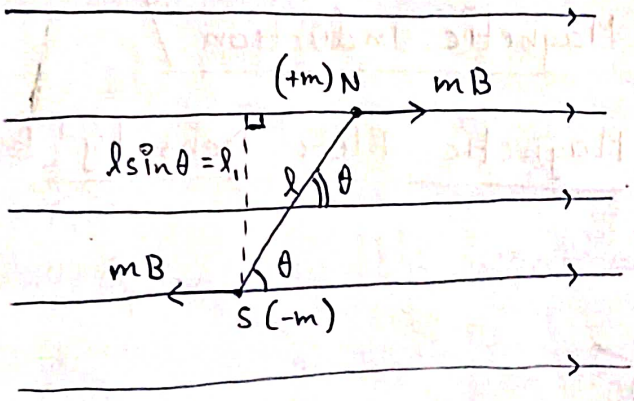




$$\vec{F_m} = m\vec{B}$$

↳ with sign

Magnetic Force on a Bar Magnet placed in Uniform Magnetic Field



$\theta \rightarrow$ angle between $\vec{H}$ and $\vec{B}$

$$F_{net} = 0 \qquad a_{COM} = 0$$

$$\tau = (m \cdot B\, l \sin\theta)$$

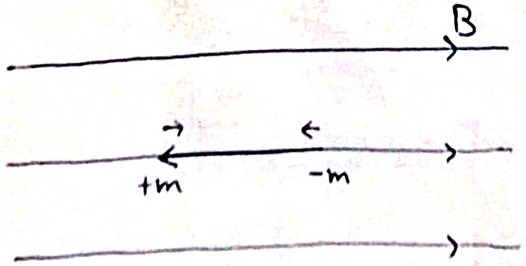
$$= MB \sin\theta$$

magentic dipole moment M
M=ml

$$\vec{\tau} = \vec{H} \times \vec{B}$$

$$U = -MB\cos\theta$$

Case - I ; $\theta = 180^\circ = \pi$ radians



$$\tau = MB \sin 180^\circ$$

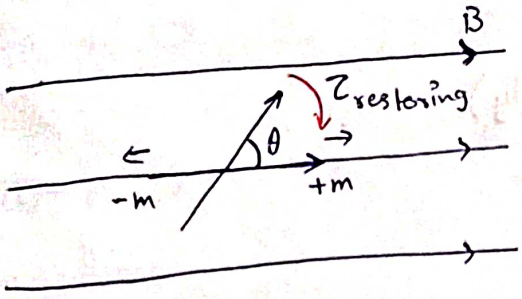
$$= 0$$

$$U = -MB\cos 180^\circ$$

$$= MB = U_{max}$$

∴ Unstable equilibrium

Case - II : $\theta = 0^\circ$

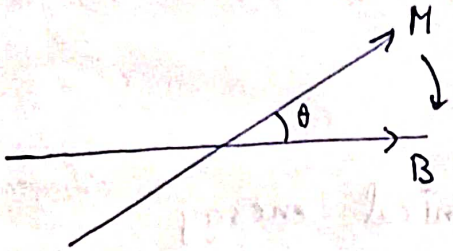


$$\tau = MB \sin 0^\circ = 0$$

$$U = -MB \cos 0^\circ = -MB = U_{min}$$

∴ Stable equilibrium

*



$$\tau_{restoring} = MB \sin\theta$$

$\theta \longrightarrow$ small

$\sin\theta \approx \theta$

$I\alpha = \tau = -MB\theta$ ⇒ Angular SHM (Proved)

$$\alpha = \frac{MB}{I}\theta = \frac{d^2\theta}{dt^2}$$

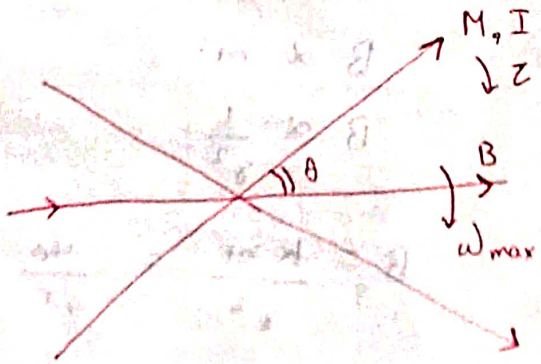
$\frac{MB}{I} \rightarrow \omega^2$

$$\omega = \sqrt{\frac{MB}{I}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{MB}}$$

$\omega^2 \rightarrow$ (I → I_{Hinge} : Hinged; I → I_{com} : not Hinged)

Q,



Find ω_{max}

Sol Total mechanical energy conservation,

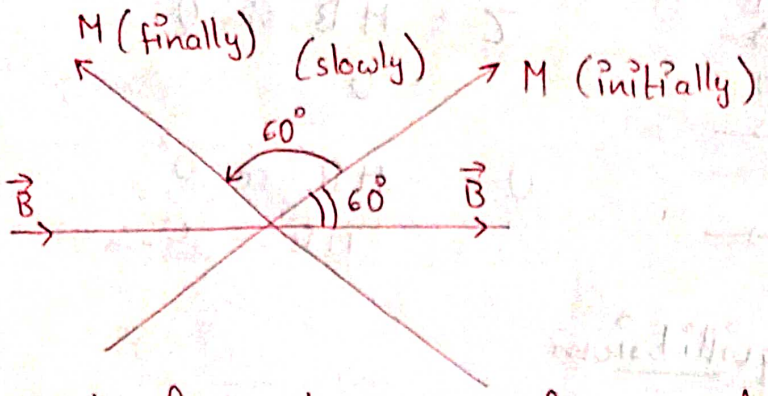
$$K_i + U_i = K_f + U_f$$

$$0 - MB\cos\theta = \frac{1}{2}I\omega^2 - MB\cos 0^\circ$$

$$\frac{1}{2}I\omega^2 = MB(1-\cos\theta)$$

$$\omega_{max} = \sqrt{\frac{2MB}{I}(1-\cos\theta)}$$

Q/



Find work done by external agent to rotate the magnet by angle 60° slowly.

Sol/

W_{ext} = change in total mechanical energy

$$W_{ext} = (K_f^{0} + U_f) - (K_i^{0} + U_i)$$

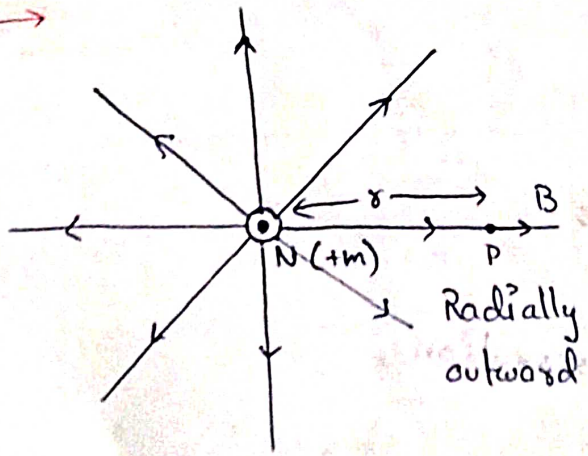
$$= U_f - U_i$$

$$= -MB\cos 120^\circ - (-MB\cos 60^\circ)$$

$$= \frac{MB}{2} + \frac{MB}{2}$$

$$= MB$$

$\vec{B}$ due to a Monopole



$B \propto m$

$B \propto \frac{1}{r^2}$

$$B = \frac{km}{r^2} = \frac{\mu_0}{4\pi}\frac{m}{r^2}$$

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{m}{r^2}\hat{r}$$

$$= \frac{\mu_0}{4\pi}\cdot\frac{m}{r^2}\cdot\frac{\vec{r}}{r}$$

$$\boxed{\vec{B} = \frac{\mu_0}{4\pi}\frac{m}{r^3}\vec{r}}$$ → with sign

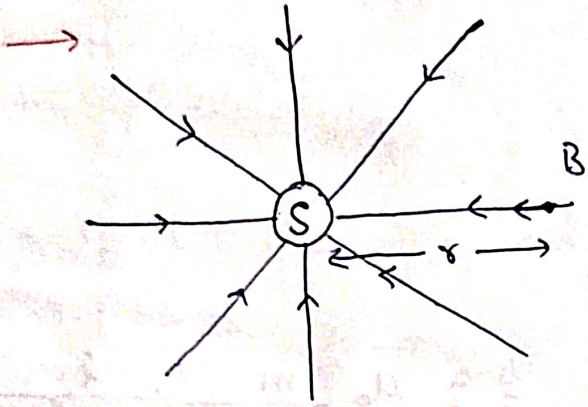
μ_0 → permeability of free space

$\frac{\mu_0}{4\pi} = 10^{-7} \frac{N}{A^2}$ remember $\Rightarrow \mu_0 = 4\pi \times 10^{-7} \frac{N}{A^2}$ remember

$\mu_0 = 1.256637 \times 10^{-6} \frac{N}{A^2} / \frac{\text{Henry}}{\text{meter}}$

don't remember

$K = 10^{-7}$ remember



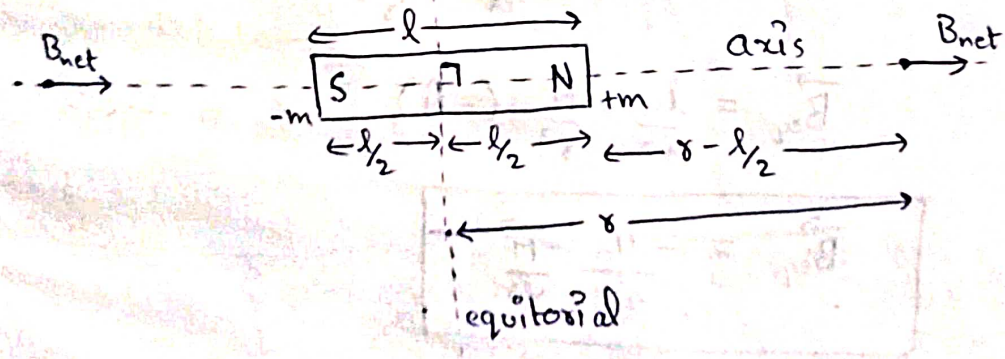
$B = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^2}$

$\vec{B} = \frac{\mu_0}{4\pi} \frac{m}{r^3} \vec{r}$

$\vec{B}$ due to (Bar Magnet: (Short)

→ Magnetic dipole

$M = m l$



$B_{axis} = \frac{\mu_0}{4\pi} \frac{m}{\left(r - \frac{l}{2}\right)^2} - \frac{\mu_0}{4\pi} \frac{m}{\left(r + \frac{l}{2}\right)^2}$

$= \frac{\mu_0}{4\pi} m \frac{rl + rl}{\left(r^2 - \frac{l^2}{4}\right)^2}$

$= \frac{\mu_0}{4\pi} \frac{m \cdot 2rl}{\left(r^2 - \frac{l^2}{4}\right)^2}$

if $l << r$ for a short magnet

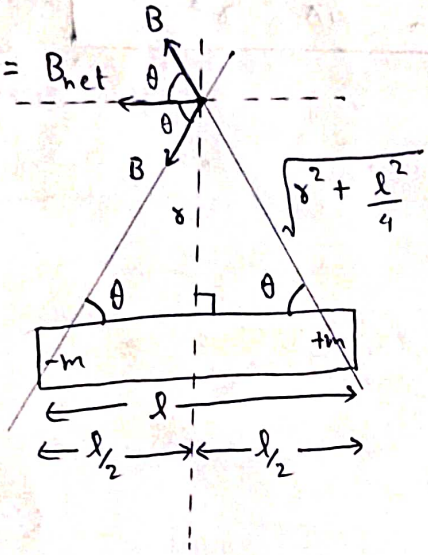
$$B_{axis} = \frac{\mu_0}{4\pi} \frac{2mrl}{r^4}$$

$$= \frac{\mu_0}{4\pi} \frac{2ml}{r^3}$$

$$= \frac{\mu_0}{4\pi} \frac{2M}{r^3}$$

$$\vec{B}_{axis} = 2\left(\frac{\mu_0}{4\pi}\right) \frac{M}{r^3} \hat{M}$$

$$\boxed{\vec{B}_{axis} = 2\left(\frac{\mu_0}{4\pi}\right) \frac{\vec{M}}{r^3}}$$



$2B\cos\theta = B_{net}$

$$B_{eq} = 2 \cdot \frac{\mu_0}{4\pi} \cdot \frac{m}{\left(r^2 + \frac{l^2}{4}\right)} \cdot \frac{l}{2\sqrt{r^2 + \frac{l^2}{4}}}$$

$$B_{eq} = \left(\frac{\mu_0}{4\pi}\right) \frac{M}{\left(r^2 + \frac{l^2}{4}\right)^{3/2}}$$

if $l << r$

$$B_{eq} = \frac{\mu_0}{4\pi} \frac{M}{r^3}$$

$$\vec{B}_{eq} = \frac{\mu_0}{4\pi} \frac{M}{r^3} (-\hat{M})$$

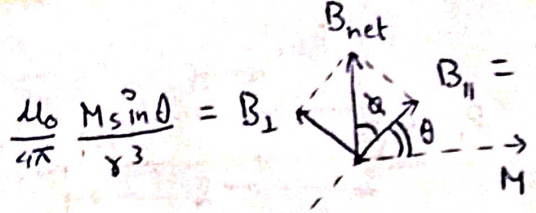
$$\boxed{\vec{B}_{eq} = -\frac{\mu_0}{4\pi} \frac{\vec{M}}{r^3}}$$

at any general point,

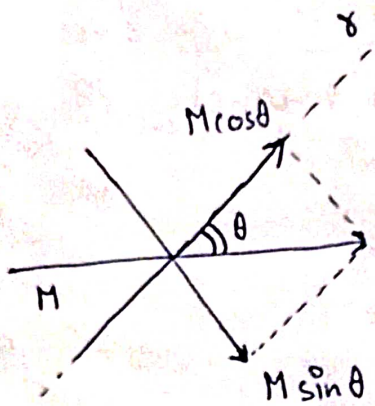
$$B_{net} = \sqrt{B_{\parallel}^2 + B_{\perp}^2}$$

$$= \frac{\mu_0}{4\pi} \frac{M}{r^3} \sqrt{\sin^2\theta + 4\cos^2\theta}$$

$$\boxed{B_{net} = \frac{\mu_0}{4\pi} \frac{M}{r^3} \sqrt{1 + 3\cos^2\theta}}$$



$$\frac{\mu_0}{4\pi}\frac{M\sin\theta}{r^3} = B_\perp \qquad B_\parallel = 2\cdot\frac{\mu_0}{4\pi}\frac{M\cos\theta}{r^3}$$



$$\tan\phi = \frac{B_\perp}{B_\parallel} = \frac{\sin\theta}{2\cos\theta} = \frac{\tan\theta}{2}$$

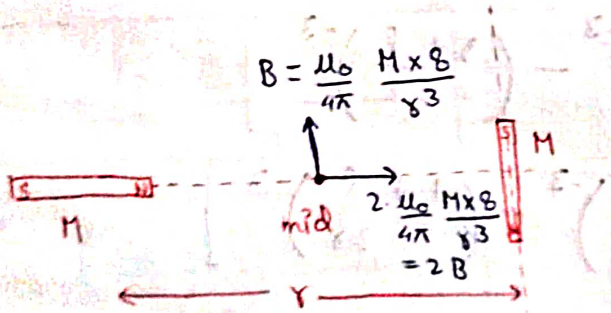
Angle between $\vec{B}_{net}$ and $\vec{r} = \phi = \tan^{-1}\left(\frac{\tan\theta}{2}\right)$

Angle between $\vec{M}$ and $\vec{B}_{net} = \theta + \phi$

$$= \theta + \tan^{-1}\left(\frac{\tan\theta}{2}\right)$$

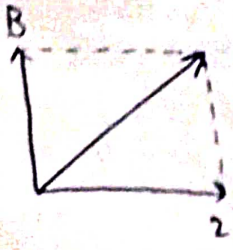
$\theta \rightarrow$ angle between $\vec{M}$ and $\vec{r}$

Q)



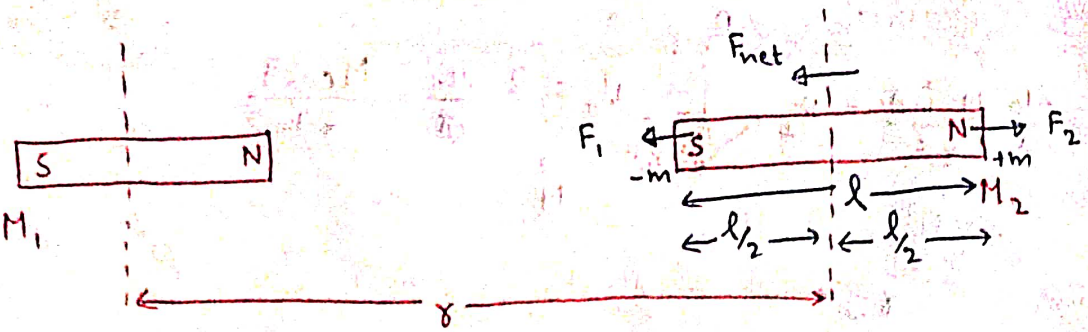
Find B at midpoint.

Sol



$$B_{net} = \sqrt{B^2 + (2B)^2} = \sqrt{5}\,B = 8\sqrt{5}\,\frac{\mu_0}{4\pi}\frac{M}{r^3}$$

Q1



Find magnetic force between these bar magnets.

Sol

$$M_2 = m\,l$$

$$F_1 = m\,B_1$$

$$F_2 = m\,B_2$$

$$F_{net} = F_1 - F_2$$

$$= m B_1 - m B_2$$

$$= m\left[2\cdot\frac{\mu_0}{4\pi}\frac{M_1}{\left(r-\frac{l}{2}\right)^3} - 2\frac{\mu_0}{4\pi}\cdot\frac{M_1}{\left(r+\frac{l}{2}\right)^3}\right]$$

$$= 2m\,\frac{\mu_0}{4\pi}\,M_1\left[\left(r-\frac{l}{2}\right)^{-3} - \left(r+\frac{l}{2}\right)^{-3}\right]$$

$$= 2m\,\frac{\mu_0}{4\pi}\,\frac{M_1}{r^3}\left[\left(1-\frac{l}{2r}\right)^{-3} - \left(1+\frac{l}{2r}\right)^{-3}\right]$$

$$= 2m\cdot\frac{\mu_0}{4\pi}\,\frac{M_1}{r^3}\left[\left(1+\frac{3l}{2r}\right) - \left(1-\frac{3l}{2r}\right)\right]$$

$$= 2m\cdot\frac{\mu_0}{4\pi}\,\frac{M_1}{r^3}\,\frac{6l}{2r}$$

$$\boxed{F_{net} = 6\left(\frac{\mu_0}{4\pi}\right)\frac{M_1 M_2}{r^4}}$$

II - Method

$$F = M\,\frac{dB}{dr}$$

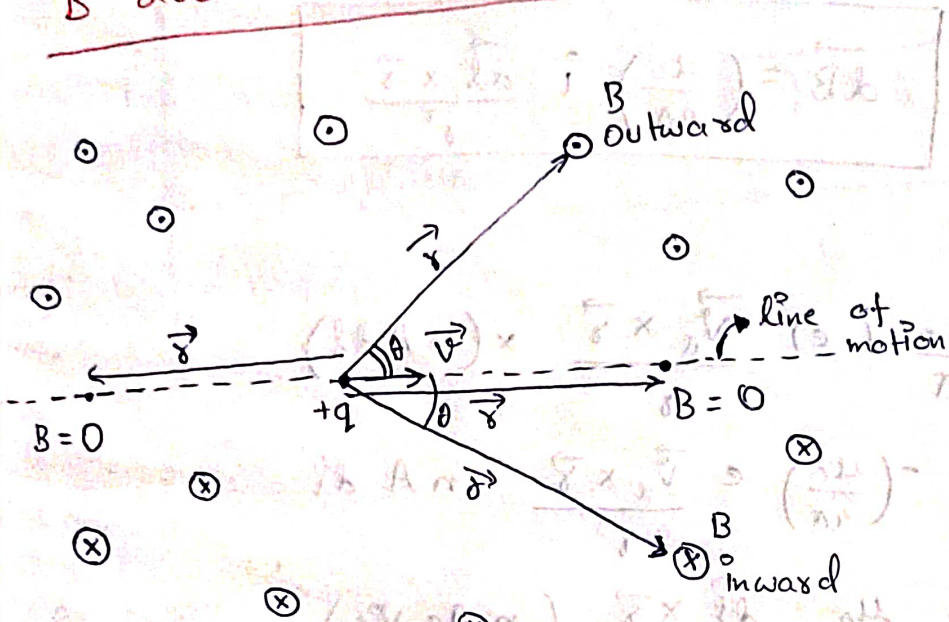
$$F = M_2\,\frac{dB_1}{dr} = M_1\,\frac{dB_2}{dr}$$

$$F = M_2 \frac{d}{dr}\left(2 \cdot \frac{\mu_0}{4\pi} \frac{M_1}{r^3}\right)$$

$$= 2\left(\frac{\mu_0}{4\pi}\right) M_1 M_2 \frac{d}{dr}(r^{-3})$$

$$= -6 \frac{\mu_0}{4\pi} \frac{M_1 M_2}{r^4}$$

$\vec{B}$ due to Moving Charge (experimental)



$\theta \rightarrow$ Angle between $\vec{r}$ and $\vec{v}$

$\theta = 0^\circ \Rightarrow B = 0$

$\theta = 180^\circ \Rightarrow B = 0$

$$B \propto q$$
$$\propto v$$
$$\propto \sin\theta$$
$$\propto \frac{1}{r^2}$$

$$B = \frac{\mu_0}{4\pi} \frac{qv \sin\theta}{r^2}$$

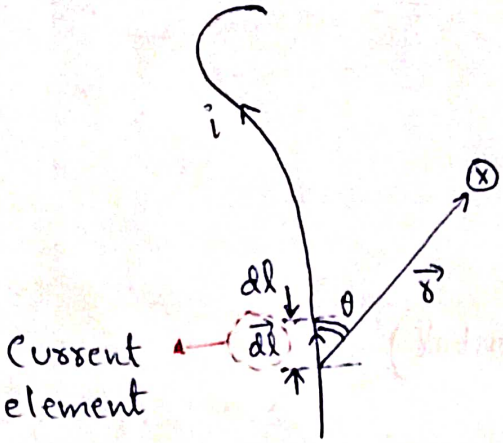
$$\vec{B} = \left(\frac{\mu_0}{4\pi}\right) q \frac{\vec{v} \times \vec{r}}{r^3}$$

q → with sign

$\vec{B} \perp \vec{v}$, $\vec{B} \perp \vec{r}$ $\Rightarrow$ $\vec{B} \perp$ to the plane containing $\vec{v}$ and $\vec{r}$

* $\vec{B}$ due to point charge depends on reference frame or observer since velocity depends on the the reference frame

$\vec{B}$ due to Current Carrying Wire / Biot Savart's Law

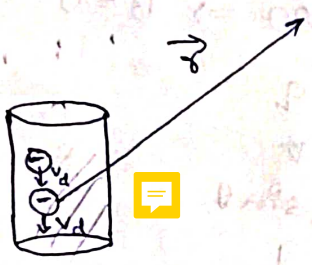


$$dB \propto (i\,dl)$$
$$\propto \sin\theta$$
$$\propto \frac{1}{\gamma^2}$$

$$dB = \left(\frac{\mu_0}{4\pi}\right) \frac{i\,dl\,\sin\theta}{\gamma^2}$$

$$\boxed{\vec{dB} = \left(\frac{\mu_0}{4\pi}\right) i \frac{\vec{dl} \times \vec{\gamma}}{\gamma^3}}$$

head of r at the pt. due to which MF is to be found
tail at the element due to which MF is to be found

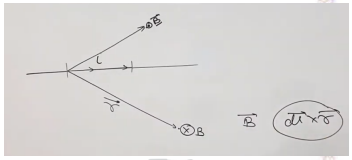


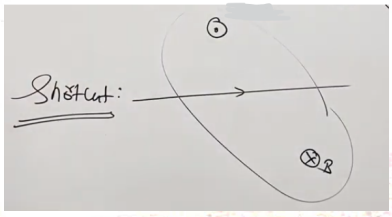
$$\frac{\mu_0}{4\pi} (-e) \frac{\vec{v}_d \times \vec{\gamma}}{\gamma^3} \times (n A\,dl)$$

$$\vec{dB} = -\left(\frac{\mu_0}{4\pi}\right) e \frac{\vec{v}_d \times \vec{\gamma}}{\gamma^3} n A\,dl$$

$$= \frac{\mu_0}{4\pi} \frac{\vec{dl} \times \vec{\gamma}}{\gamma^3} (n A e v_d)$$

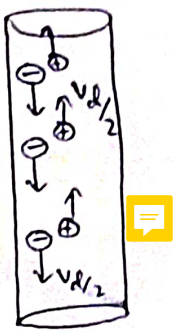
$$\boxed{\vec{dB} = \left(\frac{\mu_0}{4\pi}\right) i \frac{\vec{dl} \times \vec{\gamma}}{\gamma^3}}$$

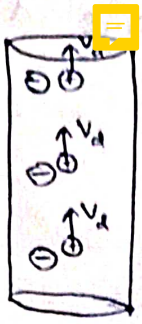




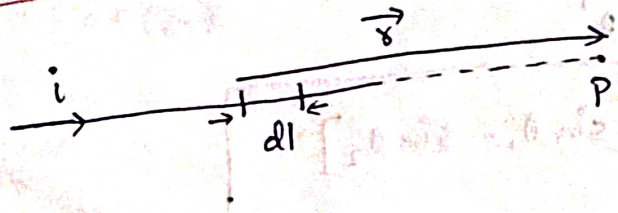
* $\vec{dl}$: direction same as current
* Direction of $\vec{B}$ is same of $\vec{dl} \times \vec{\gamma}$

Note : $\vec{B}$ due to current carrying element doesn't depend on observer.



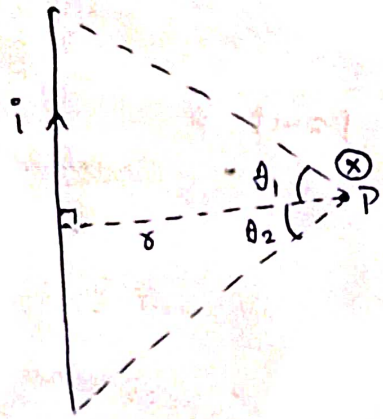


$\vec{B}$ due to a Straight Wire



$$dB = \frac{\mu_0}{4\pi} \frac{i\, dl \sin 0}{r^2} = 0$$

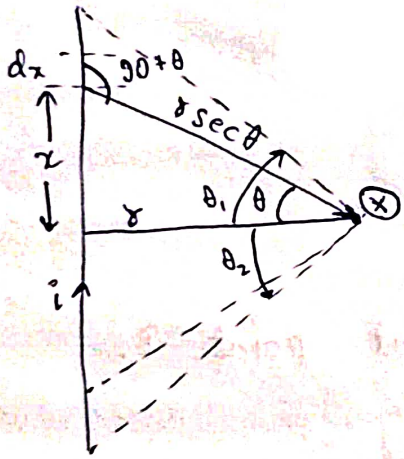
$$\boxed{B_p = 0}$$



$$\boxed{B = \frac{\mu_0 i}{4\pi\gamma} \left[\sin\theta_1 + \sin\theta_2\right]}$$

where, Q1 and Q2 are the angles made by the ends of the wire with the perpendicular to the wire from the point

Derivation :



$$dB = \left(\frac{\mu_0}{4\pi}\right) \frac{i\, dx \sin(90+\theta)}{\gamma^2 \sec^2\theta}$$

$$B = \int dB = \frac{\mu_0}{4\pi} \frac{i}{\gamma^2} \int \frac{dx \cos\theta}{\sec^2\theta}$$

$$= \frac{\mu_0}{4\pi} \cdot \frac{i}{\gamma^2} \int \frac{\gamma \sec^2\theta \cos\theta}{\sec^2\theta}\, d\theta$$

$$B = \frac{\mu_0 i}{4\pi\gamma} \int_{-\theta_2}^{\theta_1} \cos\theta\, d\theta$$

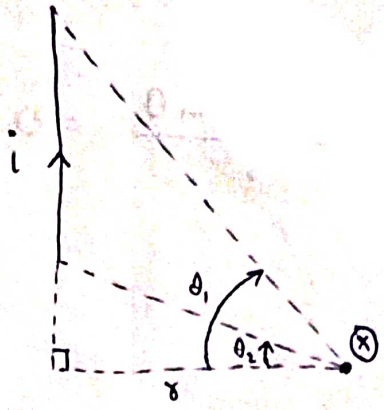
$$= \frac{\mu_0 i}{4\pi\gamma} \left(\sin\theta_1 - \sin(-\theta_2)\right)$$

$$= \frac{\mu_0 i}{4\pi\gamma} \left(\sin\theta_1 + \sin\theta_2\right)$$

$$\tan\theta = \frac{x}{\gamma}$$

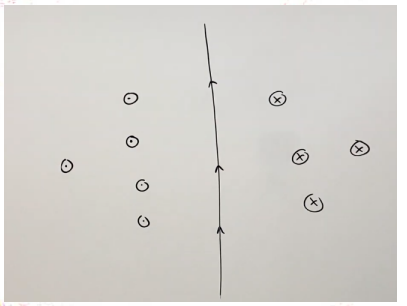
$$x = \gamma \tan\theta$$

$$dx = \gamma \sec^2\theta\, d\theta$$

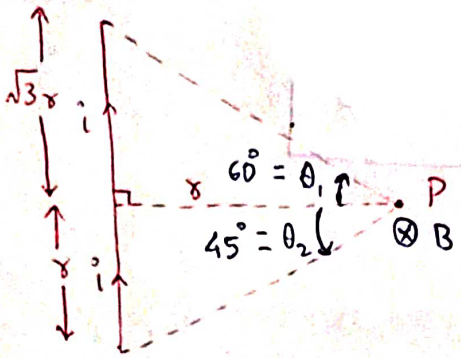


$$B = \frac{\mu_0 i}{4\pi r} \int_{\theta_2}^{\theta_1} \cos\theta \, d\theta$$

$$\boxed{B = \frac{\mu_0 i}{4\pi r} [\sin\theta_1 - \sin\theta_2]}$$



Q)



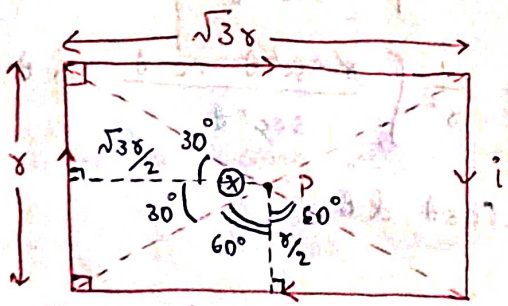
Find B at point P

Sol)

$$B = \frac{\mu_0 i}{4\pi r} (\sin 60° + \sin 45°)$$

$$= \frac{\mu_0 i}{4\pi r} \left(\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \right)$$

Q)

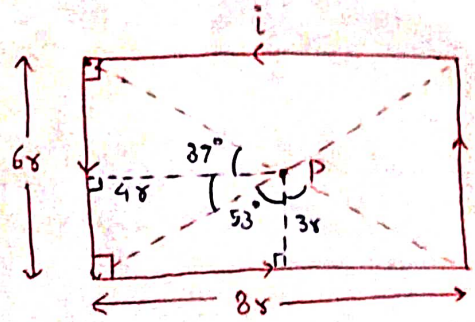


Find B at point P.

Sol)

$$\otimes \; B = \left[\frac{\mu_0 i}{4\pi\left(\frac{r}{2}\right)} \times 2\sin 60° + \frac{\mu_0 i}{4\pi\left(\frac{\sqrt{3}r}{2}\right)} \times 2\sin 30° \right] \times 2$$

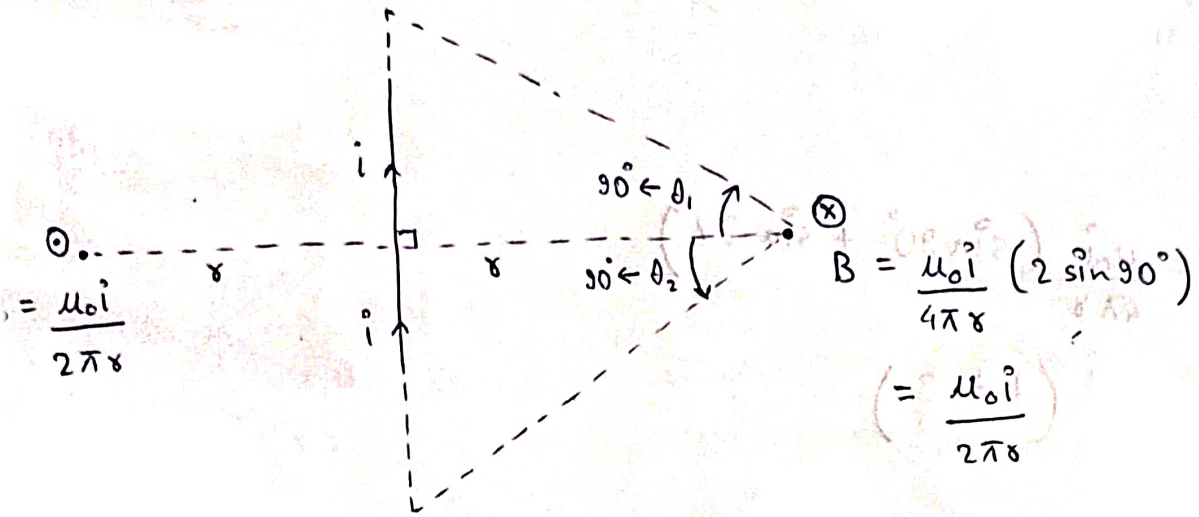
Q)



Find B at point P.

Sol $B = \left[\frac{\mu_0 i}{4\pi(3r)} \times 2\sin 53^\circ + \frac{\mu_0 i}{4\pi(4r)} \times 2\sin 37^\circ \right] \times 2$

$\vec{B}$ due to a Infinite Wire

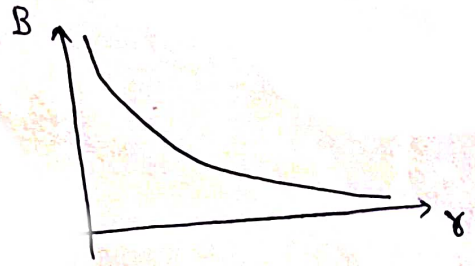


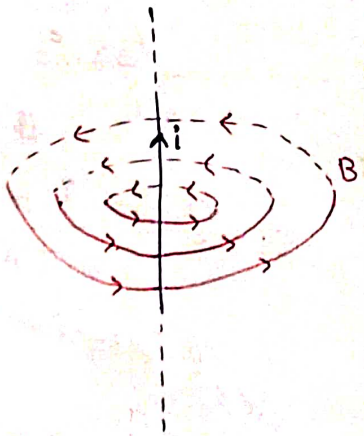
$= \frac{\mu_0 i}{2\pi r}$

$B = \frac{\mu_0 i}{4\pi r}(2\sin 90^\circ)$

$= \frac{\mu_0 i}{2\pi r}$

$B \propto \frac{1}{r}$

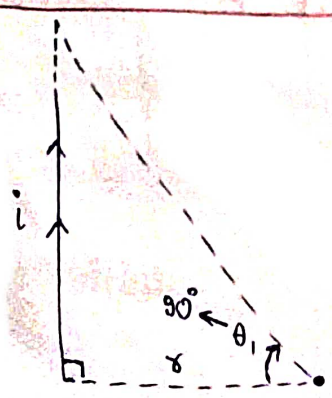




magnetic lines of forces due to an infinite wire

$\vec{B}$ due to semi-Infinite Wire

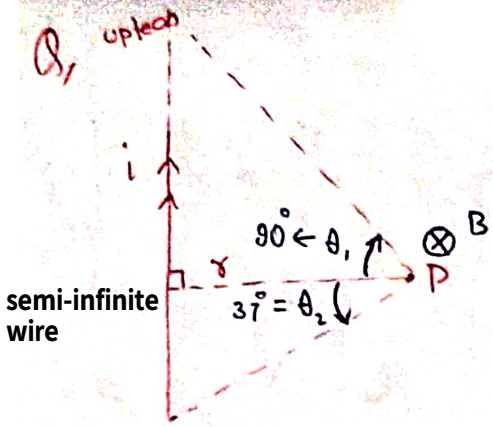
wire of infinite length, with one end at finite distance



$B = \frac{\mu_0 i}{4\pi r}(\sin 90^\circ + \sin 0^\circ)$

$= \frac{\mu_0 i}{4\pi r}$

Q, Find B at P.

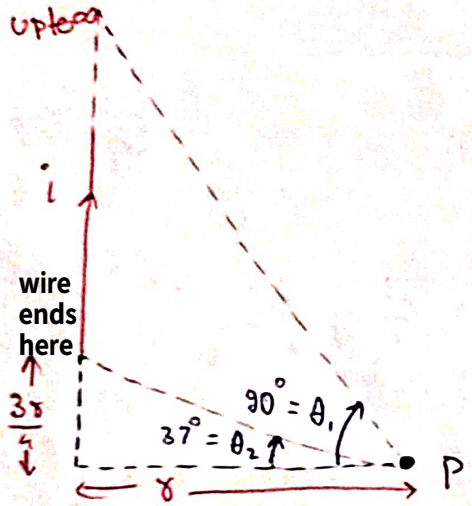


Sol

$$B = \frac{\mu_0 i}{4\pi\gamma}(\sin 90^\circ + \sin 37^\circ)$$

$$= \frac{\mu_0 i}{4\pi\gamma}\left(1 + \frac{3}{5}\right)$$

Q, Find B at point P.

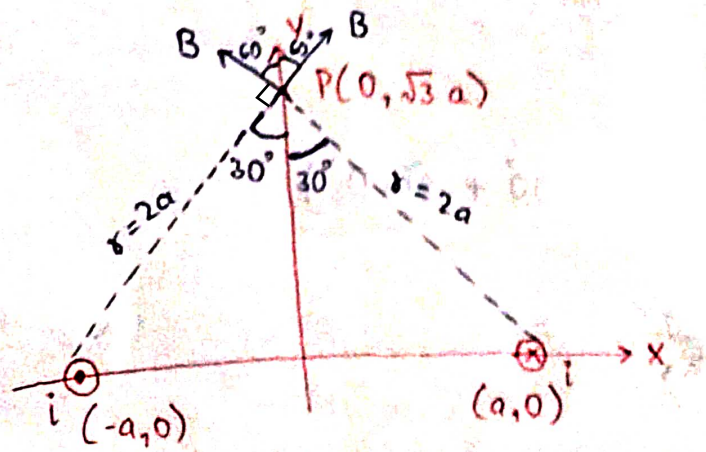


Sol

$$B = \frac{\mu_0 i}{4\pi\gamma}(\sin 90^\circ - \sin 37^\circ)$$

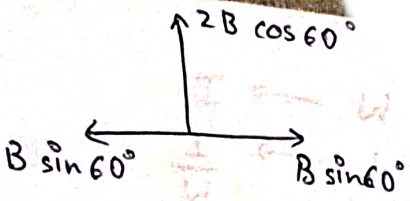
$$= \frac{\mu_0 i}{4\pi\gamma}\left(1 - \frac{3}{5}\right)$$

Q, Two infinite wires. Find B at P.



Sol

$$B = \frac{\mu_0 i}{2\pi(2a)}$$

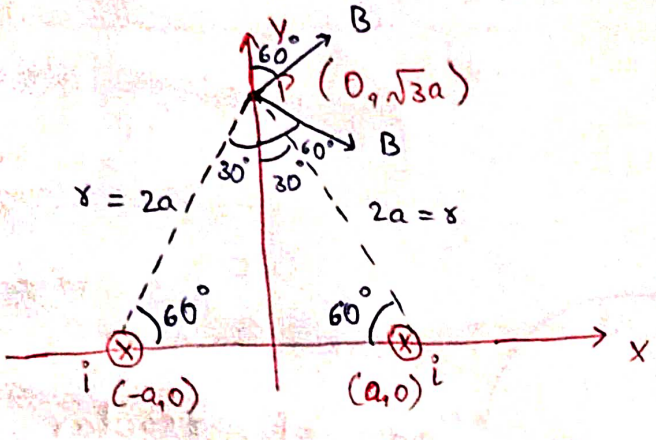


$$B_{net} = 2B\cos 60^\circ$$

$$= \frac{\mu_0 i}{4\pi a}$$

$$\vec{B}_{net} = \frac{\mu_0 i}{4\pi a}\hat{j}$$

Q)

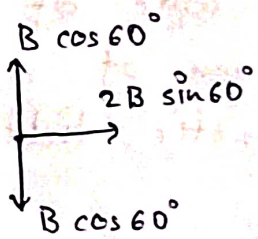


Two infinite wires. Find B at point P.

Sol

$$B = \frac{\mu_0 i}{2\pi(2a)}$$

$$= \frac{\mu_0 i}{4\pi a}$$



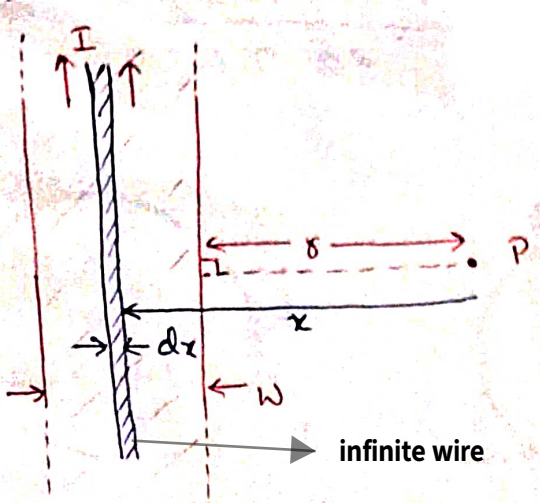
$$B_{net} = 2B\sin 60^\circ$$

$$= \sqrt{3}\,B$$

$$= \sqrt{3}\,\frac{\mu_0 i}{4\pi a}$$

$$\vec{B}_{net} = \sqrt{3}\,\frac{\mu_0 i}{4\pi a}\hat{i}$$

Q)



Infinitely long rectangular sheet. Find B at point P. I is uniformly distributed

Point P is in the same plane as this current carrying sheet

Sol

$$W \longrightarrow I$$

$$1 \longrightarrow \frac{I}{W}$$

$$dx \longrightarrow \frac{I}{W}\,dx$$

$$i = \frac{I}{W}\,dx$$

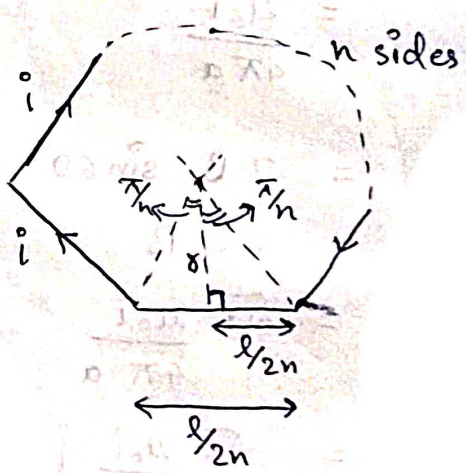
$$dB = \frac{\mu_0 i}{2\pi r} = \frac{\mu_0 \frac{I}{W}\,dx}{2\pi x}$$

$$B = \int dB = \frac{\mu_0 I}{2\pi W}\int_{r}^{r+W}\frac{dx}{x}$$

$$\boxed{B = \frac{\mu_0 I}{2\pi W}\ln\left(\frac{r+W}{r}\right)}$$

Q, A wire of length l is in a shape of regular polygon of n sides. Current 'i' is in the wire. Find magnetic field at centre of polygon.

Sol



$$\tan\frac{\pi}{n} = \frac{l/2n}{r}$$

$$r = \frac{l}{2n\tan\frac{\pi}{n}}$$

$$B = \frac{\mu_0 i}{4\pi r}(\sin\theta_1 + \sin\theta_2)$$

$$= \frac{\mu_0 i\,(2n\cdot\pi/n)}{4\pi l} \times 2\sin\frac{\pi}{n}$$

$$B_{net} = nB$$

$$= n^2\frac{\mu_0 i}{\pi l}\cdot\tan\frac{\pi}{n}\cdot\sin\frac{\pi}{n}$$

* $n \longrightarrow \infty$ Polygon $\longrightarrow$ circle

$$\tan\theta = \sin\theta \simeq \theta$$

$$B = n^2\frac{\mu_0 i}{\pi l} \times \frac{\pi}{n}\cdot\frac{\pi}{n}$$

$$B_{circle} = \frac{\mu_0 i n}{l}$$

$$= \frac{\mu_0 i n}{2\pi R}$$

$$\boxed{B_{circle} = \frac{\mu_0 i}{2R}}$$

$\vec{B}$ due to a Current Carrying Loop

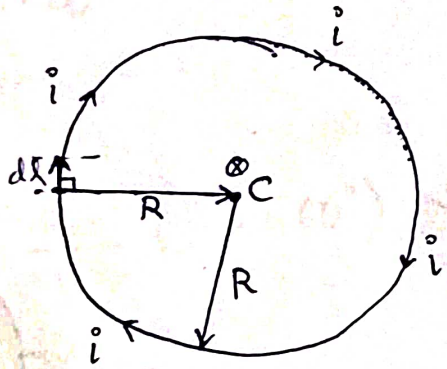
↳ Circular loop

At centre :

$l = n\, 2\pi R$

↳ no. of turns

can be fraction too
(for semicircle, quarter etc.)



$$dB = \frac{\mu_0}{4\pi} \frac{i\, dl \sin\theta}{r^2}$$

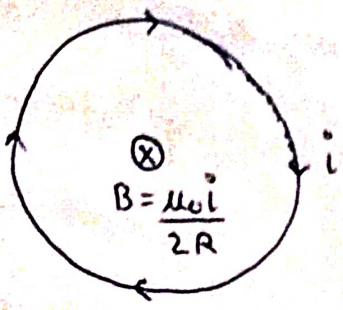
$$dB = \frac{\mu_0}{4\pi} \frac{i\, dl \sin 90^\circ}{R^2}$$

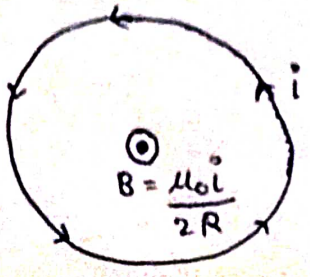
$$B = \int dB$$

$$= \frac{\mu_0}{4\pi} \frac{i}{R^2} \int dl$$

$$\boxed{B_{centre} = \frac{\mu_0 i l}{4\pi R^2}}$$

$$B_{centre} = \frac{\mu_0 i n\, 2\pi R}{4\pi R^2} \Rightarrow \boxed{B_{centre} = \frac{\mu_0 n i}{2R}}$$





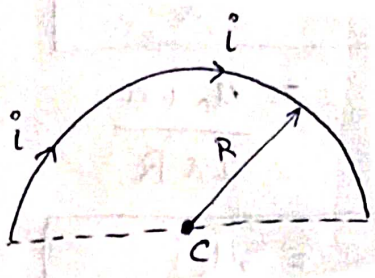
No. of turns = 1

curl your right hand in the direction of flow of current, thumb gives the direction of MF

2. Semi Circular Wire

$$B_c = \frac{\mu_0 i}{2R}\left(\frac{1}{2}\right) = \frac{\mu_0 i \pi R}{4\pi R^2}$$

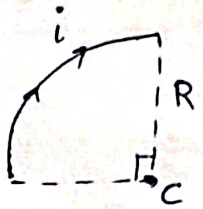
$$= \frac{\mu_0 i}{4R}$$



3. Quarter Wire

$$B_c = \frac{\mu_0 i}{4\pi R^2}\left(\frac{\pi}{2} R\right)$$

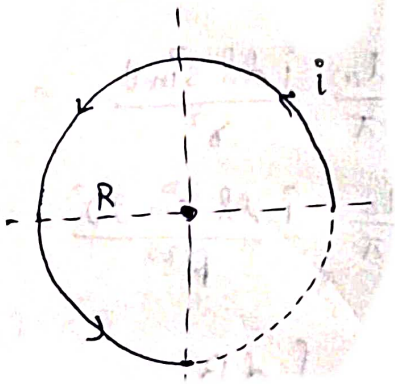
$$= \frac{\mu_0 i}{8R}$$



4. Three - Fourth Loop

$$B = \frac{\mu_0 i}{2R}\left(\frac{3}{4}\right)$$

$$= \frac{3\mu_0 i}{8R}$$



5. Arbitary Arc

$2\pi \longrightarrow 1$

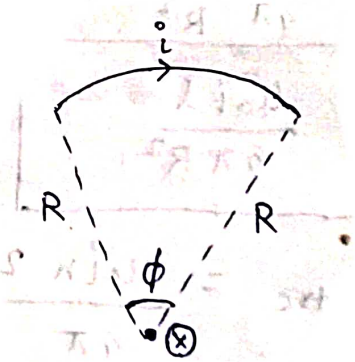
no. of turns

$1 \longrightarrow \frac{1}{2\pi}$

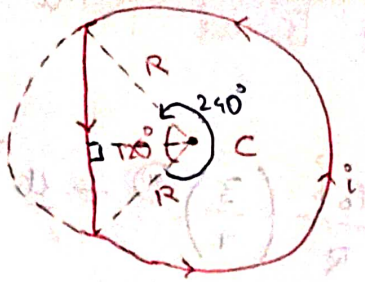
$\phi \longrightarrow \frac{1}{2\pi}\phi$

$$B = \frac{\mu_0 i \phi R}{4\pi R^2}$$

$$= \frac{\mu_0 i \phi}{4\pi R}$$



Q, Find B at point C.



Sol

no. of turns

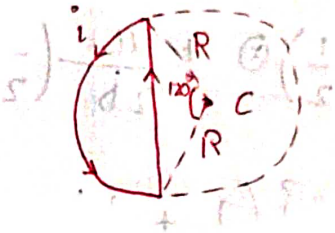
$360^\circ \longrightarrow 1$

$240^\circ \longrightarrow \frac{1}{360} \times 240$

$= \frac{2}{3}$

$$\odot B = \frac{\mu_0 i}{2R}\left(\frac{2}{3}\right)\odot + \frac{\mu_0 i}{4\pi(R/2)}\, 2\sin 60^\circ \odot$$

Q, Find B at point C.



Sol

no. of turns

$360^\circ \longrightarrow 1$

$120^\circ \longrightarrow \frac{1}{360^\circ} \times 120$

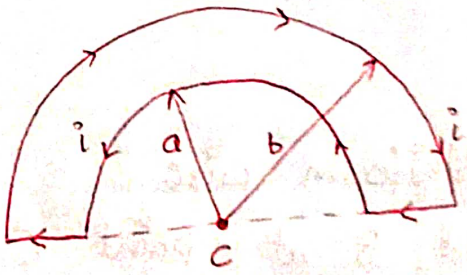
$= \frac{1}{3}$

$$B = \frac{\mu_0 i}{2R}\left(\frac{1}{3}\right)\odot + \frac{\mu_0 i}{4\pi R/2}\left(2\sin 60^\circ\right)\otimes$$

$$= \left(\frac{\mu_0 i}{6R} - \frac{\mu_0 i}{2\pi R}\sqrt{3}\right)\odot$$

$$= \left(\frac{\mu_0 i}{2\pi R}\sqrt{3} - \frac{\mu_0 i}{6R}\right)\otimes$$

Q, Find B at point C.



Sol

$$B = \frac{\mu_0 i}{4a}\odot + \frac{\mu_0 i}{4b}\otimes$$

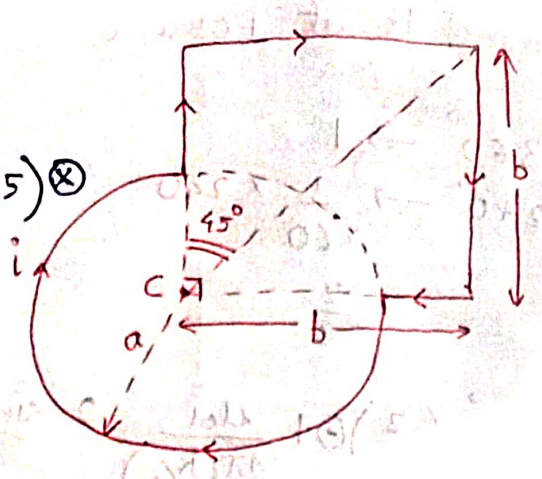
$$= \frac{\mu_0 i}{4a} - \frac{\mu_0 i}{4b}$$

$$= \frac{\mu_0 i}{4}\left(\frac{1}{a} - \frac{1}{b}\right)\odot$$

Q, Find B at point C.

Sol,

$$B = \frac{\mu_0 i}{2R}\left(\frac{3}{4}\right)\otimes + \frac{\mu_0 i}{4\pi b}(\sin 0 + \sin 45)\otimes + \frac{\mu_0 i}{4\pi b}(\sin 45 + \sin 0)\otimes$$

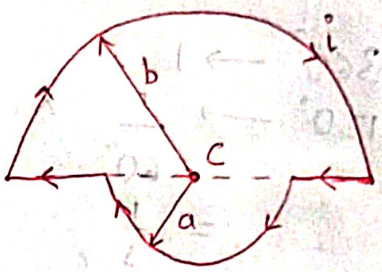


Q, Find B at point C.

Sol,

$$\frac{\mu_0 i}{2a}\left(\frac{1}{2}\right)\otimes + \frac{\mu_0 i}{2b}\left(\frac{1}{2}\right)\otimes = B$$

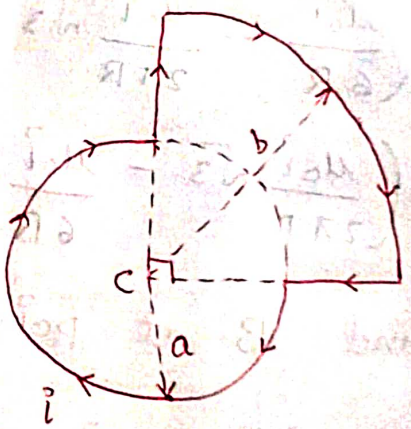
$$\frac{\mu_0 i}{4}\left(\frac{1}{a} + \frac{1}{b}\right) = B\otimes$$



Q, Find B at point C.

Sol,

$$B = \frac{\mu_0 i}{2a}\left(\frac{3}{4}\right)\otimes + \frac{\mu_0 i}{2b}\left(\frac{1}{4}\right)\otimes$$



Q, Uniform wire.

Find B at point C.

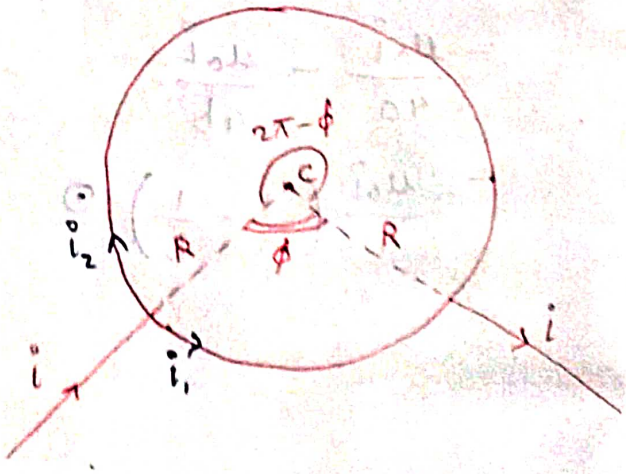
Sol,

no. of turns

$$2\pi \longrightarrow 1$$

$$2\pi - \phi \longrightarrow \frac{1}{2\pi}(2\pi - \phi)$$

$$\phi \longrightarrow \frac{1}{2\pi}\phi$$



$$i_1 + i_2 = i$$

$$\frac{i_1}{i_2} = \frac{R_2}{R_1} = l1/l2$$

$$= \frac{R(2\pi - \phi)}{R\phi}$$

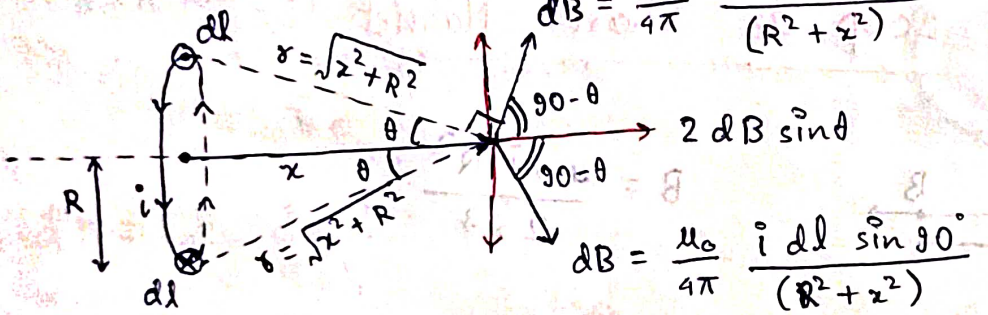
$$i_1 \phi = i_2 (2\pi - \phi)$$

$$B_c = \frac{\mu_0 i_2}{2R}\left(\frac{2\pi - \phi}{2\pi}\right)\otimes + \frac{\mu_0 i_1}{2R}\left(\frac{\phi}{2\pi}\right)\odot$$

$$= \frac{\mu_0}{4\pi R}\left(i_2(2\pi - \phi) - i_1\phi\right)\otimes$$

$$= 0$$

At Axis:



$$B = \int 2\,dB\,\sin\theta$$

$$= 2\frac{\mu_0}{4\pi}\frac{i}{(R^2+x^2)}\int \frac{dl \cdot R}{\sqrt{R^2+x^2}}$$

$$= \frac{\mu_0}{2\pi}\frac{iR}{(R^2+x^2)^{3/2}}\int_0^{l/2} dl$$

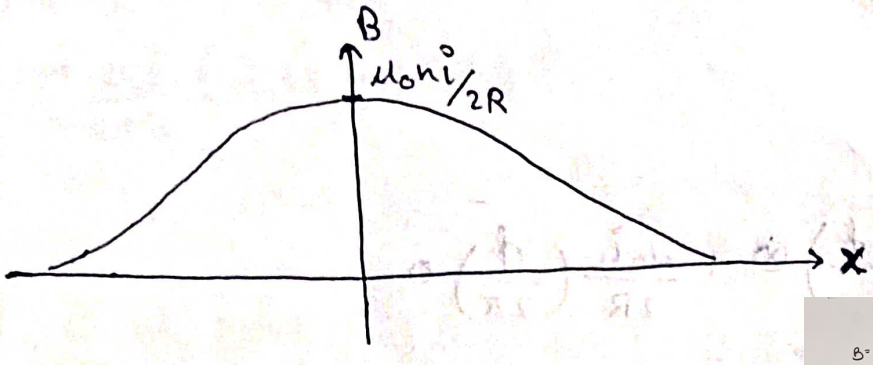
$$= \frac{\mu_0}{2\pi}\frac{iR}{(R^2+x^2)^{3/2}} \cdot \frac{l}{2}$$

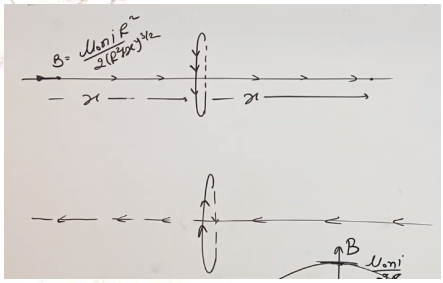
$$= \frac{\mu_0 i R}{4\pi(R^2+x^2)^{3/2}} \quad n \cdot 2\pi R$$

$$\boxed{B = \frac{\mu_0 n i R^2}{2(R^2+x^2)^{3/2}}}$$

* for $x = 0$, $B_c = \frac{\mu_0 n i}{2R}$

* n is an integer. **hence, this formula is not valid for semi-circle, quarter etc.**



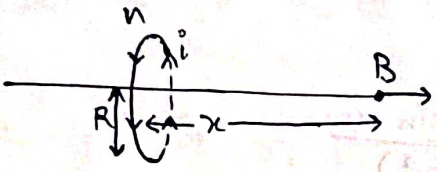


* if $R << x$

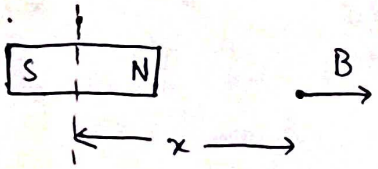
$$B = \frac{\mu_0 n i R^2}{2x^3}$$

when loop is small

Current Carrying Loop as Short Magnet



$$B = \frac{\mu_0 n i R^2}{2x^3}$$

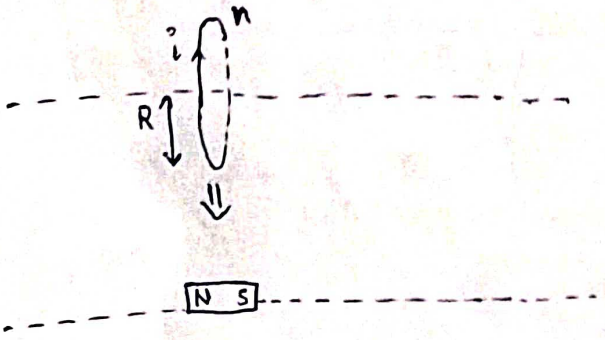


$$B = 2\frac{\mu_0}{4\pi} \cdot \frac{M}{x^3}$$

$$\Rightarrow \frac{\mu_0 n i R^2}{2x^3} = 2\frac{\mu_0}{4\pi} \cdot \frac{M}{x^3}$$

as both vary similarly with x, we can replace a bar magnet with current carrying loop and vice versa

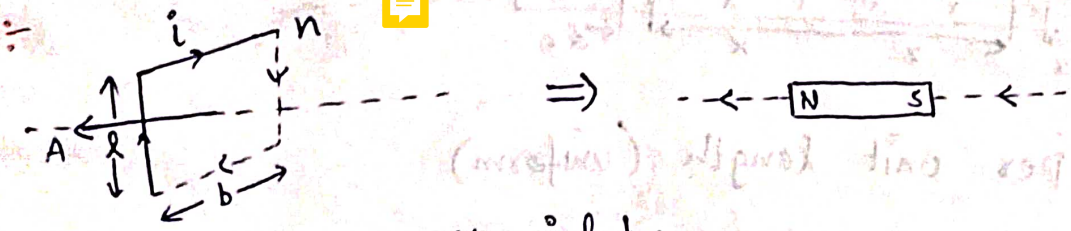
$$M = i(n\pi R^2) = iA$$



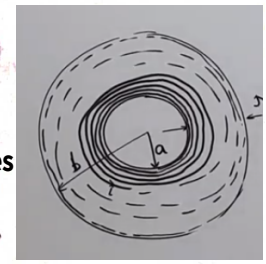
$$M = i(n\pi R^2)$$

$$\vec{M} = i\vec{A}$$

eg :-



$$M = i\,l\,b\,n$$



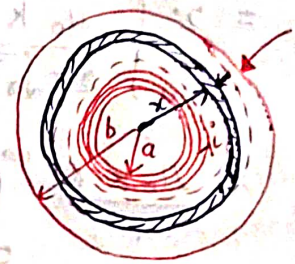
tightly wound wires

Q, Find magnetic dipole moment of this system.

Sol

$b-a \longrightarrow N$ turns

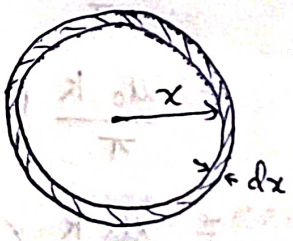
$dx \longrightarrow \dfrac{N}{b-a}\,dx$ turns



$$dM = \frac{N}{b-a}\,dx\; i\,\pi\, x^2$$

$$M = \frac{N}{(b-a)}\, i\,\pi \int_a^b x^2\,dx$$

$$= \frac{N i \pi}{3(b-a)}\,(b^3-a^3)$$



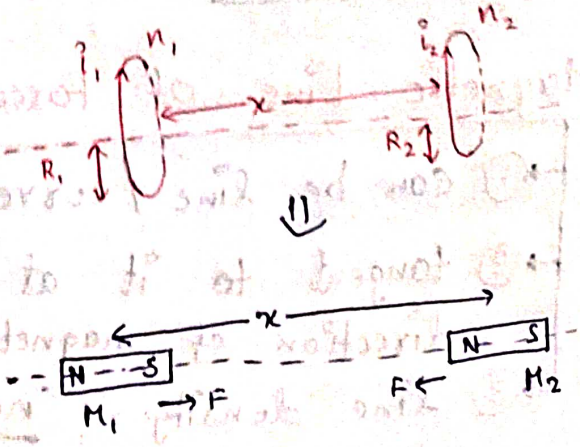
Q, $x >> R_1$ or R_2

Find magnetic force between the loops.

Sol

$$M_1 = (n_1 \pi R_1^2)\, i_1$$

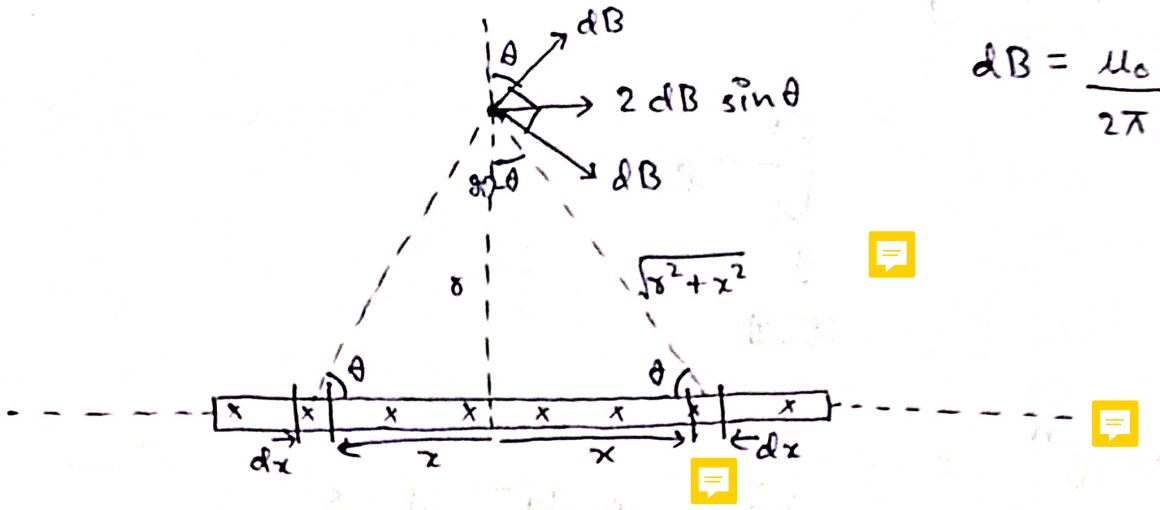
$$M_2 = (n_2 \pi R_2^2)\, i_2$$



$$F = 6\,\frac{\mu_0}{4\pi}\,\frac{M_1 M_2}{x^4}$$

$$= 6\,\frac{\mu_0}{4\pi}\,\frac{(n_1 \pi R_1^2 i_1)(n_2 \pi R_2^2 i_2)}{x^4}$$

$\vec{B}$ due to Current Carrying Infinite Sheet



$$dB = \frac{\mu_0 (k\,dx)}{2\pi\sqrt{\gamma^2+x^2}}$$

$k \rightarrow$ current per unit length (uniform)

$$B_{net} = 2\int_0^\infty dB \sin\theta$$

$$= 2\int_0^\infty \frac{\mu_0 k\,dx}{2\pi\sqrt{\gamma^2+x^2}} \frac{\gamma}{\sqrt{\gamma^2+x^2}}$$

$$= \frac{\mu_0 k \gamma}{\pi}\int_0^\infty \frac{dx}{\gamma^2+x^2}$$

$$= \frac{\mu_0 k \gamma}{\pi}\cdot\frac{1}{\gamma}\left(\tan^{-1}\frac{x}{\gamma}\right)_0^\infty$$

$$= \frac{\mu_0 k}{\pi}\left(\tan^{-1}\infty - \tan^{-1}0\right)$$

$$= \frac{\mu_0 k}{\pi}\left(\frac{\pi}{2} - 0\right)$$

$$\boxed{B = \frac{\mu_0 k}{2}}$$

this magnetic field is uniform & independent of distance from the sheet

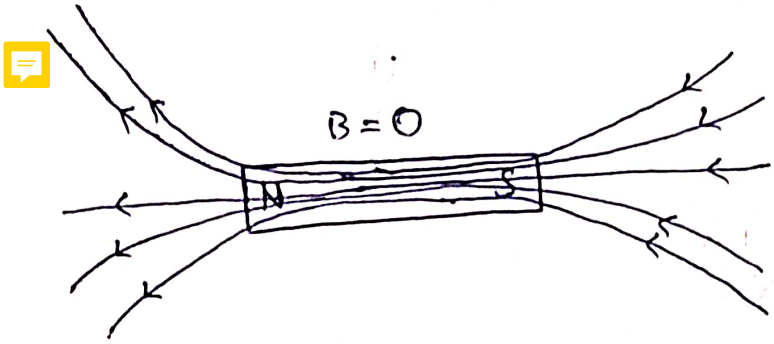
(just like EF in an infinitely large sheet)

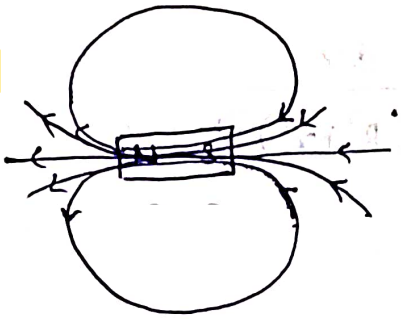
Magnetic Line of Force

1. can be line / curve
2. tangent to it at any point on it, will provide direction of magnetic field.
3. line density = $\frac{\text{no. of MLOF}}{\text{area}}$

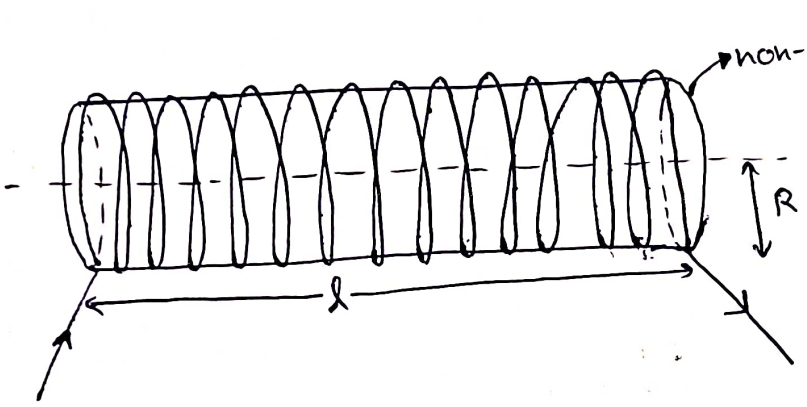
line density ↑ ⇒ M·F ↑

line density ↓ ⇒ M·F ↓





Solenoid



t → thickness of wire / diameter of wire (in meter)

N → total no. of turns

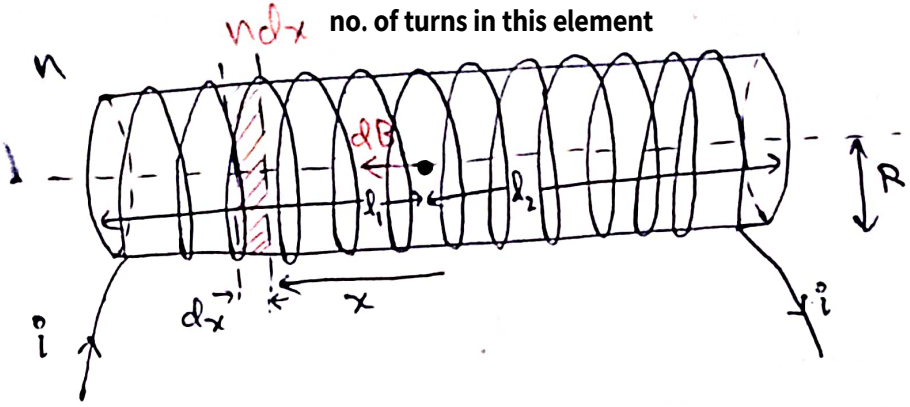
$$N t = l$$

$$n = \frac{\text{no. of turns}}{\text{length}}$$

$$= \frac{N}{l}$$

$$= \frac{l}{t\,l}$$

$$\boxed{n = \frac{1}{t}}$$



$$dB = \frac{\mu_0 (n\,dx)\, i\, R^2}{2(R^2+x^2)^{3/2}}$$

$$B = \int dB = \frac{\mu_0 n i R^2}{2} \int_{-l_1}^{l_2} \frac{dx}{(R^2+x^2)^{3/2}}$$

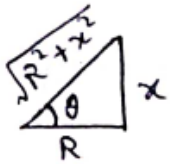
$$\int \frac{dx}{(R^2+x^2)^{3/2}}$$

$$x = R\tan\theta$$
$$dx = R\sec^2\theta\, d\theta$$

$$\int \frac{R\sec^2\theta\, d\theta}{R^3\sec^3\theta}$$

$$\frac{1}{R^2}\int \cos\theta\, d\theta$$

$$\frac{1}{R^2}(\sin\theta)$$

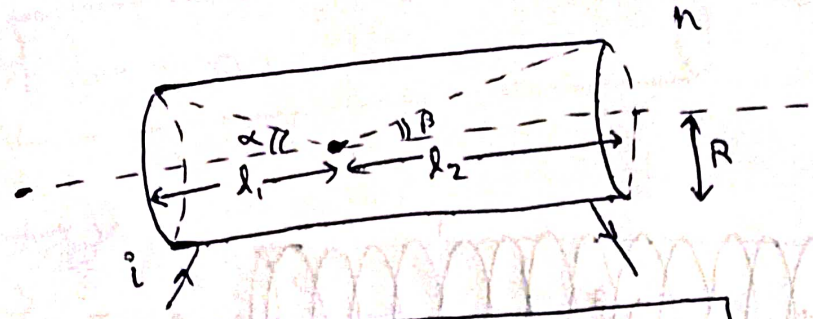


$$\frac{1}{R^2}\left(\frac{x}{\sqrt{R^2+x^2}}\right)$$

$$B = \frac{\mu_0 n i}{2}\left(\frac{x}{\sqrt{R^2+x^2}}\right)_{-l_1}^{l_2}$$

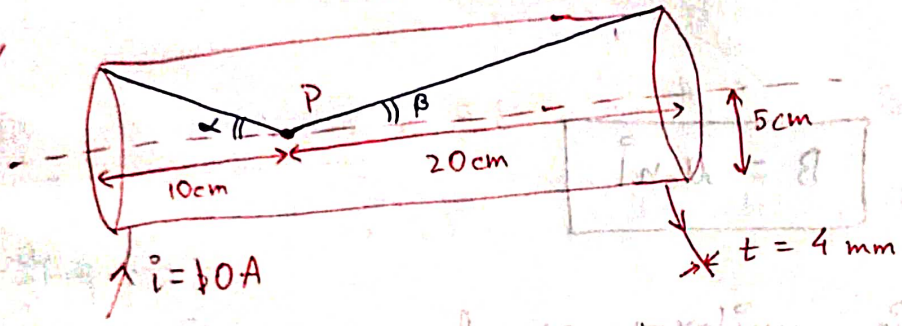
$$\boxed{B = \frac{\mu_0 n i}{2}\left(\frac{l_2}{\sqrt{R^2+l_2^2}} + \frac{l_1}{\sqrt{R^2+l_1^2}}\right)}$$

Formula



$$B = \frac{\mu_0 n i}{2}(\cos\beta + \cos\alpha)$$

Q,



Find B at point P

Sol,

$$n = \frac{1}{t} = \frac{1}{4\times10^{-3}} = 250\ /m$$

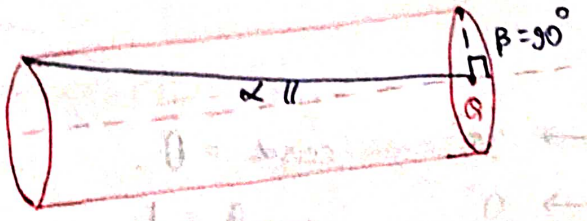
$$B_p = \frac{\mu_0 \times 250 \times 10}{2}\left(\frac{10}{\sqrt{125}} + \frac{20}{\sqrt{425}}\right)$$

$$B_p = \frac{4\pi \times 10^{-7} \times 25 \times 10}{2}\left(\frac{10}{\sqrt{125}} + \frac{20}{\sqrt{425}}\right)$$

Q, In previous question, find B at point Q

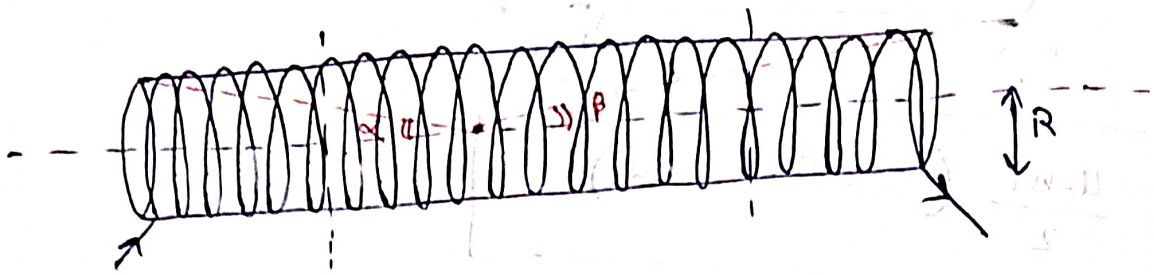
Sol,

$$B_q = \frac{\mu_0 \times 25 \times 10}{2}\left(\frac{30}{\sqrt{925}} + 0\right)$$



Ideal Solenoid

- very long
- $\ell >> R$



$\alpha, \beta \longrightarrow 0$

$\cos\alpha \simeq \cos\beta \simeq 1$

$$B = \frac{\mu_0 n i}{2}(1+1) \Rightarrow \boxed{B = \mu_0 n i}$$

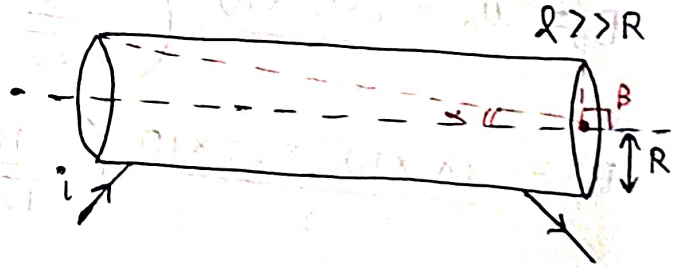
*This magnetic field is uniform as long as distance of that point to the end is very larger than R. As you move towards the ends α and β becomes larger and hence magnetic field becomes non-uniform.

$\alpha \longrightarrow 0^\circ \quad \cos\alpha = 1$

$\beta \longrightarrow 90^\circ \quad \cos\beta = 0$

$$B_{end} = \frac{\mu_0 n i}{2}(1+0)$$

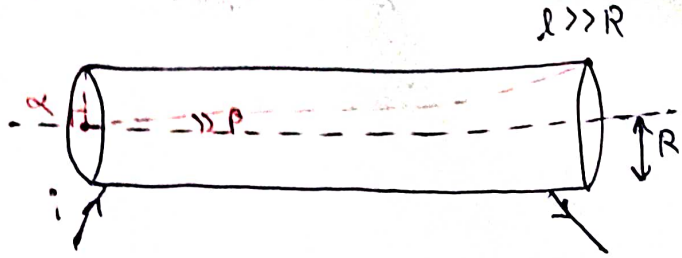
$$= \frac{\mu_0 n i}{2}$$



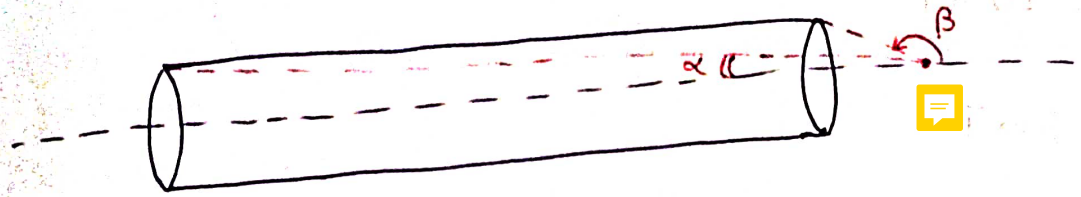
$\alpha \longrightarrow 90^\circ \quad \cos\alpha = 0$

$\beta \longrightarrow 0^\circ \quad \cos\beta = 1$

$$B_{end} = \frac{\mu_0 n i}{2}(1+0) = \frac{\mu_0 n i}{2}$$



$\ell >> R$



$\alpha \longrightarrow 0 \qquad \cos\alpha = 1$

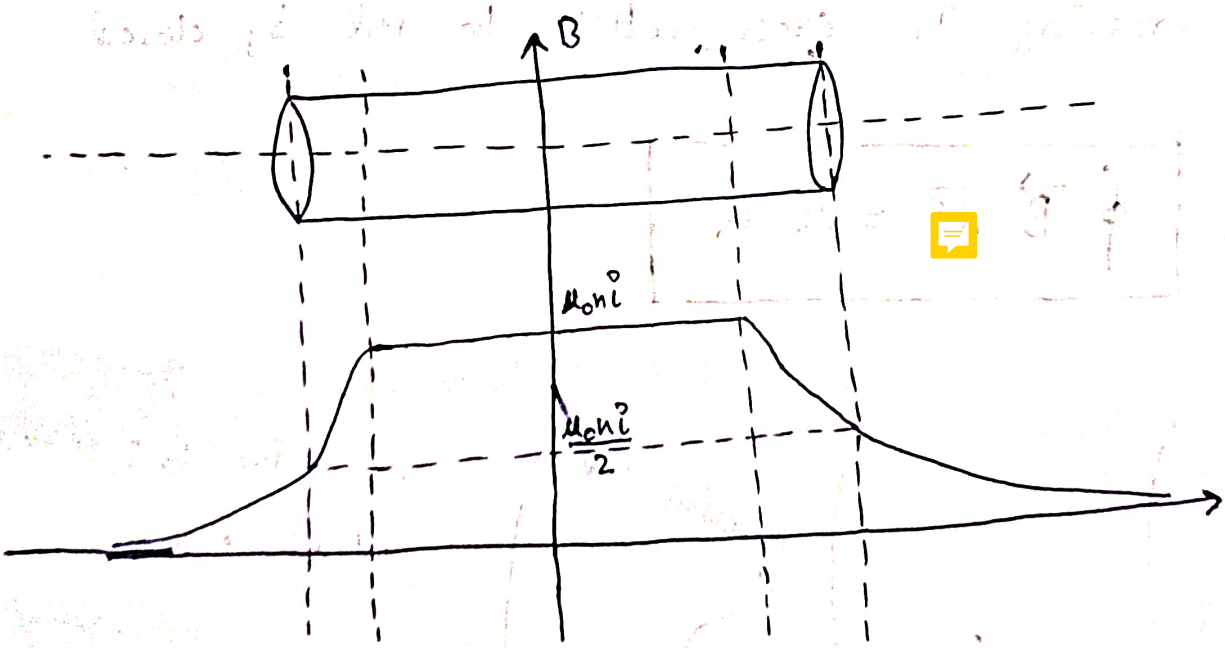
$\beta > 90 \qquad \cos\beta = (-)$

$$B_{out} = \frac{\mu_0 n i}{2}\left(1 + (-)\right)$$

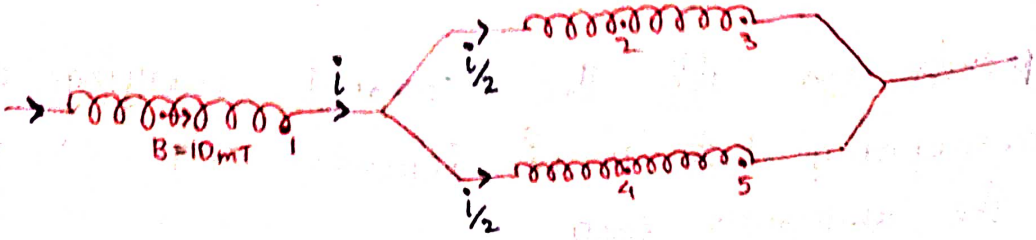
$$\boxed{B_{out} < \frac{\mu_0 n i}{2}}$$

As you move far : $\beta \uparrow \quad \cos\beta \downarrow \quad B \downarrow$

at ∞ : $\beta \longrightarrow 180^\circ \quad \cos\beta = -1 \quad B = 0$



Q/ 3 ideal identical solenoids



Find B at points 1, 2, 3, 4 and 5.

Sol $B = \mu_0 n i = 10\,mT$

$$B_1 = \frac{\mu_0 n i}{2} = 5\,mT$$

$$B_2 = \mu_0 n\left(\frac{i}{2}\right) = 5\,mT = B_4$$

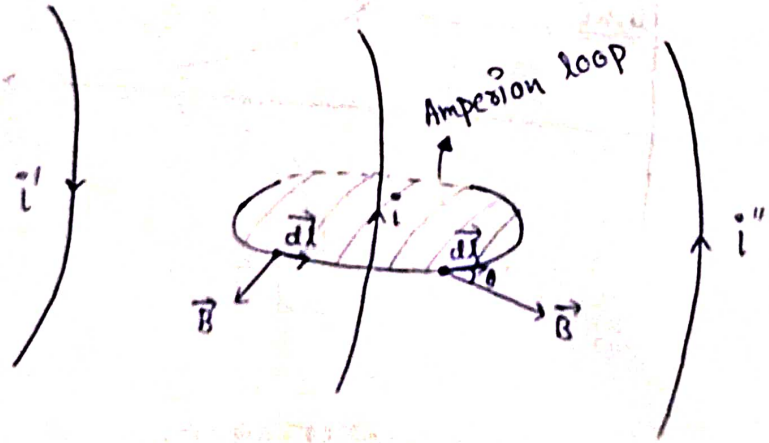
$$B_3 = \frac{\mu_0 n(i/2)}{2} = \frac{5}{2}\,mT = B_5$$

Ampere's Law

electrostatics : symmetric system : Gauss law $\Rightarrow$ EF
magnetism : symmetric system : Ampere's law $\Rightarrow$ M.F

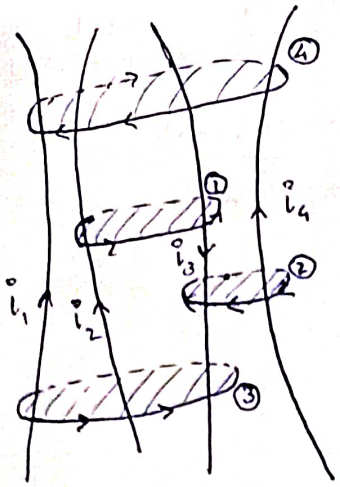
Line integral of magnetic field in a closed loop is always equal to permeability of free space (μ_0) times current crossing the cross-section, bounded by closed loop.

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 i_{in}$$



$\vec{B}$ → due to i, i', i''
$\vec{B} \cdot \vec{dl}$ → due to i

* $\vec{B}$ depends on all the current carrying elements in the sorrounding but $\vec{B} \cdot \vec{dl}$ depends only on current inside the amperian loop.



① $\oint \vec{B} \cdot \vec{dl} = \mu_0 (i_2 - i_3)$

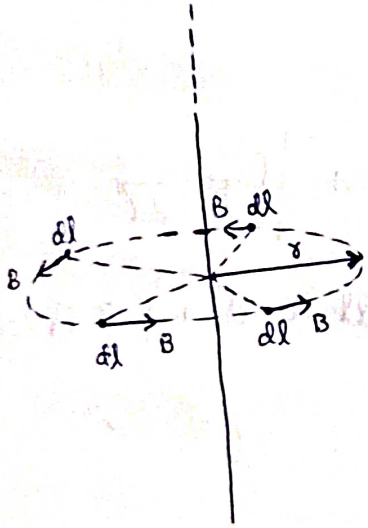
② $\oint \vec{B} \cdot \vec{dl} = \mu_0 (i_3 - i_4)$

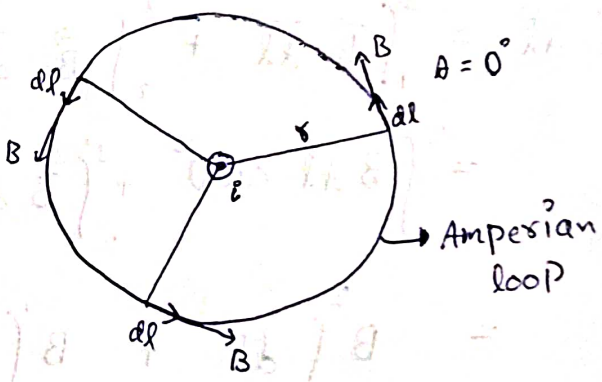
③ $\oint \vec{B} \cdot \vec{dl} = \mu_0 (i_1 + i_2 - i_3)$

④ $\oint \vec{B} \cdot \vec{dl} = \mu_0 (i_1 + i_2 - i_3 + i_4)$

$\vec{B}$ due to Infinite Wire

top view :





$$\oint \vec{B} \cdot \vec{dl} = \oint B\, dl \cos 0^\circ = \mu_0 i$$

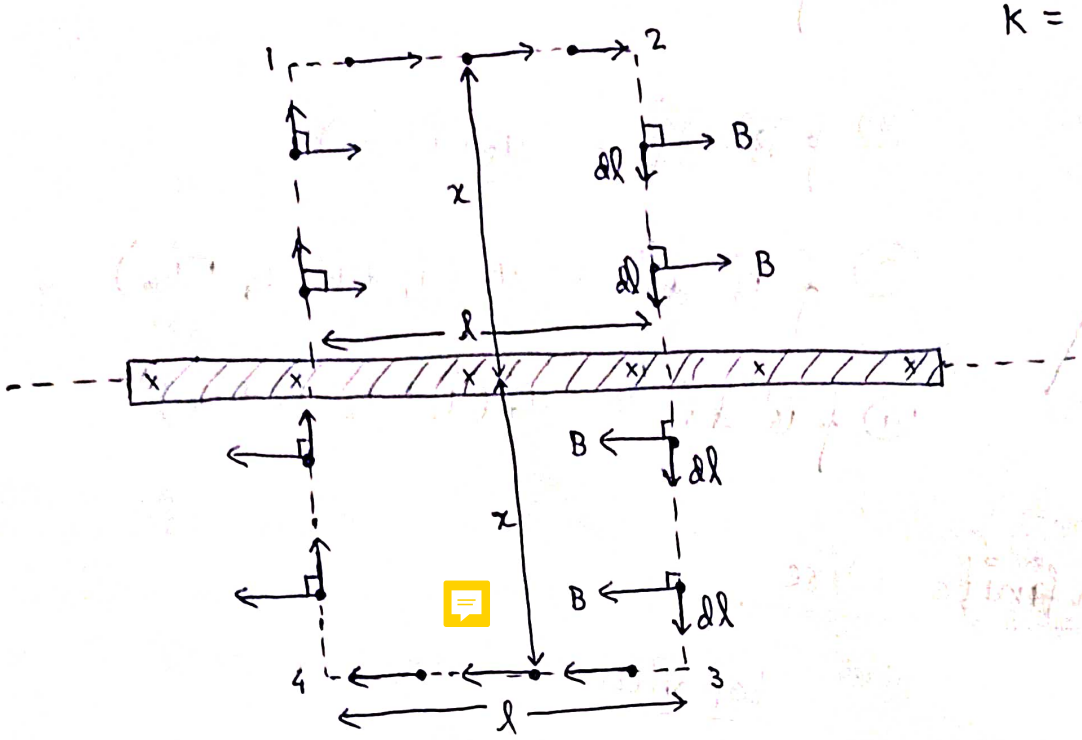
$$= B \oint dl = \mu_0 i$$

$$= B \cdot 2\pi r = \mu_0 i$$

$$\boxed{B = \frac{\mu_0 i}{2\pi r}}$$

$\vec{B}$ due to Infinitely Large Current Carrying Sheet

$K = \frac{\text{current}}{\text{length}}$



$$\oint \vec{B} \cdot \vec{dl} = \int_1^2 \vec{B} \cdot \vec{dl} + \int_2^3 \vec{B} \cdot \vec{dl} + \int_3^4 \vec{B} \cdot \vec{dl} + \int_4^1 \vec{B} \cdot \vec{dl}$$

$$= \int_1^2 B\, dl \cos 0^\circ + \int_2^3 B\, dl \cos 90^\circ + \int_3^4 B\, dl \cos 0^\circ + \int_4^1 B\, dl \cos 90^\circ$$

$$= B \int_1^2 dl + B \int_3^4 dl$$

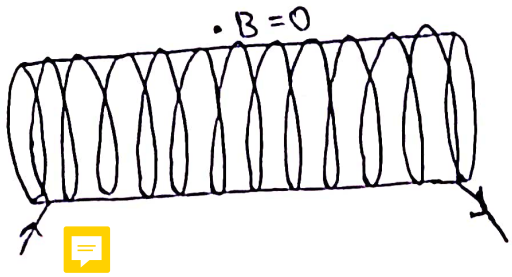
$$= B\,l + B\,l$$

$$= 2\,B\,l$$

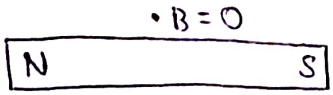
$$2\,B\,l = \mu_0 (k\,l)$$

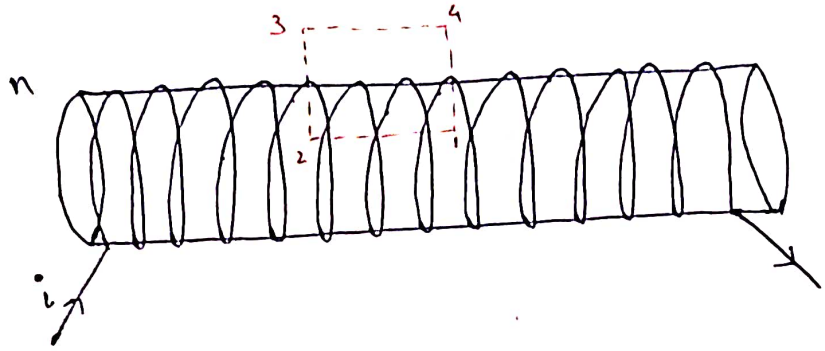
$$\boxed{B = \frac{\mu_0 k}{2}}$$

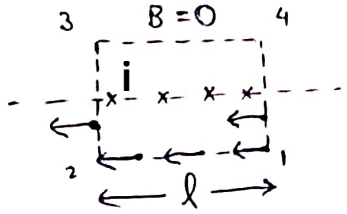
$\vec{B}$ due to Ideal Solenoid



$\Rightarrow$







by Ampere's Law,

$$\oint \vec{B}\cdot\vec{dl} = \int_1^2 \vec{B}\cdot\vec{dl} + \int_2^3 \vec{B}\cdot\vec{dl} + \int_3^4 \vec{B}\cdot\vec{dl} + \int_4^1 \vec{B}\cdot\vec{dl}$$

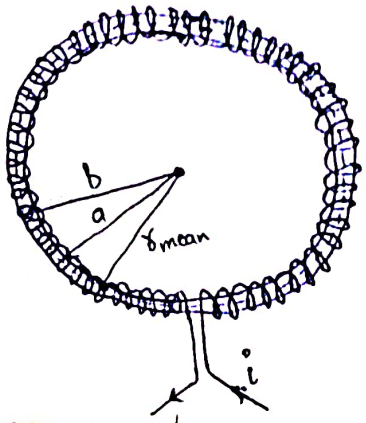
$$= \int B\,dl\cos 0^\circ + 0 + 0 + 0$$

$$= B\,l$$

$$B\,l = \mu_0\, i\, n\, l$$

$$\boxed{B = \mu_0\, n\, i}$$

Ideal Toroid



$$r_{mean} = \frac{a+b}{2} = r$$

N → total no. of turns

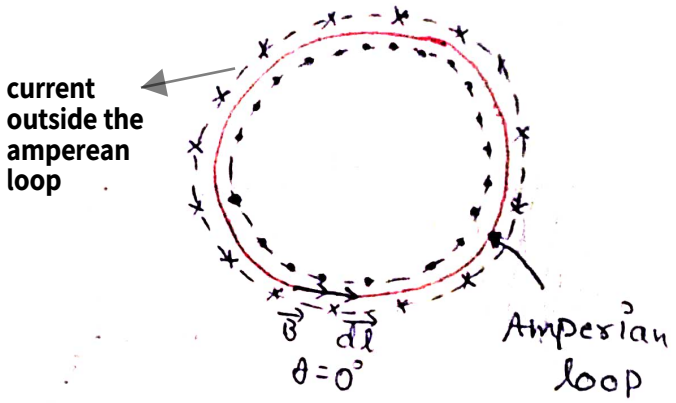
$$n = \frac{N}{2\pi r}$$

$$\oint \vec{B} \cdot \vec{dl} = \oint B\, dl \cos 0^\circ$$

$$= B \oint dl$$

$$= B \cdot 2\pi r$$

$$= \mu_0 (Ni)$$

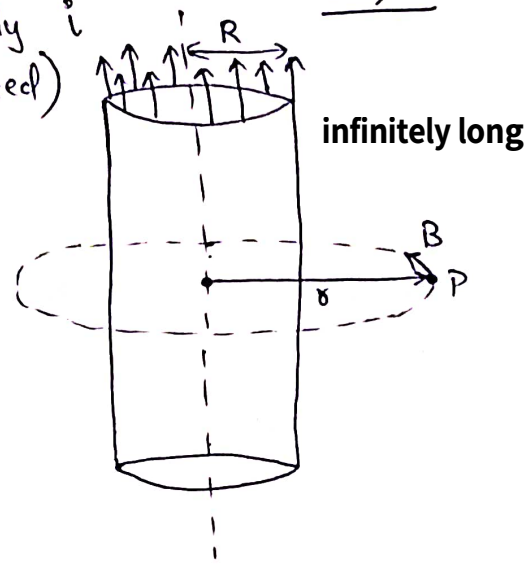


$$B = \frac{\mu N i}{2\pi r}$$ = μ0(ni)

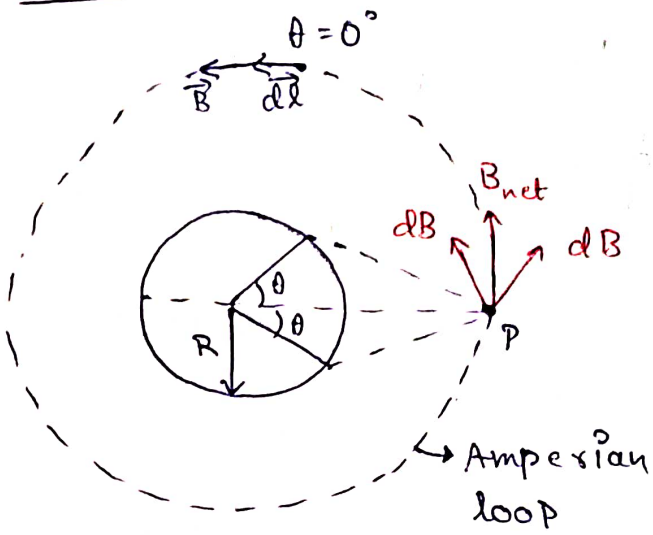
Magnetic Field due to Current Carrying Long Hollow Cylinder

(uniformly i distributed)

$r \geq R$.



top view :



$$\oint \vec{B} \cdot \vec{dl} = \oint B\, dl \cos 0^\circ$$

$$= B \oint dl$$

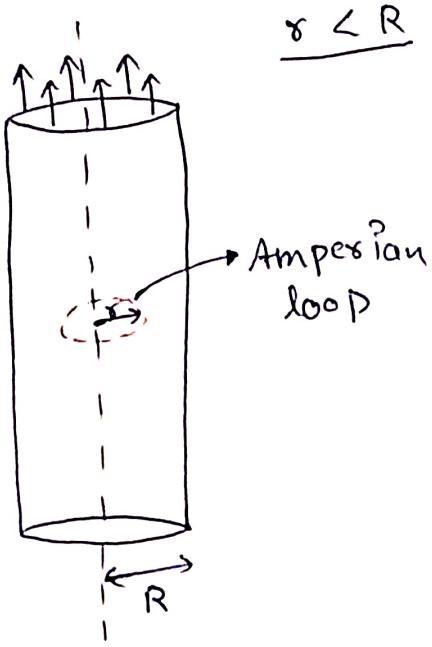
$$= B \cdot 2\pi r$$

$$B \cdot 2\pi r = \mu_0 i$$

$$B_{out} = \frac{\mu_0 i}{2\pi r}$$

* For outside points, this infinitely long current carrying hollow cylinder is behaving like infinite wire placed at axis.

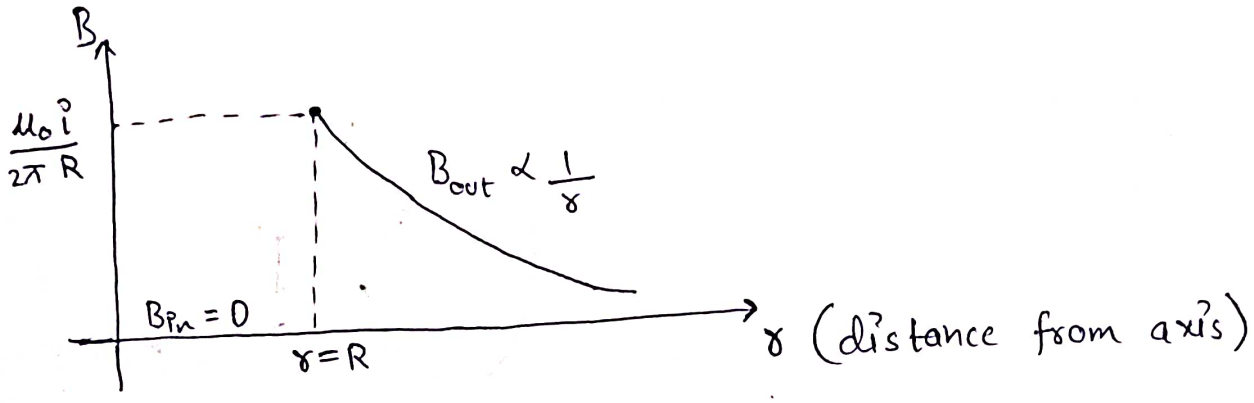
$r < R$



$$\oint \vec{B} \cdot \vec{dl} = B \cdot 2\pi r$$

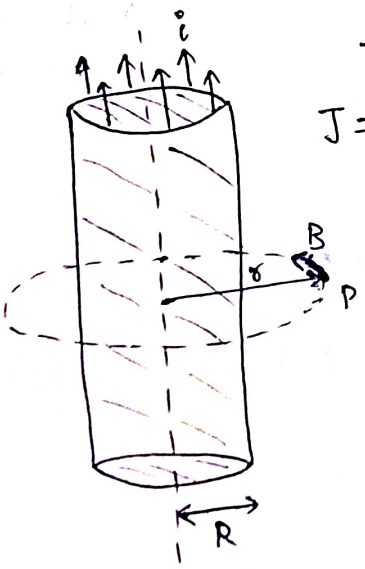
$$\Rightarrow \quad B \cdot 2\pi r = \mu_0 (0)$$

$$\boxed{B_{in} = 0}$$



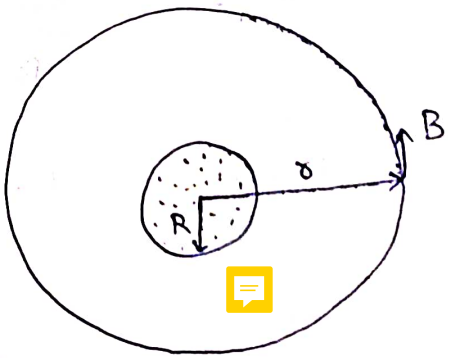
Magnetic Field due to Long Solid Cylinder

$r \geq R$



$J = \frac{i}{\pi R^2}$ → Current density

top view :



$$\oint \vec{B} \cdot \vec{dl} = B \cdot 2\pi r$$

$$\Rightarrow \quad B \cdot 2\pi r = \mu_0 i = \mu_0 J \pi R^2$$

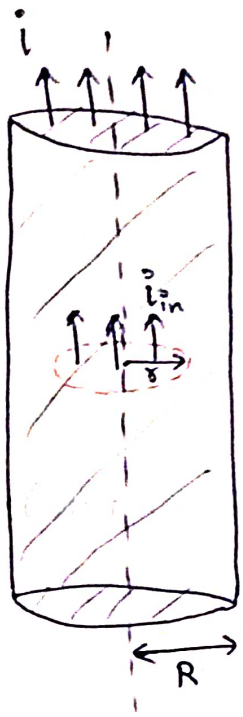
$$\boxed{B_{out} = \frac{\mu_0 i}{2\pi r} = \frac{\mu_0 J R^2}{2r}}$$

thus, behaves like a hollow cylinder/ infinitely long wire placed on the axis

$$\vec{B}_{out} = \frac{\mu_0 R^2 (\vec{J} \times \vec{r})}{2r^2}$$

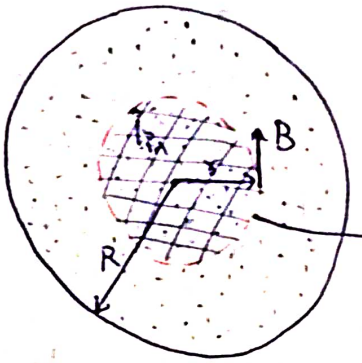
* $B_{out} \propto \frac{1}{r}$

$r \leq R$



$J = \frac{i}{\pi R^2}$

top view :



$\pi R^2 \rightarrow i$

$\pi r^2 \rightarrow \frac{i}{\pi R^2} \times \pi r^2$

$$\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi r$$

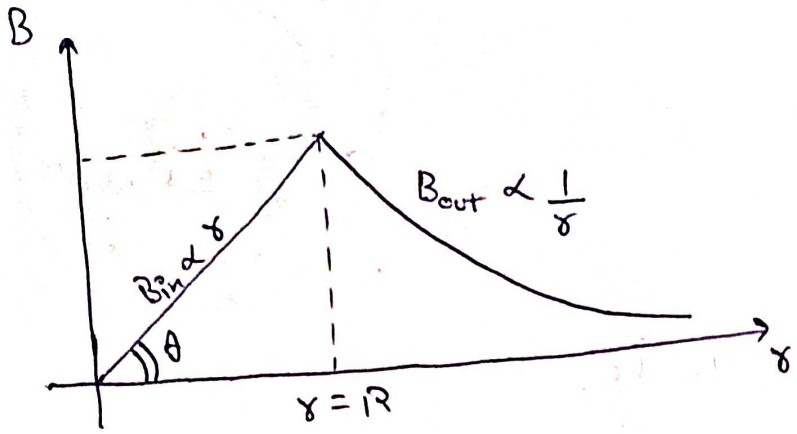
$$\Rightarrow B \cdot 2\pi r = \mu_0 i_{in}$$

$$B \cdot 2\pi r = \frac{\mu_0 i r^2}{R^2} = \mu_0 J \pi r^2$$

$$B_{in} = \frac{\mu_0 i r}{2\pi R^2} = \frac{\mu_0 J r}{2}$$

$$\vec{B}_{in} = \frac{\mu_0}{2} (\vec{J} \times \vec{r})$$

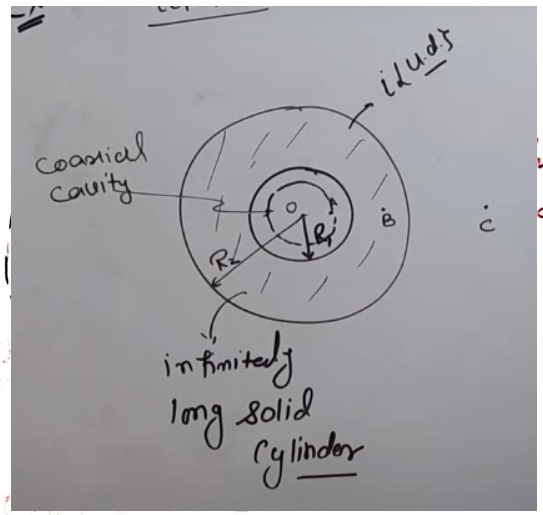
* $B_{in} \propto r$



$$\tan\theta = \text{slope} = \frac{\mu_0 i}{2\pi R^2} = \frac{\mu_0 J}{2}$$

Q/ top view :

Find B at points A, B, C.



Sol/

$$A: \oint \vec{B}\cdot\vec{dl} = B\cdot 2\pi r = \mu_0(0)$$

$$B_A = 0$$

$$B: \oint \vec{B}\cdot\vec{dl} = B\cdot 2\pi r = \mu_0(i_{in})$$

$$B\cdot 2\pi r = \mu_0 i\left(\frac{r^2 - R_1^2}{R_2^2 - R_1^2}\right)$$

$$B_B = \frac{\mu_0 i}{2r\pi}\left(\frac{r^2 - R_1^2}{R_2^2 - R_1^2}\right) \quad ; r = OB$$

$$C: \oint \vec{B}\,\vec{dl} = B\cdot 2\pi r = \mu_0 i$$

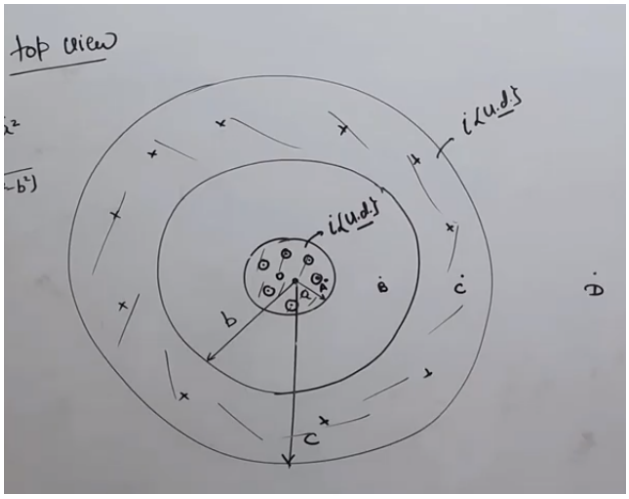
$$B_C = \frac{\mu_0 i}{2\pi r} \quad ; r = OC$$

Q, Find B at A, B, C and D

Sol

$$J_{in} = \frac{i}{\pi a^2}$$

$$J_{out} = \frac{i}{\pi(c^2 - b^2)}$$



taking appropriate amperean loops:

$$A: \oint \vec{B} \cdot \vec{dl} = B \cdot 2\pi r = \mu_0 \frac{i r^2}{a^2}$$

$$B_A = \frac{\mu_0 i r}{2\pi a^2}$$

here, r=OA

$$B: \oint \vec{B} \cdot \vec{dl} = B \cdot 2\pi r = \mu_0 i$$

$$B_B = \frac{\mu_0 i}{2\pi r}$$

here, r=OB

$$C: \oint \vec{B} \cdot \vec{dl} = B \cdot 2\pi r = \mu_0 (i_{in})$$

$$B \cdot 2\pi r = \mu_0 \left[i - \frac{i(r^2 - b^2)}{(c^2 - b^2)} \right]$$

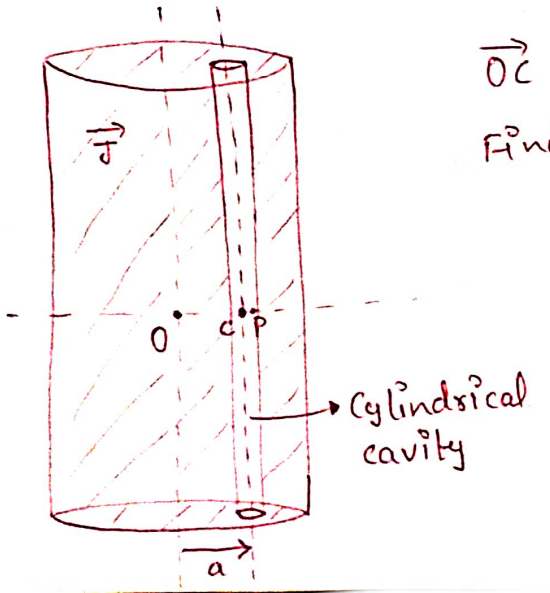
$$B_c = \frac{\mu_0 i}{2\pi r} \left(\frac{c^2 - r^2}{c^2 - b^2} \right)$$

here, r=OC

$$D: \oint \vec{B} \cdot \vec{dl} = B \cdot 2\pi r = \mu_0 (i - i)$$

$$B_D = 0.$$

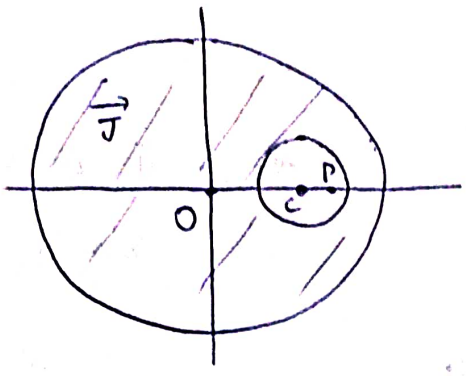
Q,

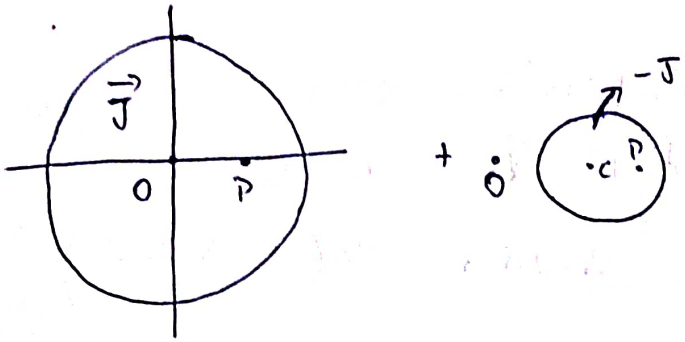


$\vec{OC} = \vec{a}$

Find magnetic field at point P.

Sol top view :





$$\vec{B}_P = \vec{B}_J + \vec{B}_{-J}$$

$$= \frac{\mu_0}{2}(\vec{J} \times \vec{OP}) + \frac{\mu_0}{2}(-\vec{J} \times \vec{CP})$$

$$= \frac{\mu_0}{2}\vec{J} \times (\vec{OP} - \vec{CP})$$

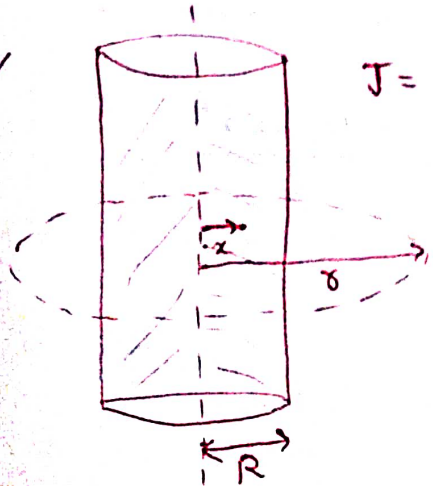
$$= \frac{\mu_0}{2}\vec{J} \times (\vec{OP} + \vec{PC})$$

$$= \frac{\mu_0}{2}(\vec{J} \times \vec{OC})$$

$$= \frac{\mu_0}{2}(\vec{J} \times \vec{a})$$

$$\boxed{B = \frac{\mu_0}{2} J a} \quad \text{(uniform)}$$

Q/



$J = J_0 x$

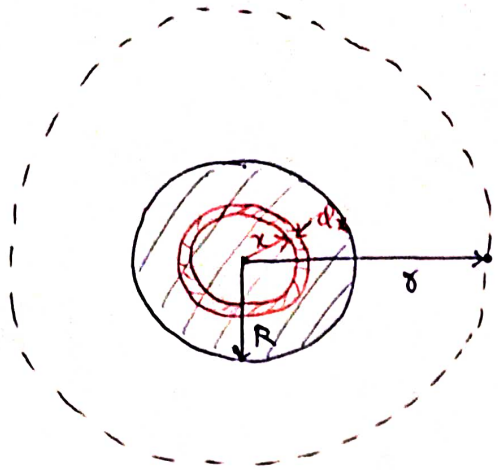
Find magnitude of magnetic field at :

① $r \geq R$

② $r \leq R$

Sol

① top view :



$$di = J_0 \, x \, 2\pi x \, dx$$

$$i = 2\pi J_0 \int_0^R x^2 \, dx$$

$$= \frac{2}{3} \pi J_0 R^3$$

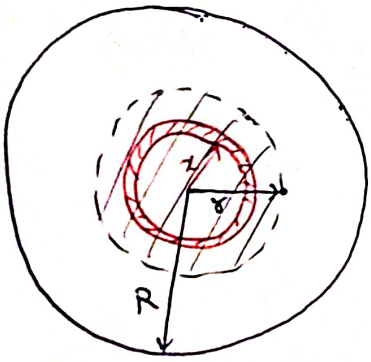
Ampere's law,

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 \, i_{in}$$

$$B \cdot 2\pi r = \mu_0 \left(\frac{2}{3} \pi J_0 R^3 \right)$$

$$B_{out} = \frac{\mu_0 J_0 R^3}{3r}$$

② top view :



$$di = J \, dA$$

$$i_{in} = \int_0^r J_0 \, x \, 2\pi x \, dx$$

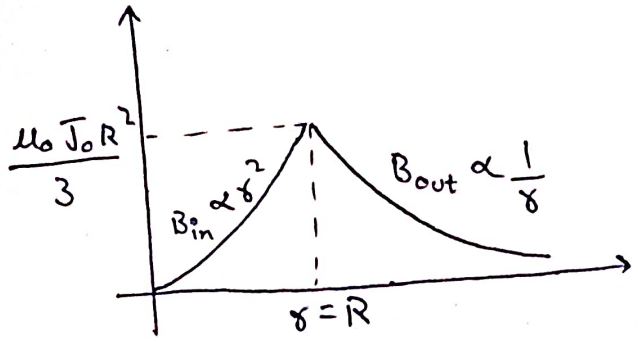
$$= 2\pi J_0 \int_0^r x^2 \, dx$$

$$= \frac{2\pi J_0 r^3}{3}$$

$$B_{in} \oint \vec{B} \cdot \vec{dl} = B \cdot 2\pi r = \mu_0 (i_{in})$$

$$B_{in} \cdot 2\pi r = \mu_0 \frac{2\pi J_0 r^3}{3}$$

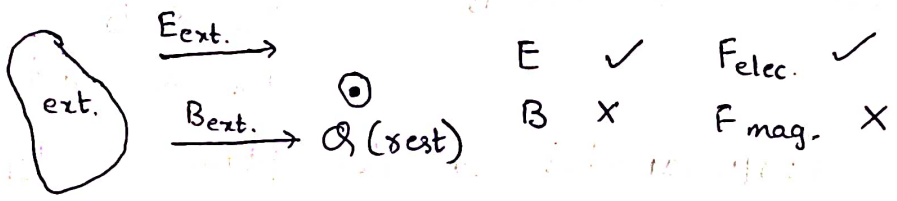
$$B_{in} = \frac{\mu_0 J_0 r^2}{3}$$



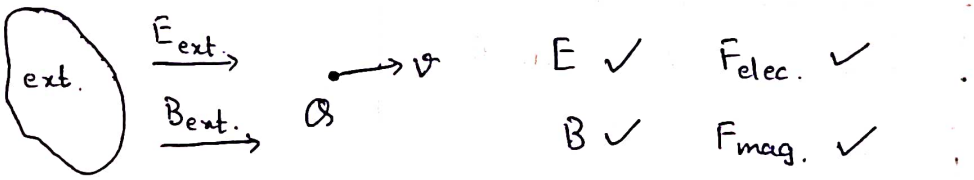
Electromagnetic Forces (EMF)

Electromagnetic Properties of a Charge

→ A charged particle at rest can create only electric field not magnetic field. Such a Charge particle can experience electric force due to external field but can not experience magnetic force



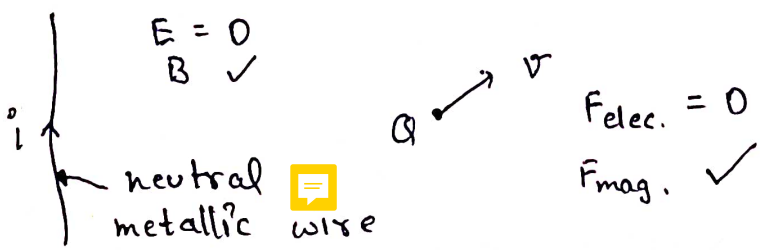
→



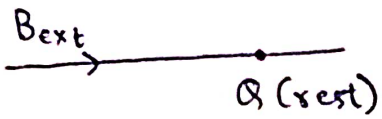
A charged particle moving with velocity 'v' can create both electric as well as magnetic field and can also experience both electric as well as magnetic force due to external electric or magnetic field.

→ If a charged particle moves with variable velocity then it emits radiations in the form of electromagnetic waves.

eg :-



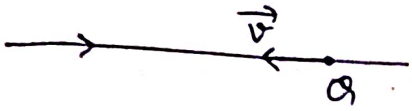
Magnetic Force on a Charged Particle (experimental)

① 

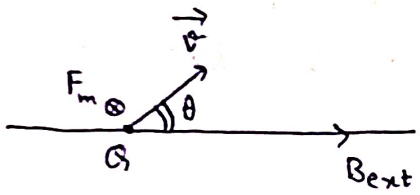
$F_m = 0$

②

$\vec{v} \parallel \vec{B} \Rightarrow F_m = 0$

③ 

$\vec{v} \uparrow\downarrow \vec{B} \Rightarrow F_m = 0$

④ 

$\theta \rightarrow$ angle between $\vec{v}$ and $\vec{B}$

$\vec{F}_m \perp \vec{v}$

$\vec{F}_m \perp \vec{B}$

$F_m \propto Q$

$F_m \propto B$

$F_m \propto \sin\theta$

* F_m is always perpendicular to the plane containing $\vec{v}$ and $\vec{B}$.

$$F_m = Q v B \sin\theta$$

$$\vec{F}_m = Q(\vec{v} \times \vec{B})$$

with sign

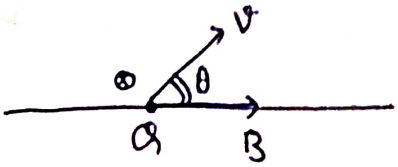
$\vec{F}_m = Q(\vec{v} \times \vec{B})$ (with sign)

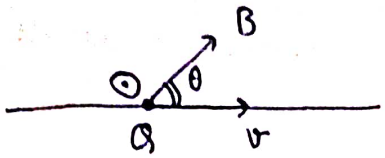
$\vec{F}_m \perp \vec{v}$

$\vec{F}_m \perp \vec{B}$

Dirn: RHTR

* Direction of magnetic force will be given by right hand thumbrule.





Q, $q = 5\,mC$, $m = 5\,mg$ has velocity $(3\hat{i} + 4\hat{j} + 5\hat{k})$ m/s and there is $\vec{B} = (2\hat{i} - 3\hat{j} + 5\hat{k})$ Tesla. Find $\vec{F}$ and $\vec{a}$ of the charge.

Sol

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 5 \\ 2 & -3 & 5 \end{vmatrix}$$

$$= 35\hat{i} - 5\hat{j} - 17\hat{k}$$

$$\vec{F}_m = q(\vec{v} \times \vec{B}) = 5 \times 10^{-3} \times (35\hat{i} - 5\hat{j} - 17\hat{k}) \text{ N}$$

$$\vec{F}_m = m \cdot \vec{a}$$

$$\vec{a} = \frac{5 \times 10^{-3} \times (35\hat{i} - 5\hat{j} - 17\hat{k})}{5 \times 10^{-6} \text{ kg}} \quad m/s^2$$

Q, A charged particle has velocity $(10\hat{i} - 20\hat{j} + 30\hat{k})$ m/s and experiencing a magnetic force of $(5\hat{i} - 3\hat{j} + F_z\hat{k})$. Find value of F_z.

Sol

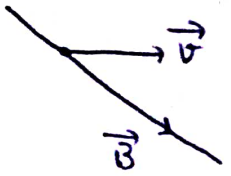
$$\vec{F}_m \perp \vec{v}$$

$$\Rightarrow \vec{F}_m \cdot \vec{v} = 0$$

$$60 + 50 + 30F_z = 0$$

$$F_z = -\frac{11}{3}$$

Work done by a magnetic Force on a Charge Particle



$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

$$\text{Power} = \vec{F}_m \cdot \vec{v}$$

$$= q(\vec{v} \times \vec{B}) \cdot \vec{v}$$

$$= 0$$

$$W = \int P\,dt = 0$$

* Magnetic force does no work on a charged particle.

by work energy theorem,

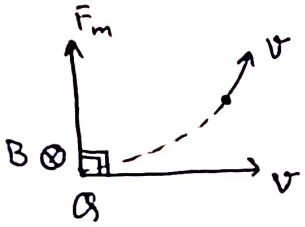
$W_{all} = \Delta K$

$W_{\vec{mag}}^{0} = \Delta K$

$\Rightarrow K_i = K_f$

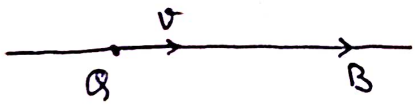
V = constant
K.E = constant

* Magnetic force can't charge K.E or speed of a charged particle.
* Magnetic force can change velocity (only direction of velocity / motion).



 # Different cases of motion of charged particle in a magnetic field:

$\vec{B} \parallel \vec{v}$

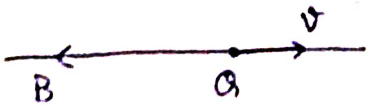


$F_m = q v B \sin 0°$

$= 0$

Motion is straight line uniform motion.

$\vec{B} \uparrow\downarrow \vec{v}$



$F_m = q v B \sin 180°$

$= 0$

Motion is straight line uniform motion.

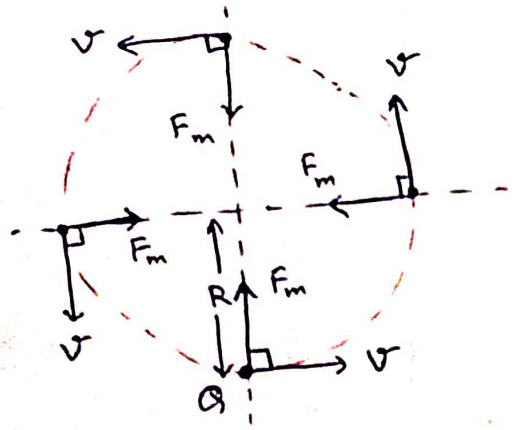
$\vec{B} \perp \vec{v}$

$\vec{B}$ = uniform

→ 2-D motion

→ Planar motion, in the plane of paper

⊗ B = uniform



$$F_m = Q v B \sin 90^\circ$$
$$= Q v B$$

→ Uniform Circular Motion

$$F_m = m a_c$$
$$Q v B = m \frac{v^2}{R}$$

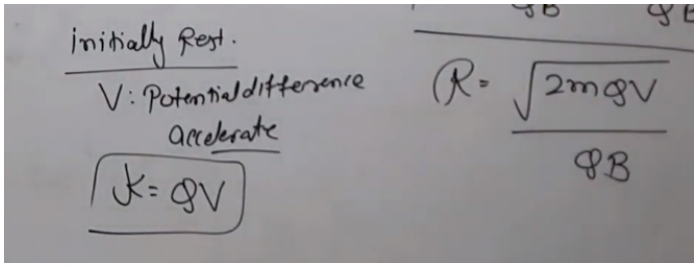
$$\boxed{R = \frac{m v}{Q B}}$$

$$\boxed{R = \frac{P}{Q B} = \frac{\sqrt{2mk}}{Q B}}$$

initially rest,

V → potential difference accelerate

$$\boxed{k = Q V}$$



ω → angular velocity

$$\omega = \frac{v}{R} = \frac{v Q B}{m v}$$

$$\boxed{\omega = \frac{Q B}{m}}$$

$T \rightarrow$ time period

$$T = \frac{2\pi}{\omega}$$

$$\boxed{T = \frac{2\pi m}{qB}}$$

* Time period is independent of speed.

$$\boxed{\text{Frequency} = f = \frac{1}{T} = \frac{qB}{2\pi m}}$$

Q, Proton, α- Particle and Deuteron. Same speed. Find the ① ratio of their radii
② ratio of their time period

⊗ B = uniform

$\longrightarrow v$

Sol ① $R_p = \frac{mv}{qB} = \frac{1 \times v}{e \times B} = \frac{v}{eB}$

$R_\alpha = \frac{4v}{2e \times B} = \frac{2v}{eB}$

$R_D = \frac{2v}{eB}$

$R_p : R_\alpha : R_D = 1 : 2 : 2$

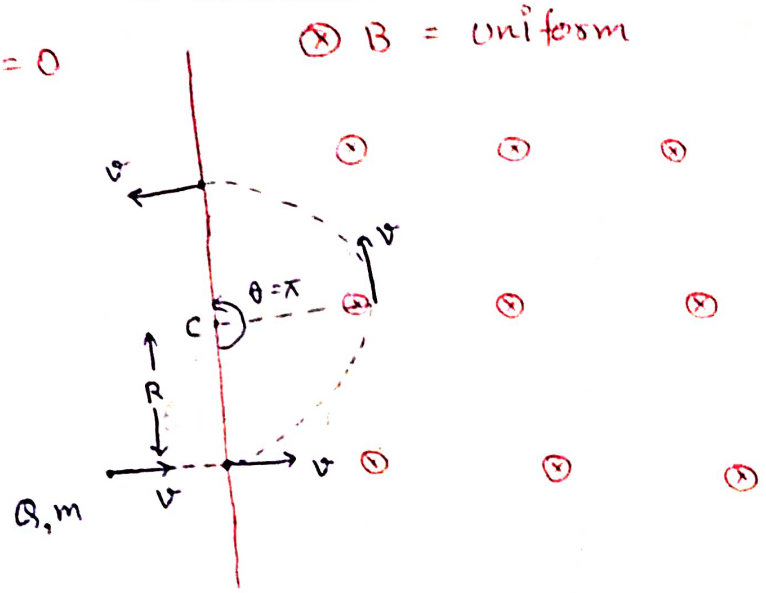
② $T_p = \frac{2\pi \times 1}{eB} = \frac{2\pi}{eB}$

$T_\alpha = \frac{2\pi \times 4}{2eB} = \frac{4\pi}{eB}$

$T_D = \frac{2\pi \times 2}{eB} = \frac{4\pi}{eB}$

$T_p : T_\alpha : T_D = 1 : 2 : 2$

Q, B = 0



Find time spent by ~~magnetic~~ charged particle inside magnetic field.

Sol

$$R = \frac{mv}{QB}$$

$$t = \frac{T}{2} = \frac{\frac{2\pi m}{QB}}{2}$$

$$t = \frac{\pi m}{QB}$$

OR

$$\theta = \omega t$$

$$\pi = \frac{QB}{m} t$$

$$t = \frac{\pi m}{QB}$$

$$\text{Impulse} = P_f - P_i$$

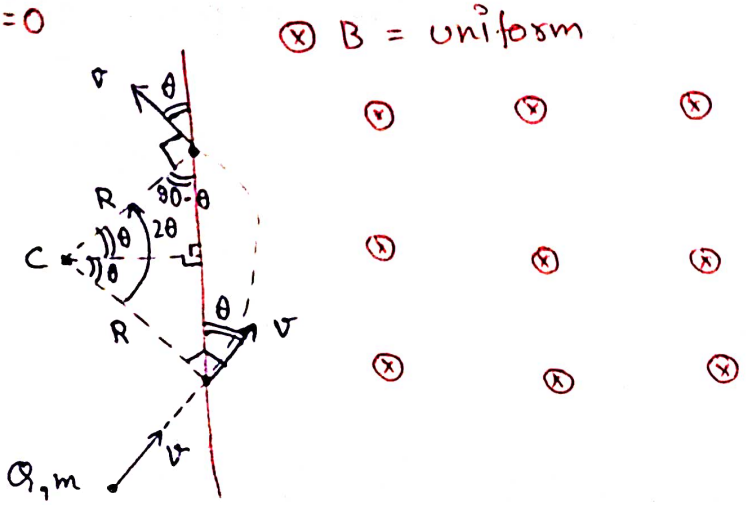
$$= (-mv\,\hat{i}) - (mv\,\hat{i})$$

$$= -2mv\,\hat{i}$$

$$J = 2mv$$

$$(F_m)_{avg} = \frac{\Delta P}{\Delta t} = -\frac{2mv\,\hat{i}}{\pi m} \times QB$$

Q, B = 0



Find time taken by charged particle inside the magnetic field.

Sol,

$2\pi \longrightarrow T$

$2\theta \longrightarrow \frac{T}{2\pi}\ 2\theta$

$= \frac{2\pi m}{qB\pi}\ \theta$

$= \frac{2\theta m}{qB}$

OR

$2\theta = \omega t$

$2\theta = \frac{qB}{m}\ t$

$t = \frac{2\theta m}{qB}$

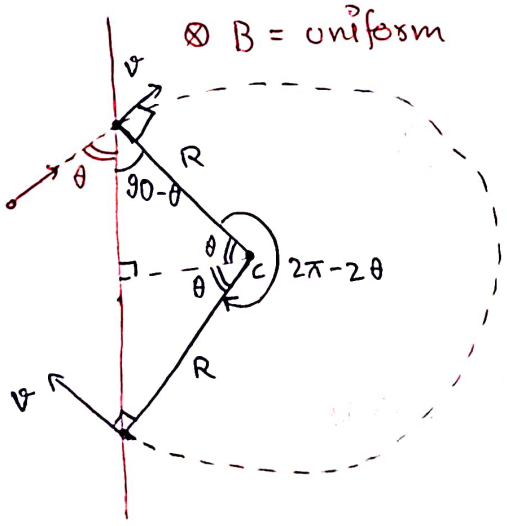
$R = \frac{mv}{qB}$

$\vec{J} = \vec{\Delta P} = \vec{P_f} - \vec{P_i}$

$= (-mv\sin\theta\ \hat{i} + mv\cos\theta\ \hat{j}) - (mv\sin\theta\ \hat{i} + mv\cos\theta\ \hat{j})$

$= -2mv\sin\theta\ \hat{i}$

Q, $B = 0$



Find time taken by electron inside magnetic field.

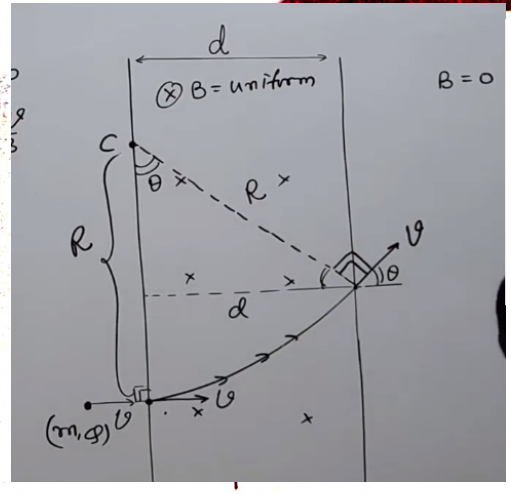
Sol

$R = \frac{mv}{eB}$

$\omega t = 2\pi - 2\theta$

$\frac{eB}{m}\ t = 2(\pi - \theta)$

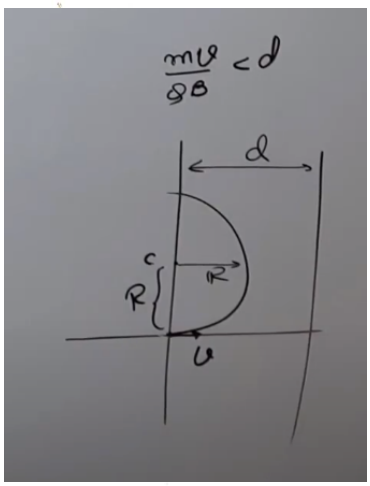
$t = \frac{2(\pi-\theta)m}{eB}$



$d < \frac{mv}{qB}$

Find time spent by charge inside magnetic field.

Sol $R = \frac{mv}{qB} > d$



$$\sin\theta = \frac{d}{R}$$

$$\theta = \sin^{-1}\frac{dqB}{mv} = \omega t$$

$$\sin^{-1}\frac{dqB}{mv} = \frac{qB}{m}\, t$$

$$t = \frac{m}{qB}\sin^{-1}\frac{dqB}{mv}$$

$\vec{J}$ = change in momentum

$$= \vec{P}_f - \vec{P}_i$$

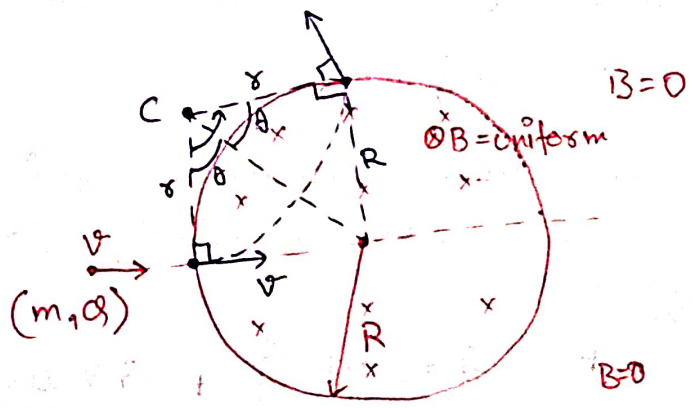
$$= (mv\cos\theta\,\hat{i} + mv\sin\theta\,\hat{j}) - mv\,\hat{i}$$

Q, Find time spent by charge inside magnetic field.

Sol

$$R = \frac{mv}{qB}$$

$$\omega = \frac{qB}{m}$$



$$\tan\theta = \frac{R}{R} = \frac{RqB}{mv}$$

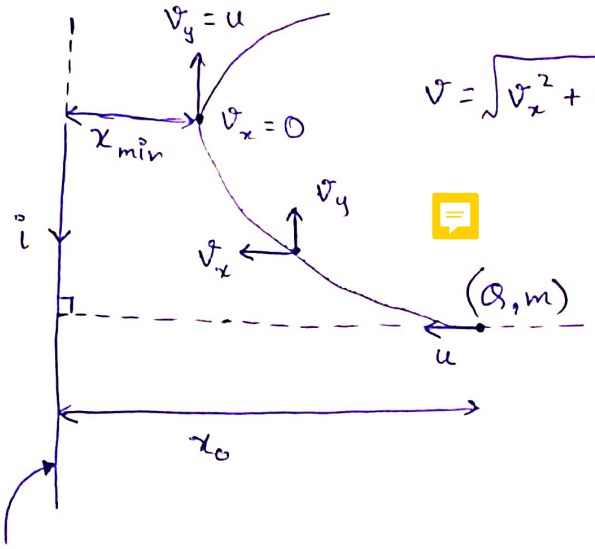
$$\theta = \tan^{-1}\frac{RqB}{mv}$$

$$2\theta = \omega t$$

$$2\tan^{-1}\frac{RqB}{mv} = \frac{qB}{m}\,t$$

$$t = \frac{2m}{qB}\tan^{-1}\frac{RqB}{mv}$$

Imp. Ques.

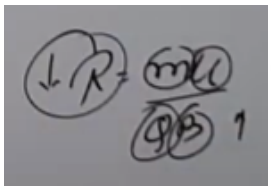


$$v = \sqrt{v_x^2 + v_y^2} = u$$

Infinitely long wire

→ 2-D motion
→ Planar motion
→ Plane of paper
→ Motion is not circular
→ Motion is along a curve in which radius of curvature will decrease as q moves towards the wire.
→ At the instant when seperation between charge and wire is minimum, then velocity of charge is

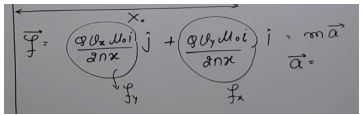
$$\vec{v} = u\,\hat{j}$$

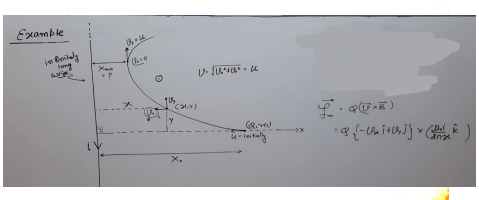


Q, In above situation, find minimum seperation between charge and infinitely long current carrying wire.

Sol

$$\vec{F} = \frac{q\,v_x\,\mu_0 i}{2\pi x}\,\hat{j} + \frac{q\,v_y\,\mu_0 i}{2\pi x}\,\hat{i} = m\vec{a}$$





$$\vec{a} = \frac{q\mu_0 i}{2\pi m}\frac{v_y}{x}\hat{i} + \frac{q\mu_0 i}{2\pi m}\frac{v_x}{x}\hat{j}$$

$$a_x = \frac{dv_x}{dt} = v_x\frac{dv_x}{dx} = \frac{q\mu_0 i}{2\pi m}\frac{v_y}{x}$$

$$v_x\frac{dv_x}{dx} = \frac{q\mu_0 i}{2\pi m}\frac{v_y}{x} \quad \text{—①}$$

$$v_x^2 + v_y^2 = u^2 = \text{constant}$$

$$2v_x\,dv_x + 2v_y\,dv_y = 0$$

$$v_x\,dv_x = -v_y\,dv_y$$

from eq ①

$$-v_y\frac{dv_y}{dx} = \frac{\mu_0 q i}{2\pi m}\frac{v_y}{x}$$

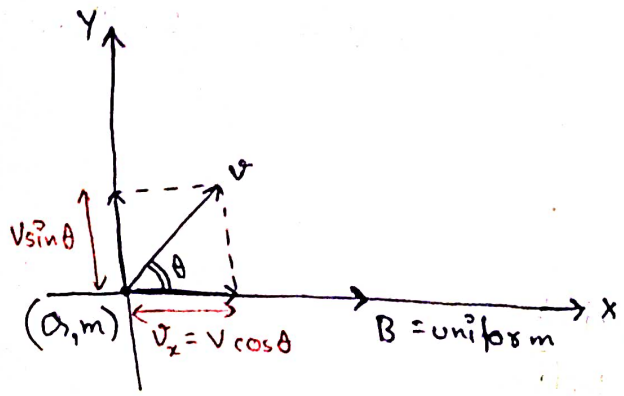
$$-\int_0^u dv_y = \frac{q\mu_0 i}{2\pi m}\int_{x_0}^{x_{min}}\frac{dx}{x}$$

$$-(v_y)_0^u = \frac{q\mu_0 i}{2\pi m}\ln\frac{x_{min}}{x_0}$$

$$-\frac{2\pi m u}{q\mu_0 i} = \ln\frac{x_{min}}{x_0}$$

$$x_{min} = x_0\, e^{-\frac{2\pi m u}{q\mu_0 i}}$$

Helical Path



$$F_x = 0$$

$$a_x = 0$$

$$v_x = \text{constant} = v\cos\theta$$

$$x = v_x t = (v\cos\theta)\,t$$

YZ - Plane

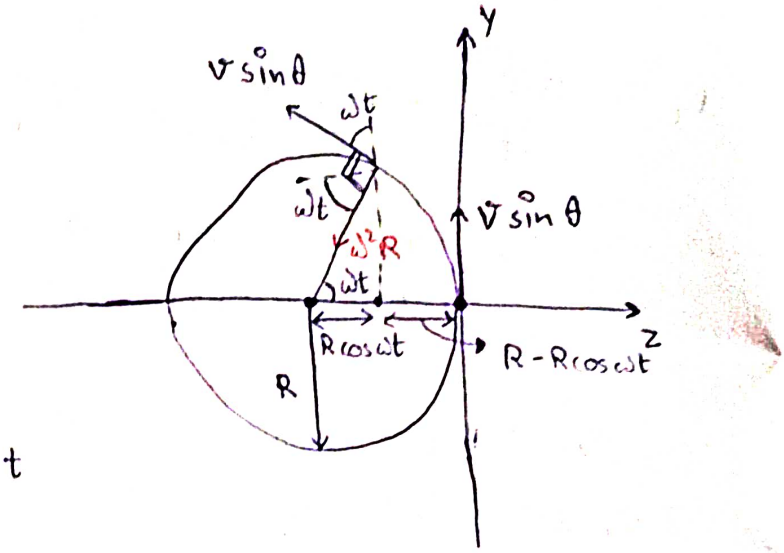
$$R = \frac{m(v\sin\theta)}{qB}$$

$$\omega = \frac{qB}{m}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

$$a_y = -\omega^2 R \sin\omega t$$

$$= -\frac{q^2B^2}{m^2}\,\frac{mv\sin\theta}{qB}\,\sin\frac{qB}{m}t$$



Y

$$Y = R\sin\omega t$$

$$= \frac{mv\sin\theta}{qB}\sin\frac{qB}{m}t$$

$$v_y = \frac{dy}{dt} = (v\sin\theta)\cos\frac{qB}{m}t$$

$$a_y = \frac{dv_y}{dt} = -\frac{v\sin\theta\cdot qB}{m}\sin\frac{qB}{m}t$$

Z

$$z = -(R - R\cos\omega t)$$

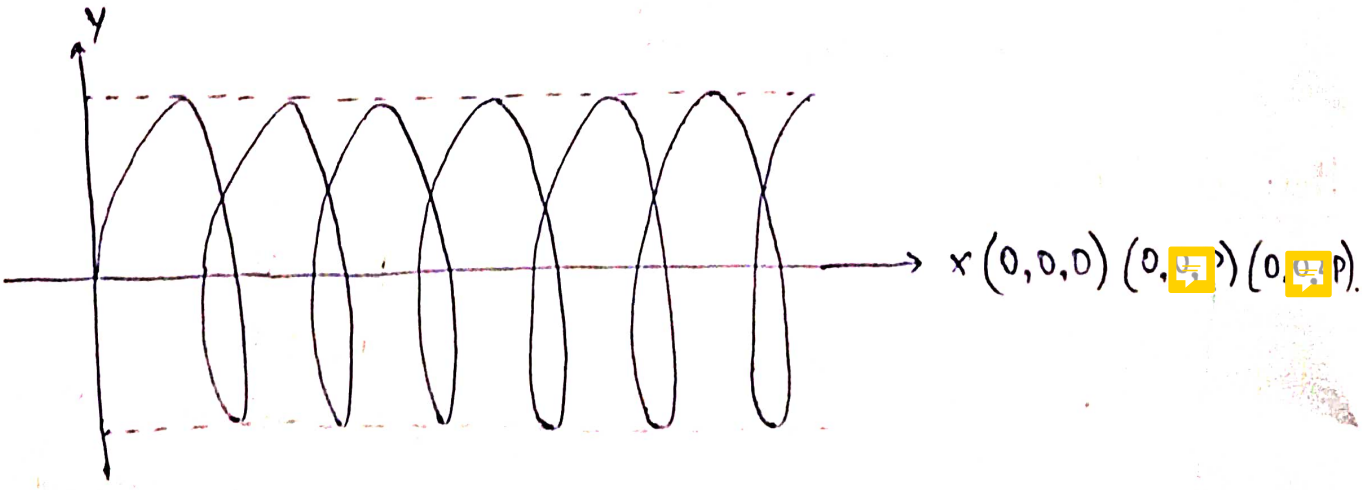
$$v_z = \frac{dz}{dt} = -R\omega\sin\omega t$$

$$= -\frac{mv\sin\theta}{qB}\,\frac{qB}{m}\sin\frac{qB}{m}t$$

$$= -v\sin\theta\,\sin\left(\frac{qB}{m}t\right)$$

$$a_z = \frac{dv_z}{dt}$$

$$= -\frac{v\sin\theta\cdot qB}{m}\cos\frac{qB}{m}t$$



→ Helical path with uniform pitch.

→ X-axis is tangent to the helix.

* Pitch length = $p = v_x T$

$= (v\cos\theta)\ \frac{2\pi m}{qB}$

* X-touch : $\left(0, 0, n\frac{2\pi m}{qB} v\cos\theta\right)$

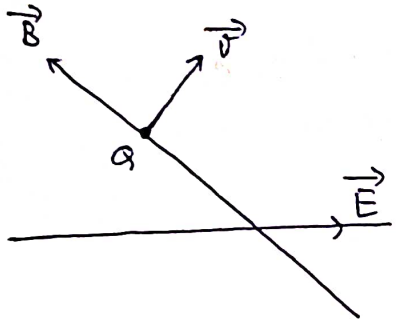
* $\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$

speed $= \sqrt{v_x^2 + v_y^2 + v_z^2}$

$= u$

* $\vec{a} = 0\,\hat{i} + a_y\hat{j} + a_z\hat{k}$
* displacement $= x\hat{i} + y\hat{j} + z\hat{k}$
* distance travelled $= ut$

Lorentz Force / Electromagnetic Force



$\vec{F} = \vec{F_e} + \vec{F_m}$

$\boxed{\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})}$

*** Case-1:**

$\vec{a} = \frac{q\vec{E}}{m}$ $\quad$ $\vec{E}, \vec{B}$

q, m $\quad$ v

$\vec{F} = \vec{F_e} + \vec{F_m}$

$= qE + q(\vec{v} \times \vec{B})$ ($\vec{v} \times \vec{B} = 0$)

uniformly accelerated motion in straight line.

$v_{final} = v + \frac{qE}{m}t$

$x = vt + \frac{1}{2}\frac{qE}{m}t^2$

*** Case-2:**

B, E

q, m $\quad$ v

$a = \frac{qE}{m}$

$\theta = 180°$

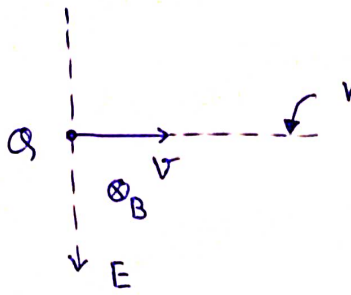
$\vec{v} \times \vec{B} = 0$

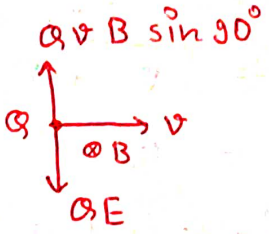
$F_m = 0$

uniformly accelerated motion along straight line with turning back.

*** Case-3:**

→ $\vec{E}$ and $\vec{B}$ are perpendicular to each other.

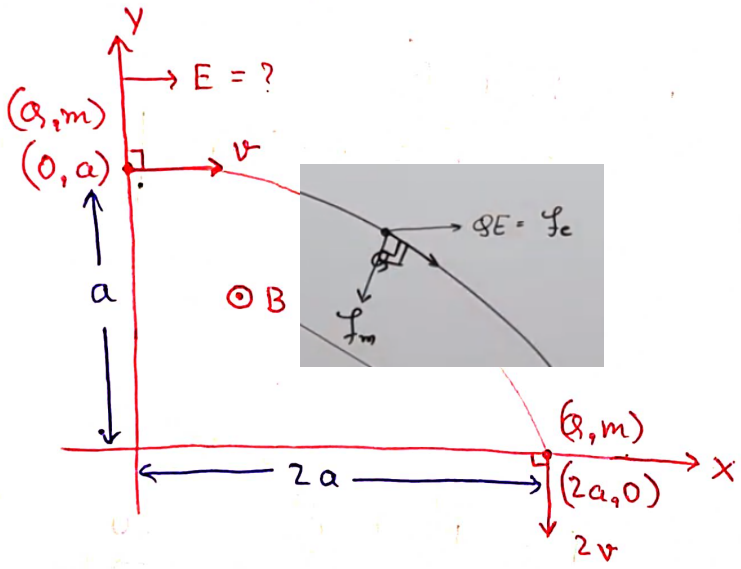




$$qvB = qE$$

$$\boxed{v = \frac{E}{B}}$$

Q/



E, B → uniform
Find value of E.

Sol/ by work energy theorem,

$$W_e + W_m = K_f - K_i$$

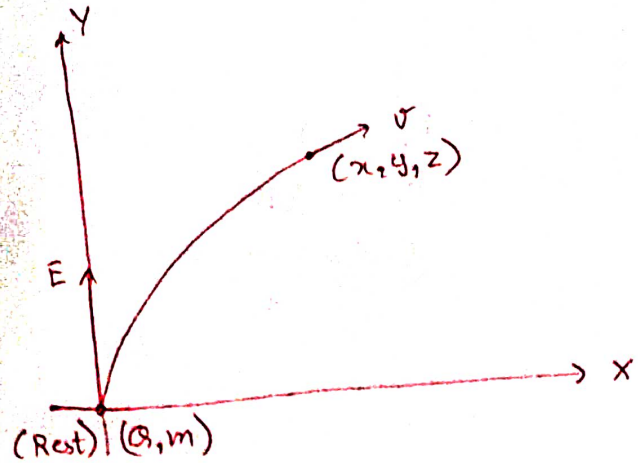
$$qE(2a) + 0 = \frac{1}{2}m(2v)^2 - \frac{1}{2}mv^2$$

$$2qEa = 2mv^2 - \frac{1}{2}mv^2$$

$$2qEa = \frac{3}{2}mv^2$$

$$E = \frac{3mv^2}{4qa}$$

Q) Find speed of charge when it arrives at (x, y, z).



Sol by work energy theorem,

$$W_e + W_m^{0} = K_f - K_i$$

$$Q E Y + 0 = \frac{1}{2} m v^2 - 0$$

$$v = \sqrt{\frac{2 Q E Y}{m}}$$

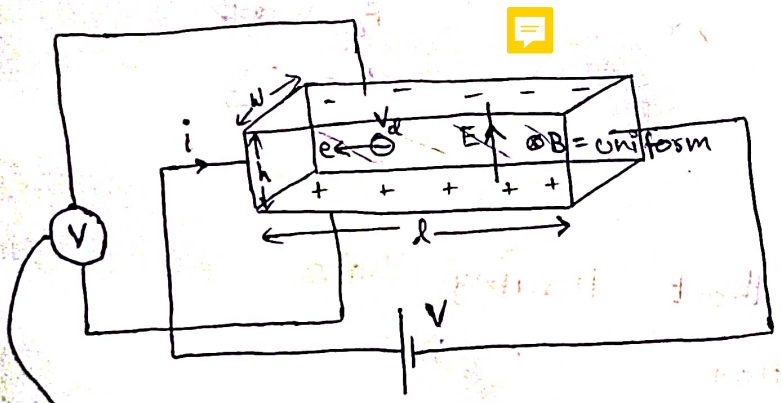
$$\vec{F}_e = Q E \hat{j}$$

$$W_e = \vec{F}_e \cdot \vec{s}$$

$$= (Q E \hat{j}) \cdot (x\hat{i} + y\hat{j} + z\hat{k})$$

$$= Q E Y$$

Hall Effect : experimental setup used to calculate the drift velocity of electrons

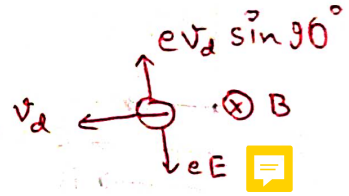


V_H = Hall voltage

$= Eh$

$$V_H = B v_d h$$

$$= \frac{B h i}{n h w e} = \frac{B i}{n w e}$$



eq^m, $e v_d B = e E$

$$v_d = \frac{E}{B}$$

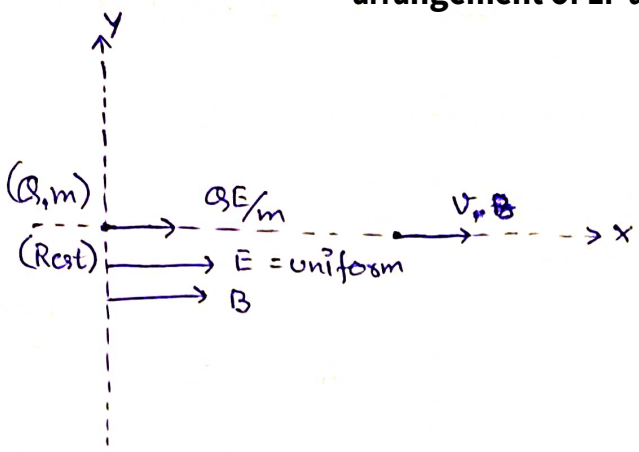
$$i = n A e v_d$$

$$= n h w e v_d$$

$$v_d = \frac{i}{n h w e}$$

 # Miscellaneous cases of arrangement of EF and MF:

Case- I :-



$F_m = 0$

$F_e = qE$

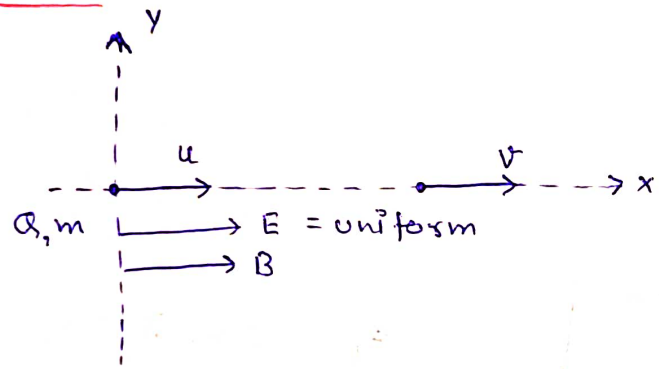
$a = \frac{qE}{m}$

Straight line motion without turning back.
Uniformly accelerated motion.

$$\vec{v} = 0 + \frac{qE}{m} t \, \hat{i}$$

$$\vec{r} = \frac{1}{2} \frac{qE}{m} t^2 \, \hat{i}$$

Case- II :-



$F_m = 0$

$F_e = qE$

$a = \frac{qE}{m}$

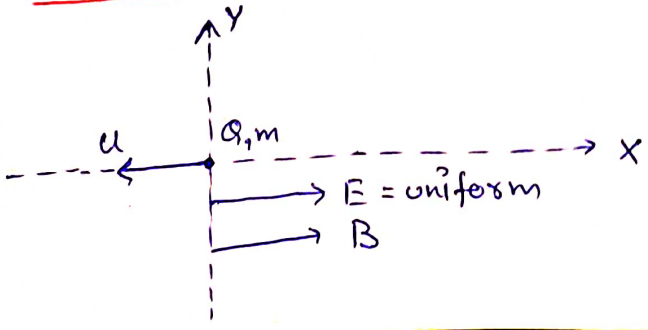
Straight line motion without turning back.
Uniformly accelerated motion.

$$\vec{v} = \left(u + \frac{qE}{m} t\right) \hat{i}$$

$$\vec{r} = x\hat{i} + 0\hat{j} + 0\hat{k}$$

$$= \left(ut + \frac{1}{2} \frac{qE}{m} t^2\right) \hat{i}$$

Case- III :-



$F_m = 0$

$F_e = qE$

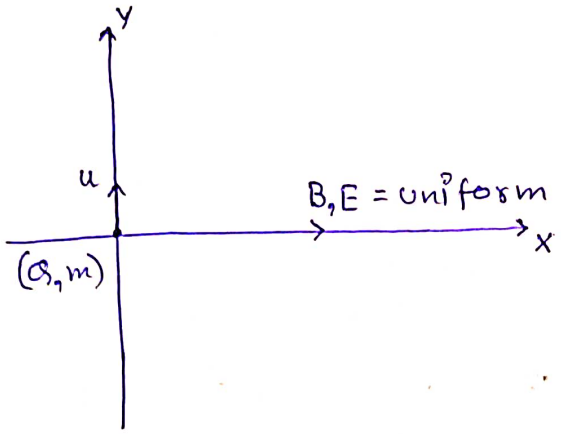
$a = \frac{qE}{m}$

straight line motion with turning back.
Uniformly accelerated motion.

$$\vec{v} = \left(-u + \frac{qE}{m}t\right)\hat{i}$$

$$\vec{r} = \left(-ut + \frac{1}{2}\frac{qE}{m}t^2\right)\hat{i}$$

Case - IV :-



$F_e \longrightarrow$ will act along x-axis

$F_m \longrightarrow$ will act ~~along~~ in yz-plane

x-axis : $F_x = qE$

$$a_x = \frac{qE}{m}$$

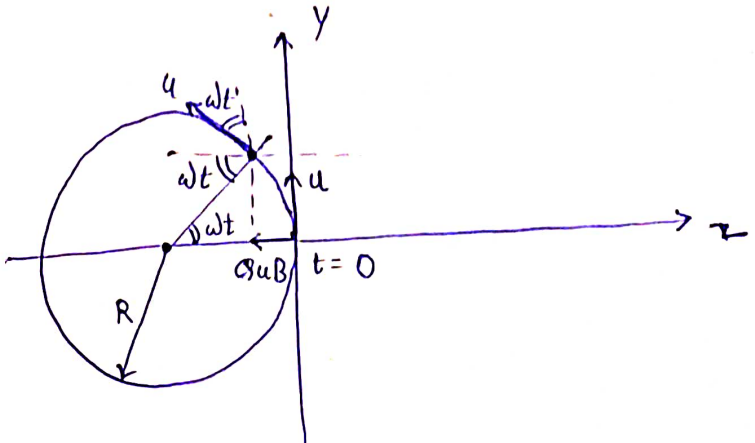
$$\boxed{v_x = 0 + \frac{qE}{m}t}$$

$$v_x^2 = u_x^2 + 2a_x x$$

$$v_x^2 = 0 + 2\frac{qE}{m}x$$

$$\boxed{v_x = \sqrt{\frac{2qEx}{m}}}$$

yz - Plane :



$$F_m = q u B \sin 90^\circ$$
$$= q u B = \frac{m u^2}{R}$$

$$\boxed{R = \frac{mu}{qB}}$$

$$\omega = \frac{u}{R} = \frac{qB}{m}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

y

$$y = R \sin \omega t$$
$$= \frac{mu}{qB} \sin \frac{qB}{m} t$$

$$v_y = \frac{dy}{dt} = u \cos \frac{qB}{m} t$$

z

$$z = -(R - R \cos \omega t)$$
$$= -\frac{mu}{qB}\left(1 - \cos \frac{qB}{m} t\right)$$

$$v_z = \frac{dz}{dt} = -u \sin \frac{qB}{m} t$$

$$\vec{v} = \frac{qE}{m} t \hat{i} + u \cos \frac{qB}{m} t \hat{j} - u \sin \frac{qB}{m} t \hat{j}$$

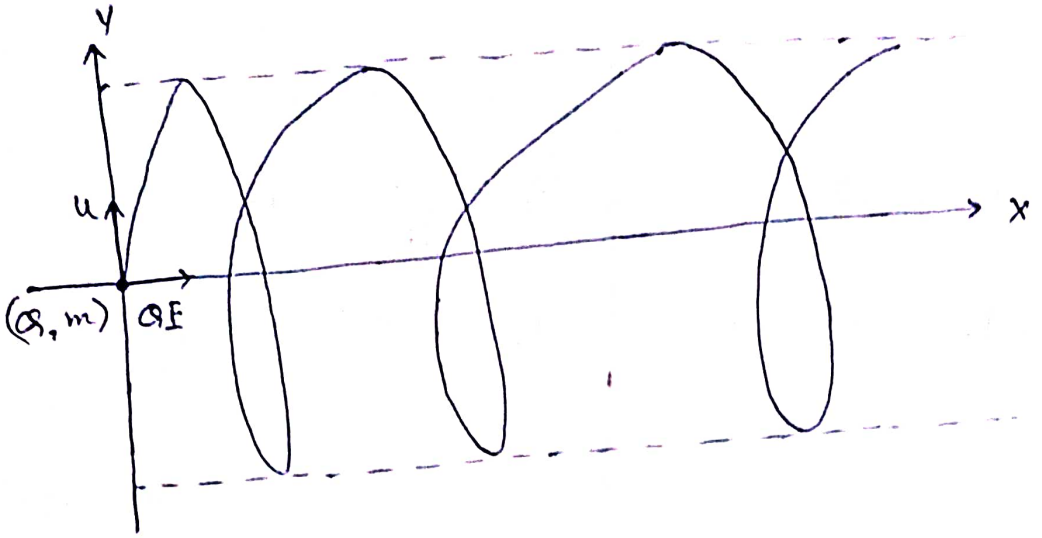
$$|\vec{v}| = \sqrt{\frac{q^2 E^2}{m^2} t^2 + u^2}$$

y

$$a_y = \frac{dv_y}{dt} = -\frac{u^2}{R} \sin \omega t$$

z

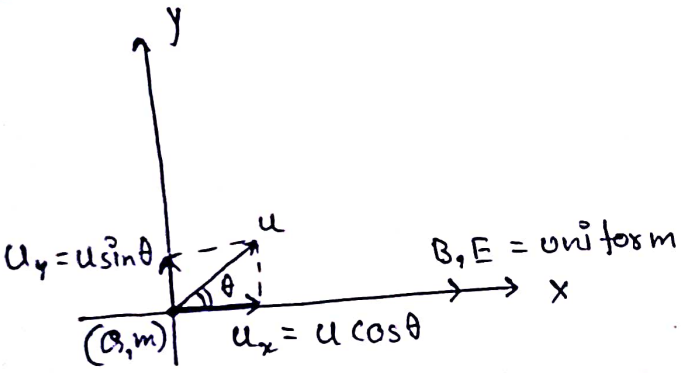
$$a_z = \frac{dv_z}{dt} = -\frac{u^2}{R} \cos \omega t$$



→ Helical path
→ Non-uniform pitch length
→ x-axis is acting like a tangent

points of contact with X-axis : $(x, 0, 0)$ $x = 0, \frac{1}{2}\frac{qE}{m}T^2, \frac{1}{2}\frac{qE}{m}(2T)^2, \frac{1}{2}\frac{qE}{m}(3T)^2 \ldots$

Case - V :-



X: motion is given by EF & F_e

$$F_x = qE$$

$$a_x = \frac{qE}{m}$$

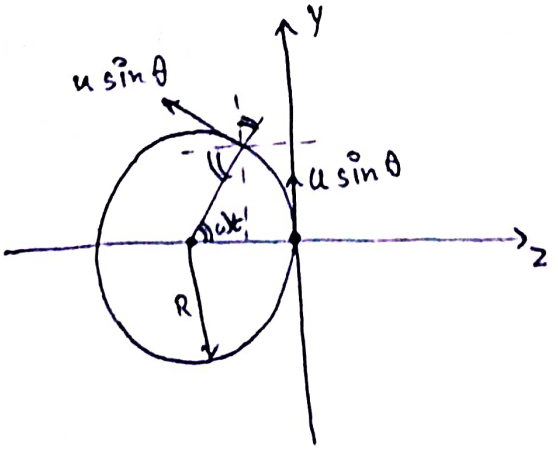
$$v_x = u_x + a_x t$$

$$= u\cos\theta + \frac{qE}{m}t$$

$$x = u_x t + \frac{1}{2}a_x t^2$$

$$= u\cos\theta \cdot t + \frac{1}{2}\frac{qE}{m}t^2$$

y-z Plane :



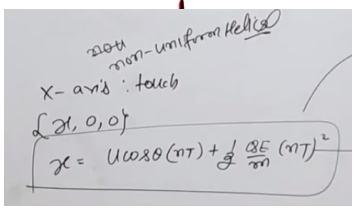
$$\omega = \frac{qB}{m}$$

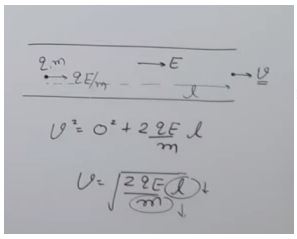
$$R = \frac{mu\sin\theta}{qB}$$

$$T = \frac{2\pi m}{qB}$$

y	z
$y = R \sin \omega t$	$z = -(R - R\cos\omega t)$
$v_y = \frac{dy}{dt} = u \sin\theta \cos\omega t$	$v_z = \frac{dz}{dt} = -u \sin\theta \sin\omega t$
$a_y = \frac{dv_y}{dt} = -\omega^2 R \sin\omega t$	$a_z = -\omega^2 R \cos\omega t$

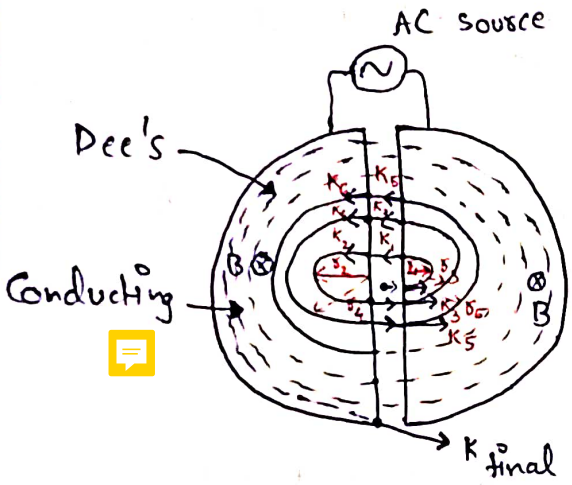
non-uniform helical motion





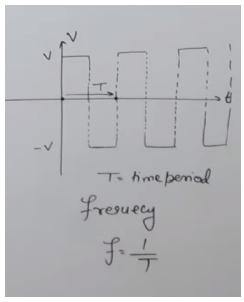
Cyclotron

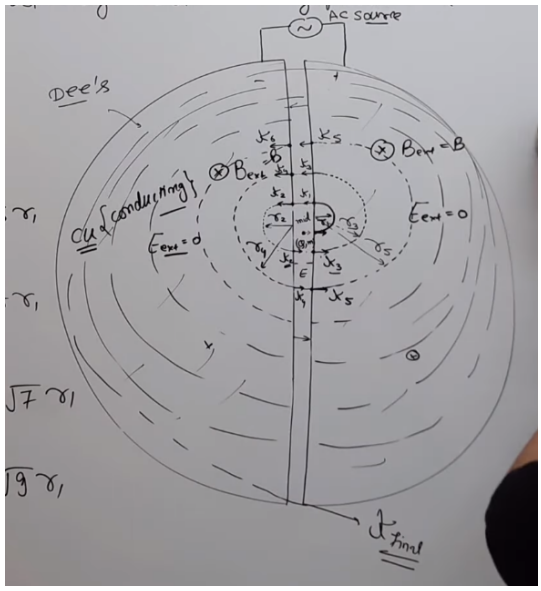
- is used to accelerated charged particles.
- specially heavier charge particle like proton, α-particle el



$E_{ext} = 0$

$\otimes B_{ext} = B$





$$\frac{qV}{2} = K_1 - 0 \Rightarrow K_1 = \frac{qV}{2}$$

$$K_2 = \frac{3}{2} qV$$

$$K_3 = \frac{5}{2} qV$$

$$K_4 = \frac{7}{2} qV$$

$$K_5 = \frac{9}{2} qV$$

$$K_6 = \frac{11}{2} qV$$

$$r_1 = \frac{\sqrt{2mK_1}}{qB}$$

$$r_2 = \frac{\sqrt{2mK_2}}{qB} = \sqrt{3}\, r_1$$

$$r_3 = \frac{\sqrt{2mK_3}}{qB} = \sqrt{5}\, r_1$$

$$r_4 = \frac{\sqrt{2mK_4}}{qB} = \sqrt{7}\, r_1$$

$$r_5 = \frac{\sqrt{2mK_5}}{qB} = \sqrt{9}\, r_1$$

→ In a complete circle, increase in K.E of charge particle is $2qV$

→ Increase in K.E in 1 sec $= 2qVf$

$$= 2qV\frac{qB}{2\pi m}$$

$$= \frac{q^2VB}{\pi m}$$

$f \rightarrow$ frequency

→ If time taken by charge particle to come out of the cyclotron is t, then

$$K_{final} = 2qVft$$

$$= \frac{q^2VBt}{\pi m}$$

→ $$r = R = \frac{\sqrt{2mK_f}}{qB}$$

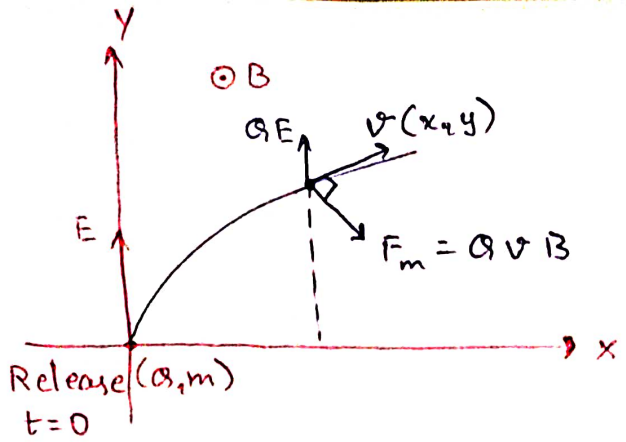
$$K_f = \frac{(qBR)^2}{2m}$$

→ Condition for tuning or resonance between cyclotron and AC source,

$$f_{AC} = f$$

$$\boxed{f_{AC} = \frac{qB}{2\pi m}}$$

Q,



$\vec{E} = E\hat{j}$ (uniform)

$\vec{B} = B\hat{k}$ (uniform)

① Find initial acceleration of charged particle.

Sol $\vec{a} = \frac{qE}{m}\hat{j}$

② Is the motion 2-D or 3-D ?

Sol 2-D motion in XY plane

③ Find speed of charge when it is at (x, y)

Sol by work energy theorem,

$$W_e + \cancel{W_m^{0}} = K_f - \cancel{K_i^{0}}$$

$$qEy = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2qEy}{m}}$$

→ As charge moves away from X-axis, its speed increases.

→ As charge moves towards X-axis, its speed decreases.

→ As charge touches X-axis, its speed becomes zero.

Y-axis : motion is governed by E as well as B

X-axis : motion is governed by B.

(4) Find $\vec{v}$ and position of charge at any time 't'.

Sol $\vec{F} = \vec{F}_e + \vec{F}_m = qE\hat{j} + q(v_x\hat{i} + v_y\hat{j}) \times B\hat{k}$

$\vec{F} = qE\hat{j} - (qv_xB)\hat{j} + (qv_yB)\hat{i} = m\vec{a}$

$\vec{a} = \underbrace{\left(\frac{qBv_y}{m}\right)}_{\frac{dv_x}{dt}}\hat{i} + \underbrace{\left(\frac{qE - qv_xB}{m}\right)}_{\frac{dv_y}{dt}}\hat{j}$

$\frac{dv_x}{dt} = \frac{qB}{m}v_y$

$\frac{dv_y}{dt} = \frac{qE}{m} - \frac{qBv_x}{m}$

$\frac{d^2v_y}{dt^2} = -\frac{qB}{m}\frac{dv_x}{dt} = -\frac{q^2B^2}{m^2}v_y$

$$\boxed{\frac{d^2v_y}{dt^2} = -\frac{q^2B^2}{m^2}v_y}$$

$v_y = v_0 \sin\left(\frac{qB}{m}t + \phi\right)$

$t = 0, \quad v_y = 0 = v_0 \sin(\phi) \Rightarrow \phi = 0$

$\therefore \quad v_y = v_0 \sin\left(\frac{qB}{m}t\right)$

$t = 0, \quad a_y = \frac{qE}{m} = \frac{dv_y}{dt} = \frac{v_0 qB}{m}\cos\left(\frac{qB}{m}t\right)$

$a_y = \frac{v_0 Bq}{m} \times 1 = \frac{qE}{m}$

$v_0 = \frac{E}{B}$

$$\boxed{v_y = \frac{E}{B}\sin\left(\frac{qB}{m}t\right)}$$

$\int_0^{v_x} dv_x = \frac{qB}{m}\cdot\frac{E}{B}\int_0^t \sin\left(\frac{qB}{m}t\right)dt$

$v_x = \frac{qE}{m}\cdot\frac{m}{qB}\left(-\cos\frac{qB}{m}t\right)_0^t$

$$v_x = \frac{E}{B}\left(1 - \cos\frac{qB}{m}t\right)$$

looking at this expression, we can say Vx always >= 0

$$v_x = \frac{dx}{dt}$$

$$\int_0^x dx = \frac{E}{B}\int_0^t \left(1 - \cos\frac{qB}{m}t\right) dt$$

$$x = \frac{E}{B}\left(t - \frac{m}{qB}\sin\frac{qB}{m}t\right)_0^t$$

$$x = \frac{E}{B}\left(t - \frac{m}{qB}\sin\frac{qB}{m}t\right)$$

$$v_y = \frac{E}{B}\sin\frac{qB}{m}t = \frac{dy}{dt}$$

$$\int_0^y dy = \frac{E}{B}\int_0^t \sin\frac{qB}{m}t \, dt$$

$$y = \frac{Em}{qB^2}\left(1 - \cos\frac{qB}{m}t\right)$$

(5) Find the time when Y-coordinate is maximum.

Sol/

$$\cos\frac{qB}{m}t = -1$$

$$\frac{qB}{m}t = \pi, 3\pi, 5\pi, 7\pi \ldots\ldots$$

$$t = (2n+1)\frac{\pi m}{qB}$$

$$y_{max} = \frac{2Em}{qB^2}$$

⑥ Find the time when charge touches X-axis.

Sol $y = 0 \Rightarrow \cos \frac{qB}{m} t = 1$

$$\frac{qB}{m} t = 0, 2\pi, 4\pi \text{----}$$

$$\boxed{t = (2n\pi)\frac{m}{qB}}$$

$$x = \frac{E}{B}\left(\frac{2n\pi m}{qB} - 0\right)$$

$$\boxed{x = \frac{2n\pi m E}{qB^2}}$$

$$\rightarrow x = \frac{E}{B}\left(t - \frac{m}{qB} \sin \frac{qB}{m} t\right)$$

$$x - \frac{Et}{B} = -\frac{mE}{qB^2} \sin \frac{qB}{m} t \quad \text{— ①}$$

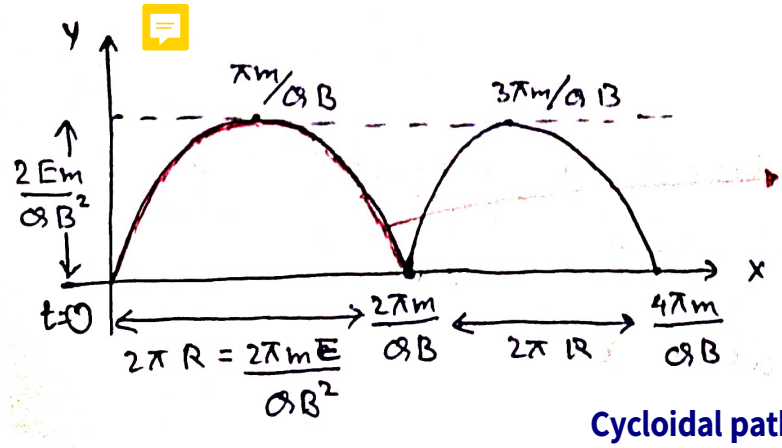
$$y = \frac{Em}{qB^2}\left(1 - \cos \frac{qB}{m} t\right)$$

$$y - \frac{Em}{qB^2} = -\frac{mE}{qB^2} \cos \frac{qB}{m} t \quad \text{— ②}$$

Square and add eq ① & ②

$$\left(x - \frac{Et}{B}\right)^2 + \left(y - \frac{Em}{qB^2}\right)^2 = \left(\frac{mE}{qB^2}\right)^2$$

Centre : $\left(\frac{Et}{B}, \frac{mE}{qB^2}\right)$, $R = \frac{mE}{qB^2}$



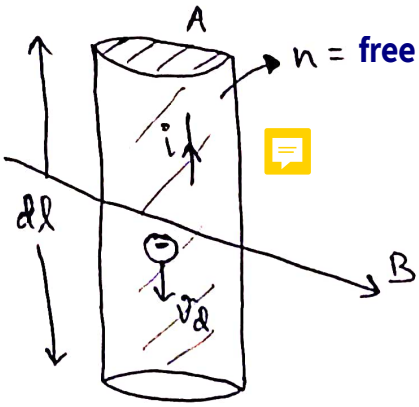
Cycloidal path

distance travelled $= 8R$

$= \frac{8mE}{qB^2}$

Cycloid path

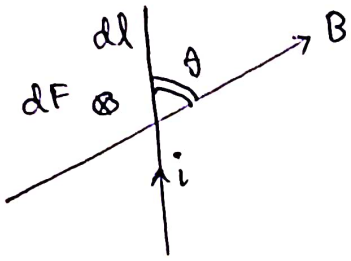
Magnetic Force on Current Carrying Wire



Force on an electron $= (-e)(\vec{v}_d \times \vec{B})$

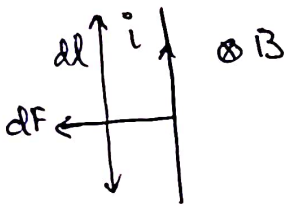
Force on all electrons collectively
$= -e(\vec{v}_d \times \vec{B})\, n A\, dl$

Force on this current element
$= d\vec{F} = i(\vec{dl} \times \vec{B})$



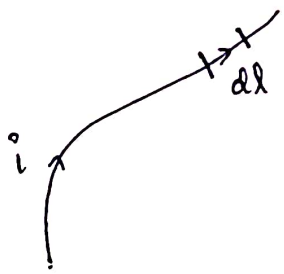
$dF = i\, dl\, B \sin\theta$

eg :-



$dF = i\, dl\, B \sin 90$
$= i\,(dl)\, B$

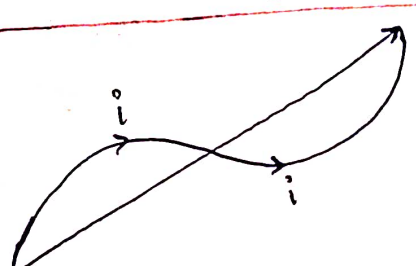
Magnetic Force on Long Wire



$\vec{F} = \int d\vec{F} = \int i(\vec{dl} \times \vec{B} = i\int(\vec{dl} \times \vec{B})$

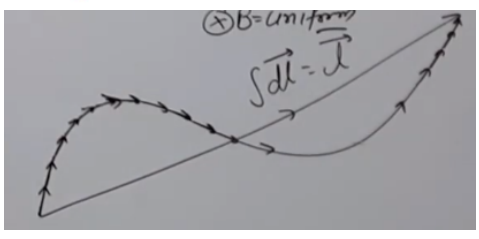
Force on Long Wire Placed in Magnetic Field

uniform



⊗ B = uniform

$\int \vec{dl} = \vec{l}$



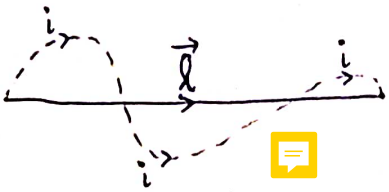
$$\vec{F} = \int d\vec{F} = \int i(d\vec{l} \times \vec{B})$$

$$= i\int (d\vec{l} \times \vec{B})$$

$$= i\int (d\vec{l}) \times \vec{B}$$

$$= i(\vec{l} \times \vec{B}) \rightarrow$$ only for uniform magnetic field

$$\boxed{F = i l B \sin\theta}$$



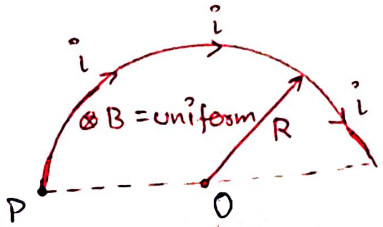
$\otimes B$ = uniform

$$\vec{F} = i(\vec{l} \times \vec{B})$$

$$F = i l B \sin\theta$$

$\theta \rightarrow$ angle between $\vec{l}$ and $\vec{B}$.

Q, Find magnetic force on wire.



Sol,

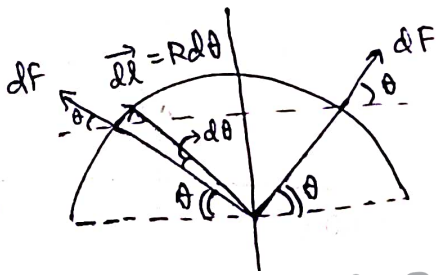
Method- I

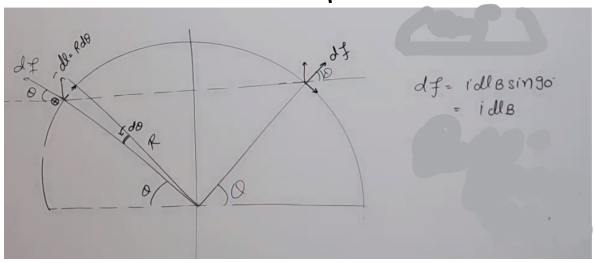
$$F_{net} = \int_0^{\pi} dF \sin\theta$$

$$= \int_0^{\pi} i\, dl\, B \sin\theta$$

$$= \int_0^{\pi} i B R\, d\theta \sin\theta$$

$$= 2 i B R$$



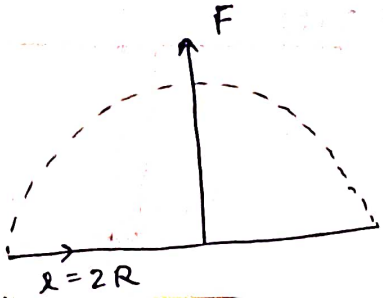


Method - II

$$F = i l B \sin 90^\circ$$

$$= i \cdot 2R\, B$$

$$= 2 i B R$$



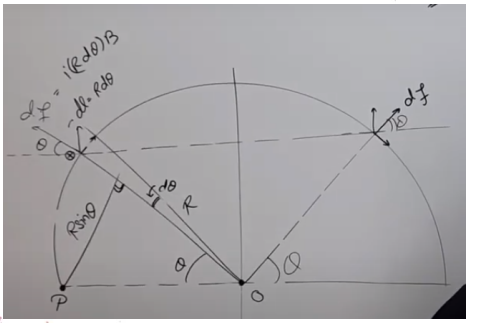
$\longrightarrow \tau_o = 0$

$\longrightarrow d\tau_p = dF \cdot l_\perp$

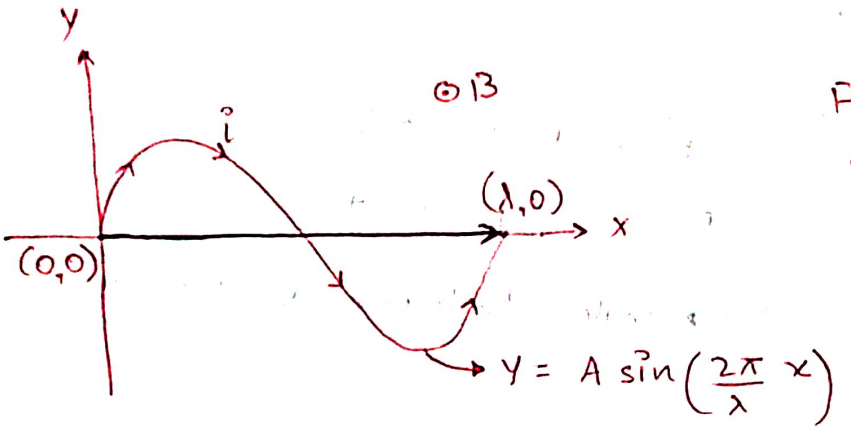
$$\tau_p = \int_0^{\pi} i B R\, d\theta \cdot R \sin\theta$$

$$= i B R^2 \int_0^{\pi} \sin\theta\, d\theta$$

$$= 2 i B R^2$$



Q/



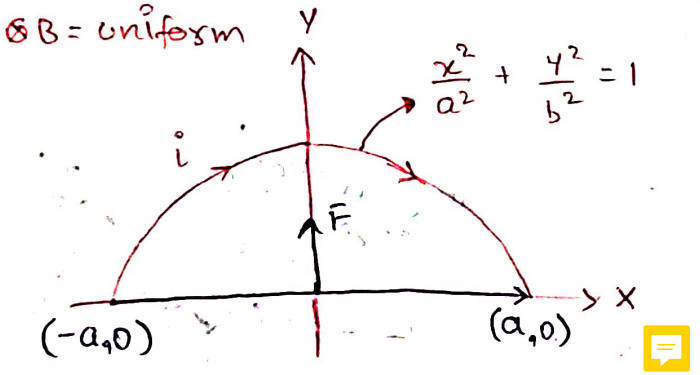
Find magnetic force on wire

Sol

$F = i \lambda B \sin 90^\circ$

$= i \lambda B$ in downwards direction

Q/



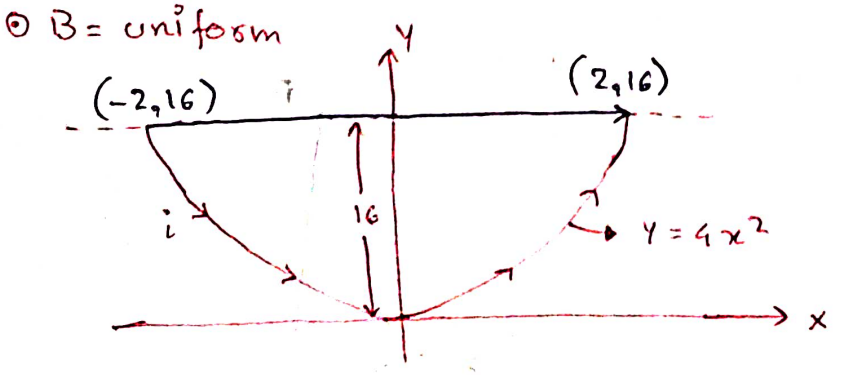
Find magnetic force on wire.

Sol

$F = i \cdot 2a \cdot B \sin 90^\circ$

$= 2 i B a$

Q/

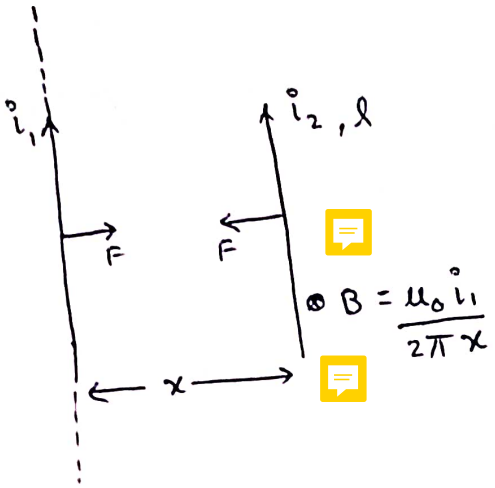


Find magnetic force on wire.

Sol $F = i(4)\,B \sin 90^\circ$

$= 4iB$ in downwards direction

Magnetic Force Between two Current Carrying Parallel Wires



$$F = i_2 \ell \left(\frac{\mu_0 i_1}{2\pi x}\right) \sin 90^\circ$$

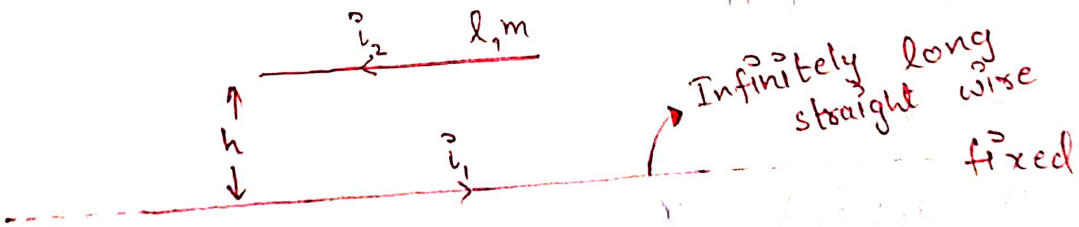
$$= \frac{\mu_0 i_1 i_2}{2\pi x}\ell$$

$$\boxed{\frac{F}{\ell} = \frac{\mu_0 i_1 i_2}{2\pi x}}$$

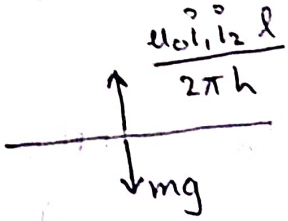
→ If current in both wires are of same direction, then force is attractive.

→ If current in both wires are of opposite direction, then force is repulsive.

Q/ Vertical parallel plane. "g". Find h such that small wire is in equilibrium.



Sol for eqm,

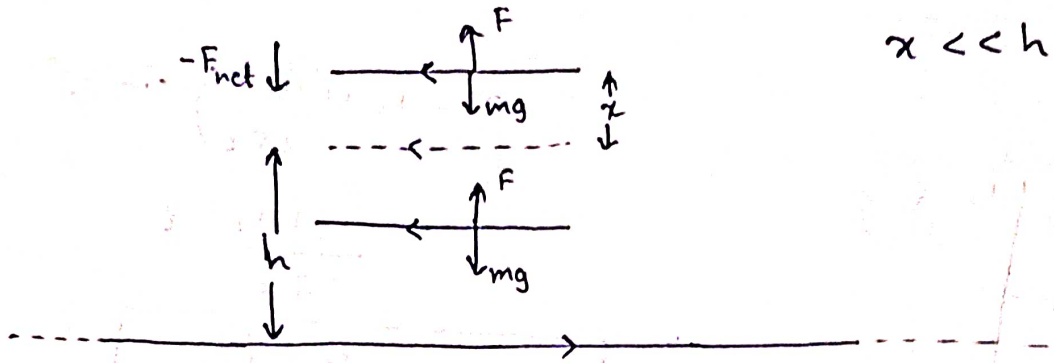


$$\frac{\mu_0 i_1 i_2 \ell}{2\pi h} = mg$$

$$h = \frac{\mu_0 i_1 i_2 \ell}{2\pi mg}$$

Q, In previous question, prove that if small wire is displaced slightly in vertical direction, it will perform SHM and also find time period of SHM.

Sol,



$x << h$

$$F_{net} = mg - \frac{\mu_0 i_1 i_2 l}{2\pi(h+x)}$$

$$= mg - \frac{\mu_0 i_1 i_2 l}{2\pi h}\left(1+\frac{x}{h}\right)^{-1}$$

$$= mg - \frac{\mu_0 i_1 i_2 l}{2\pi h}\left(1-\frac{x}{h}\right)$$

$$= mg - \frac{\mu_0 i_1 i_2 l}{2\pi h} + \left(\frac{\mu_0 i_1 i_2 l}{2\pi h^2}\right) x$$

$\therefore F_{net} \propto x$

$\therefore$ Motion is SHM

$$F_{net} = ma$$

$$a = \frac{\mu_0 i_1 i_2 l}{2\pi m h^2} x$$

In SHM,

$$a = \omega^2 x$$

$$\therefore \omega = \sqrt{\frac{\mu_0 i_1 i_2 l}{2\pi m h^2}}$$

$$\omega = \sqrt{\frac{hg}{h^2}} = \sqrt{\frac{g}{h}}$$

$$T = 2\pi\sqrt{\frac{h}{g}}$$

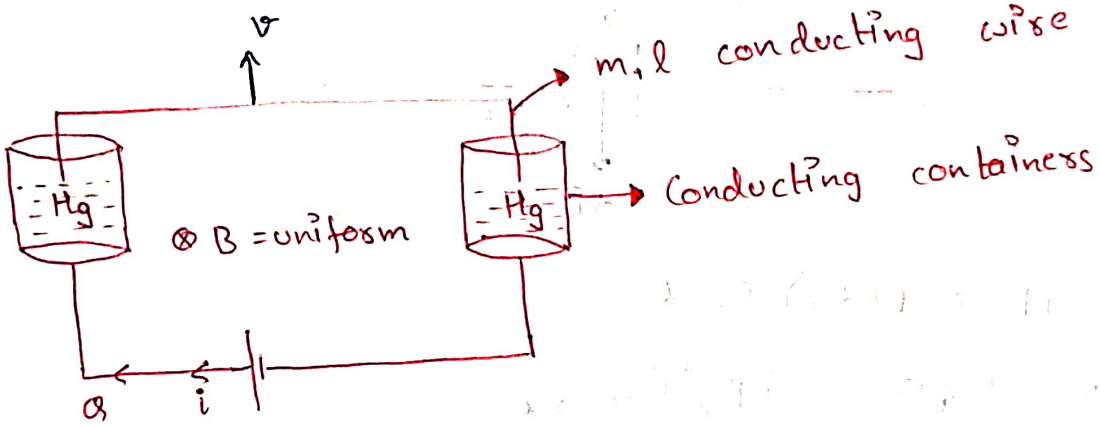
$$= 2\pi\sqrt{\frac{\mu_0 i_1 i_2 l}{2\pi m g^2}}$$

Magnetic Force on Wire

Q, When switch is turned on, charge provided by battery is Q. Magnetic force on the wire is impulsive and it lifts the wire. Find

① Velocity of wire when it loses contact with Hg

② Maximum height achieved by the curve



Sol/ ① by impulse momentum theorem,

$$\vec{J} = \vec{P}_f - \vec{P}_i$$

$$\int i l B\, dt = mv - 0$$

$$lB\int i\, dt = mv$$

$$lBQ = mv$$

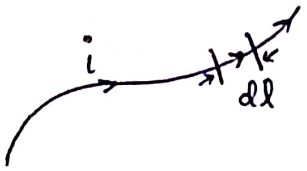
$$v = \frac{lBQ}{m}$$

②

$$H = \frac{v^2}{2g}$$

$$= \frac{l^2 B^2 Q^2}{2m^2 g}$$

Magnetic Force on Wire ($B \neq$ uniform)

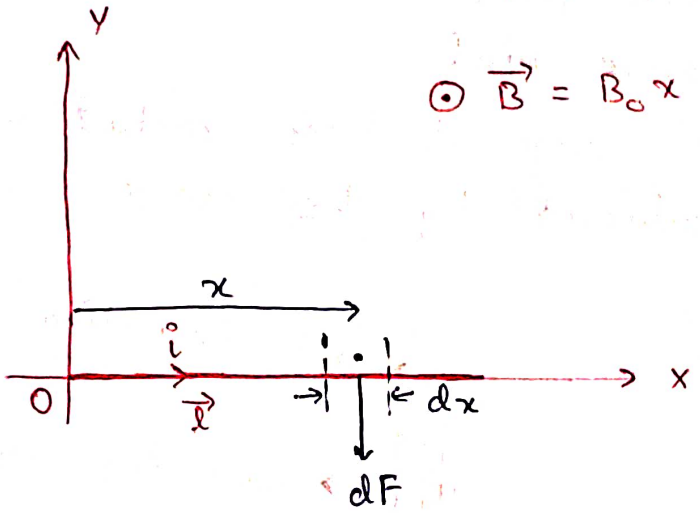


$$\vec{dF} = i(\vec{dl} \times \vec{B})$$

$$\vec{F} = \int \vec{dF} = \int i\,(\vec{dl} \times \vec{B})$$

$$= i\int (\vec{dl} \times \vec{B})$$

Q, Find magnetic force on ~~axis~~ wire. Also find torque about O.



Sol,

$$dF = i(dx)\,B_0 x$$

$$F = \int dF = iB_0 \int_0^l x\,dx$$

$$= \frac{i B_0 l^2}{2}$$

$$d\tau = dF \cdot x$$

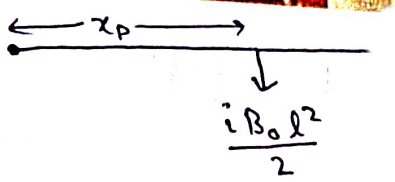
$$= (i B_0 x\,dx)\,x$$

$$\tau = i B_0 \int_0^l x^2\,dx$$

$$= \frac{i B_0 l^3}{3}$$

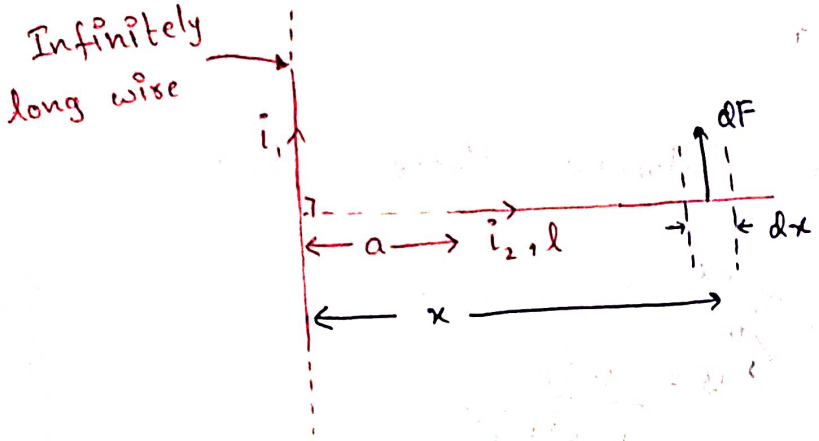
Q, In previous question, find point of application of force.

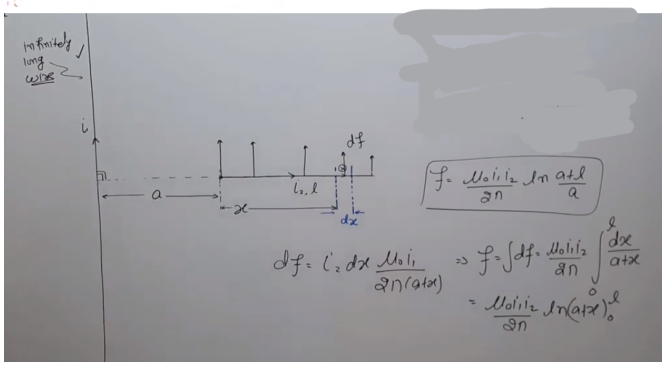
Sol $\tau = x_p \cdot \frac{i B_0 l^2}{2} = \frac{i B_0 l^3}{3}$



$$x_p = \frac{2l}{3}$$

Q, Find magnetic force between wires



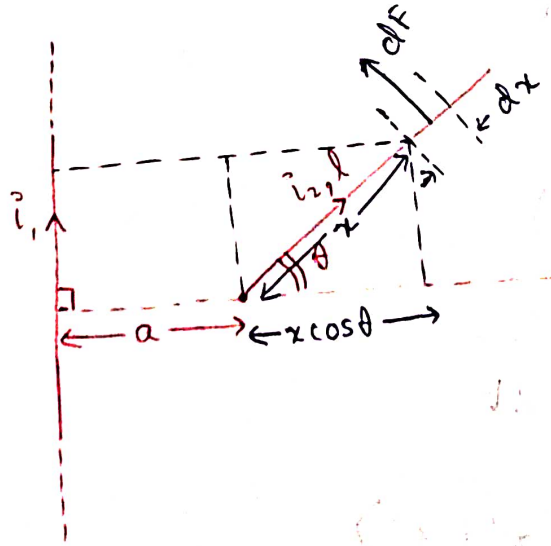


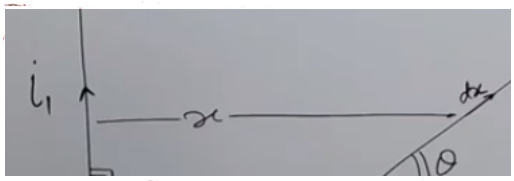
Sol $dF = i_2 (dx) \frac{\mu_0 i_1}{2\pi x}$

$$F = \int dF = \frac{\mu_0 i_1 i_2}{2\pi} \int_a^{a+l} \frac{dx}{x}$$

$$F = \frac{\mu_0 i_1 i_2}{2\pi} \ln \frac{a+l}{a}$$

Q, Find magnetic force between wires.





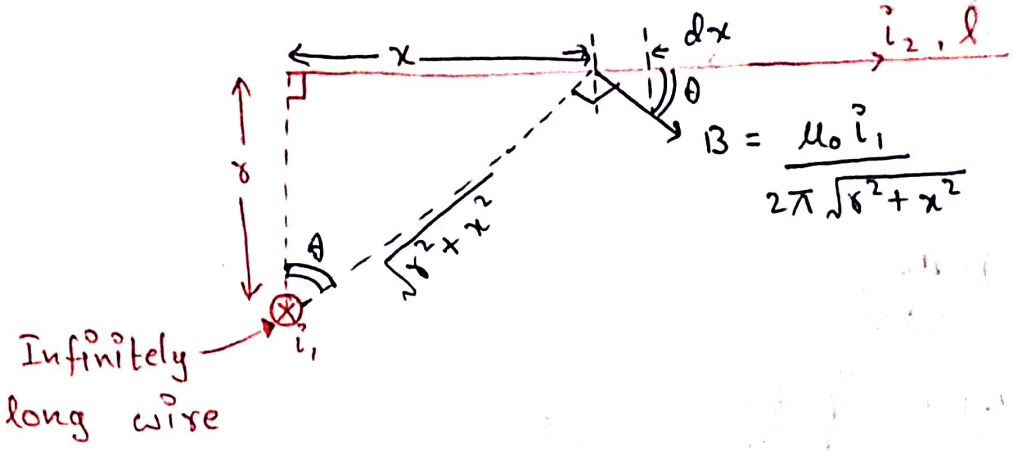
Sol $dF = i_2 \, dx \, \frac{\mu_0 i_1}{2\pi (a + x\cos\theta)}$

$$F = \int dF = \frac{\mu_0 i_1 i_2}{2\pi} \int_0^{l} \frac{dx}{a + x\cos\theta}$$

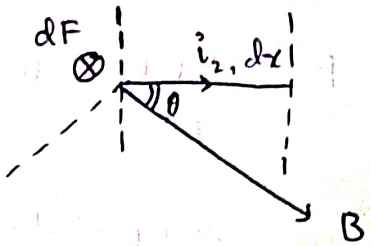
$$= \frac{\mu_0 i_1 i_2}{2\pi \cos\theta} \ln (a + x\cos\theta)_0^{l}$$

$$F = \frac{\mu_0 i_1 i_2}{2\pi \cos\theta} \ln \frac{(a + l\cos\theta)}{a}$$

Q. Find magnetic force between these wires.



Sol



$$dF = \frac{i_2\, dx\, \mu_0 i_1}{2\pi\sqrt{r^2+x^2}} \sin\theta$$

$$= \frac{\mu_0 i_1 i_2}{2\pi\sqrt{r^2+x^2}} \cdot \frac{x}{\sqrt{r^2+x^2}}\, dx$$

$$F = \frac{\mu_0 i_1 i_2}{2\pi} \int_0^{l} \frac{x\, dx}{(r^2+x^2)}$$

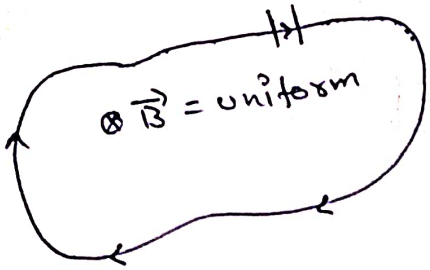
let $r^2 + x^2 = t^2$

$2x\, dx = 2t\, dt$

$$\int \frac{t\, dt}{t^2} = \frac{1}{2} \ln (r^2 + x^2)$$

$$F = \frac{\mu_0 i_1 i_2}{4\pi} \left(\ln \frac{r^2+l^2}{r^2} \right)$$

Force and Torque on Current Carrying Loop placed in ($\vec{B}$ = uniform)



$$\vec{dF} = i(\vec{dl} \times \vec{B})$$

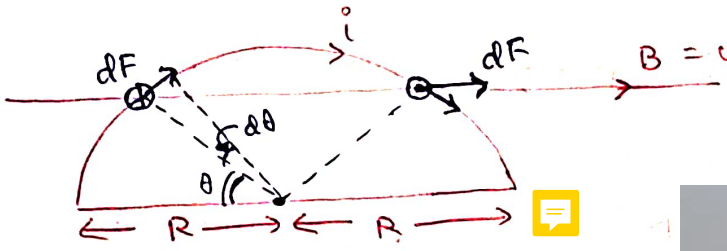
$$\vec{F} = \int \vec{dF} = i\int \vec{dl} \times \vec{B}$$

$$= i\left(\int \vec{dl}\right) \times \vec{B}$$

$$= i(\vec{0} \times \vec{B})$$

$$\boxed{\vec{F} = 0}$$

Q, Find force and torque on loop.

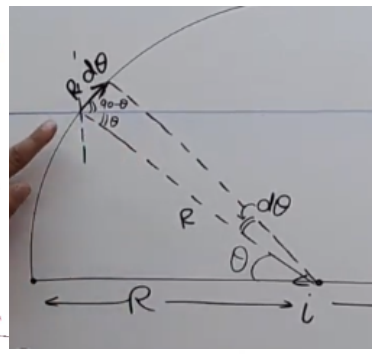


Sol

$$dF = i(R\,d\theta)\,B\,\sin(90-\theta)$$

$$dF = i\,R\,B\,\cos\theta\,d\theta$$

$$F_{net} = 0$$



$$dz = (dF)\,2R\cos\theta$$

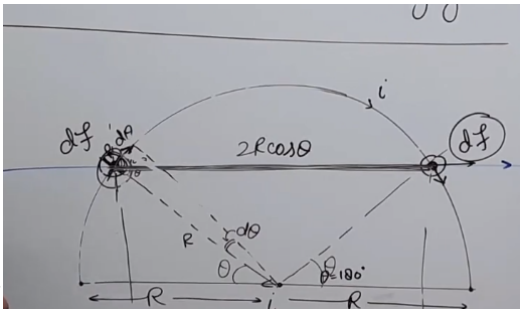
$$dz = (iRB\cos\theta\,d\theta)\,2R\cos\theta$$

$$z = 2iBR^2\int_0^{\pi/2} \cos^2\theta\,d\theta$$

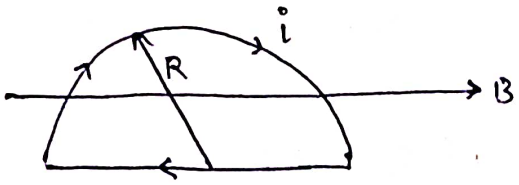
$$= \frac{2iBR^2}{2}\left(\theta + \frac{\sin 2\theta}{2}\right)_0^{\pi/2}$$

$$= iBR^2\left(\frac{\pi}{2} + \frac{\sin\pi}{2}\right)$$

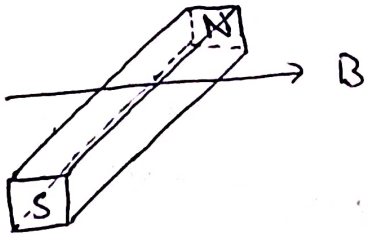
$$z = \frac{\pi}{2}R^2 iB = \left(\frac{\pi R^2}{2} i\right)B$$



II nd - Method



$\Rightarrow$



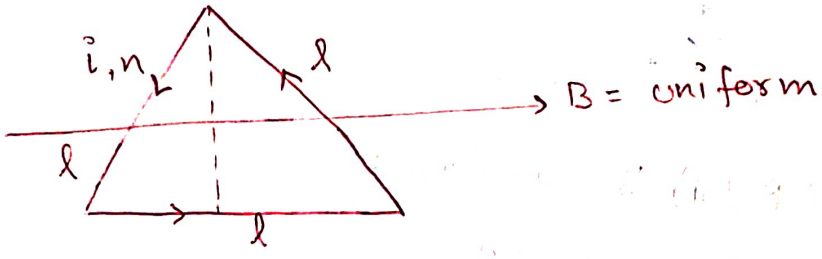
$$M = \frac{i\pi R^2}{2}$$

$$\tau = MB\sin\theta$$

$$= \frac{i\pi R^2}{2} B \sin 90^\circ$$

$$= \frac{i\pi R^2 B}{2}$$

Q, Find force and torque on loop



Sol

$$M = i \times \left(\frac{1}{2} \times l \cdot \frac{l\sqrt{3}}{2}\right) \times n$$

$$\tau = MB\sin\theta$$

$$= \frac{Bil^2n}{4} \sin 90^\circ$$

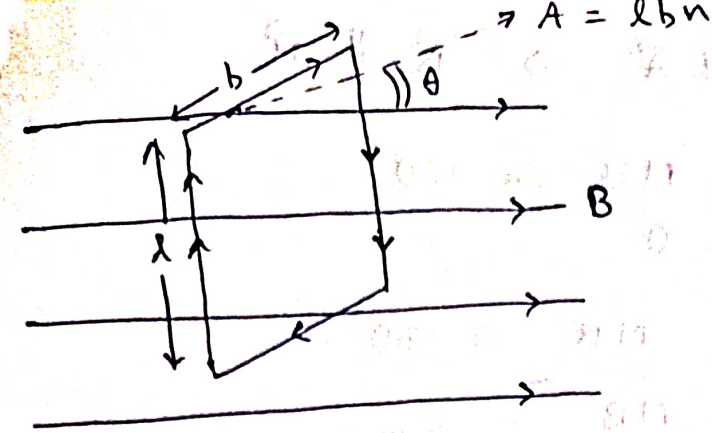
$$= \frac{\sqrt{3}\, i\, l^2 n B}{4}$$

Force and Torque on Current Carrying Loop

$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$\tau = MB\sin\theta$$

$$\boxed{\tau = i(lbn)\sin\theta}$$



$\Rightarrow A = lbn$

$n \rightarrow$ no. of turns

Interaction potential energy,

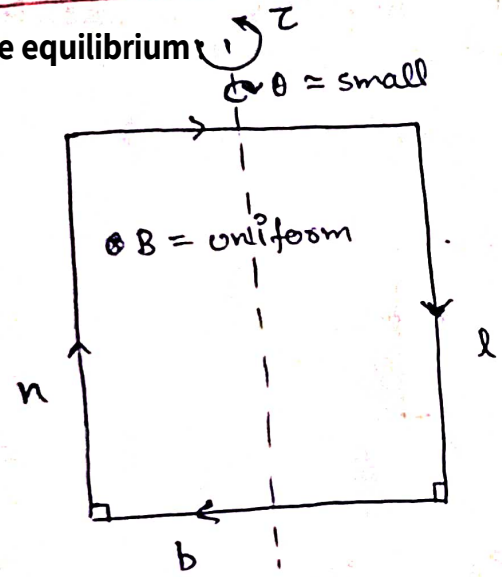
$U = -MB\cos\theta$.

$$\boxed{U = -\vec{M}\cdot\vec{B}}$$

$$\boxed{U = -(ilbn)\,B\cos\theta}$$

Stable and Unstable Equilibrium Positions

#Stable equilibrium



$\vec{B} \parallel \vec{A} \Rightarrow \vec{B} \parallel \vec{M}$

$\tau = MB\sin 0^\circ$

$= 0$

$U = -MB\cos 0^\circ$

$= -MB = U_{min}$

$\therefore$ Stable equilibrium position

for small Q, sinQ = Q

$I\alpha = \tau_{restoring} = MB\theta$ (Angular SHM)

$$\alpha = \frac{MB}{I}\theta$$

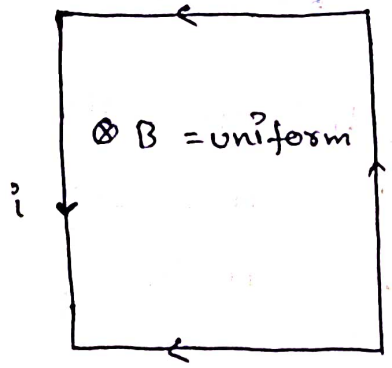
$$\therefore \omega = \sqrt{\frac{MB}{I}}$$

$$\boxed{T = 2\pi\sqrt{\frac{I}{MB}}}$$

 # Unstable equilibrium :

anti-parallel

$\vec{B} \parallel \vec{A} \Rightarrow \vec{B} \parallel \vec{M}$



$\tau = MB \sin 180^\circ$

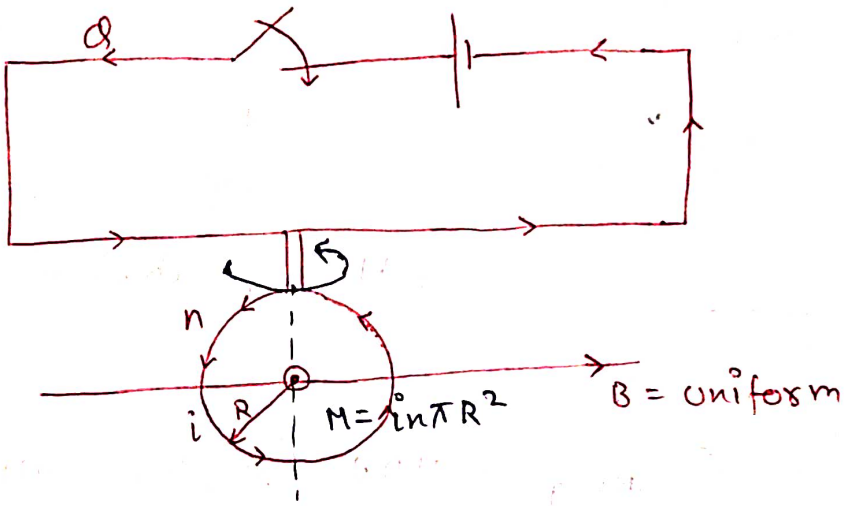
$= 0$

$U = -MB \cos 180^\circ$

$= MB = U_{max}$

$\therefore$ Unstable equilibrium position

Q, When switch is turned on battery provides charge 'Q' ist instantaneously. Find angular momentum of the circular loop



Sol

$\tau = (inπR^2) B \sin 90^\circ$

by angular momentum angular impulse theorem,

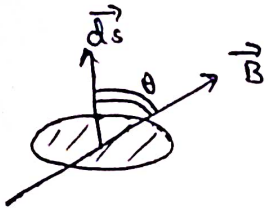
$\int \tau \, dt = L_f - \cancel{L_i}^{0}$

$\int inπR^2 B \, dt = L_f = 0$

$nπR^2 B \int i \, dt = nπR^2 BQ = L_f$

$L_f = nπR^2 BQ$

Magnetic Flux (ϕ)

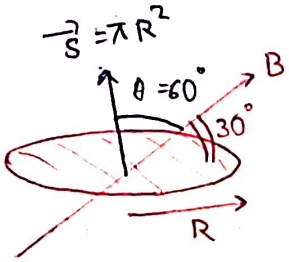


$$d\phi = \vec{B} \cdot \vec{ds}$$

$$\phi = \int d\phi = \int \vec{B} \cdot \vec{ds}$$

Surface integral of magnetic field.

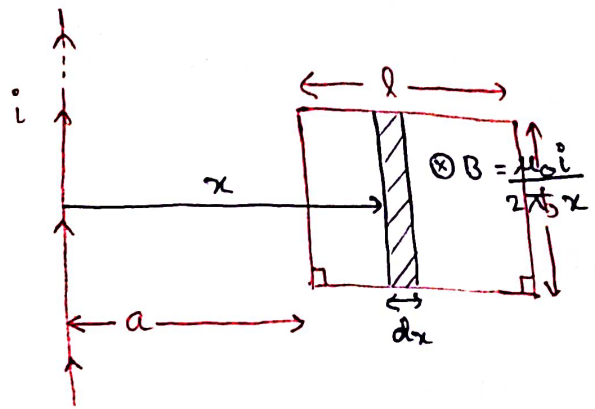
→ scaler
→ can be +ve, -ve or zero
→ Unit : Tesla-m^2 or Weber (Wb)
→ Dimension : ~~$[M L^{-2} I^{-1}]$~~ $[M L^2 I^{-1} T^{-2}]$
→ ϕ is surface integral of magnetic field

Q, 

Find flux.

Sol

$$\phi = \int \vec{B} \cdot \vec{ds}$$
$$= \vec{B} \cdot \int \vec{ds}$$
$$= \vec{B} \cdot \vec{S}$$
$$= B S \cos\theta$$
$$= B \cdot \pi R^2 \cdot \cos 60°$$
$$= \frac{B \pi R^2}{2}$$

Q, 

Find magnetic flux passing through the loop.

Sol

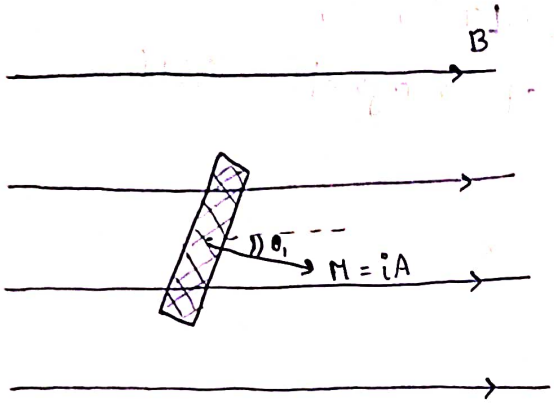
$$d\phi = B\, dA \cos\theta$$

$$= \frac{\mu_0 i}{2\pi x}\, b\, dx \cos 0^\circ$$

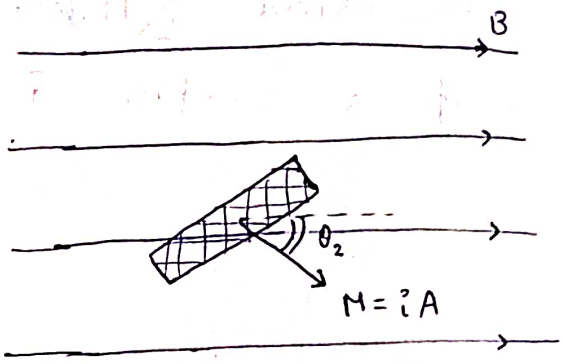
$$\phi = \int d\phi = \frac{\mu_0 i b}{2\pi} \int_a^{a+l} \frac{dx}{x}$$

$$\phi = \frac{\mu_0 i b}{2\pi} \ln\left(\frac{a+l}{a}\right)$$

Work done to rotate a Current carrying Loop



$$U_i = -MB\cos\theta_1$$



$$U_f = -MB\cos\theta_2$$

Slowly

$$W_{ext} = U_f - U_i$$

$$= -MB\cos\theta_2 + MB\cos\theta_1$$

$$= -iAB\cos\theta_2 + iAB\cos\theta_1$$

* OR

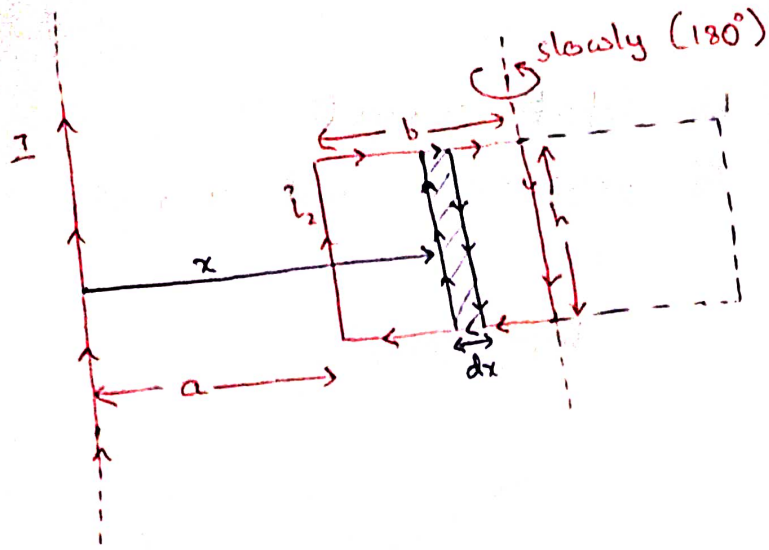
$$W_{ext} = MB(\cos\theta_1 - \cos\theta_2)$$

$$= i(AB\cos\theta_1 - AB\cos\theta_2)$$

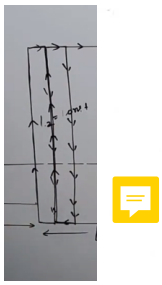
$$= i(\phi_1 - \phi_2)$$

$$\boxed{W_{ext} = -i\,\Delta\phi}$$

Q1/



Find W_{ext}



Sol

$$U = -MB\cos\theta$$

$$dU = -(i_2 h\, dx)\frac{\mu_0 i_1}{2\pi x}\cos 0^\circ$$

$$U_i = \int_a^{a+b} dU = -\frac{\mu_0 i_1 i_2 h}{2\pi}\int_a^{a+b}\frac{dx}{x}$$

$$U_i = -\frac{\mu_0 i_1 i_2}{2\pi}\ln\frac{(a+b)}{a} \quad \text{—(1)}$$

for final PE

$$du = -(i_2 h\, dx)\frac{\mu_0 i}{2\pi x}\cos\pi$$

since the angle of the area vector has now been reversed, hence, cos(pi)

$$U_f = \frac{\mu_0 i_1 i_2 h}{2\pi}\int_{a+b}^{a+2b}\frac{dx}{x}$$

$$U_f = \frac{\mu_0 i_1 i_2 h}{2\pi}\ln\frac{a+2b}{a+b} \quad \text{—(2)}$$

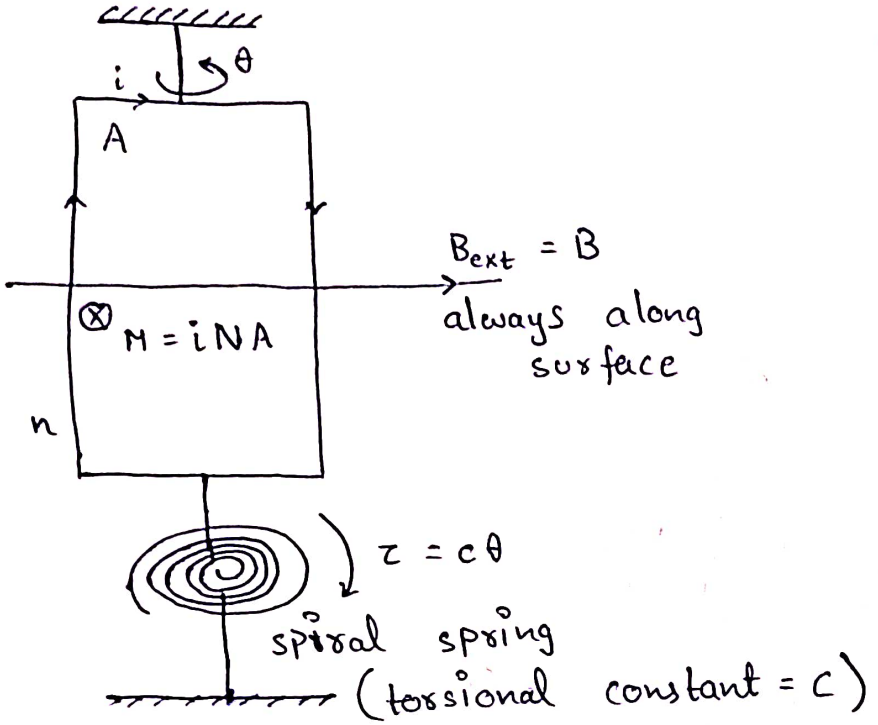
$$W_{ext} = U_f - U_i$$

$$= \frac{\mu_0 i_1 i_2 h}{2\pi}\left(\ln\frac{a+2b}{a+b} + \ln\frac{a+b}{a}\right)$$

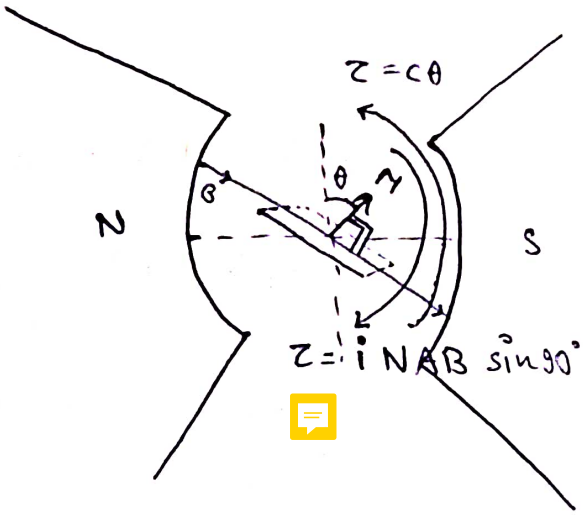
$$= \frac{\mu_0 i_1 i_2 h}{2\pi}\ln\frac{a+2b}{a}$$

Moving Coil Galvanometer

- it is used to detect current
- it is used to measure small current



top view :



$$\tau_{MF} = \tau_{SS}$$

$$iNAB = C\theta$$

$$\theta = \frac{NAB}{C} i$$

$$S = \frac{\theta}{i} = \frac{NAB}{C}$$

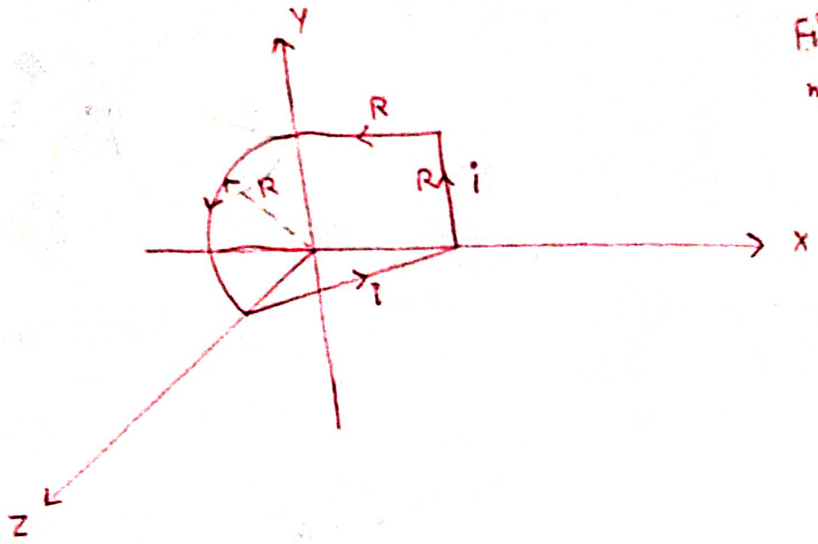
C → torsional constant

S → current sensitivity

Magnetic Moment

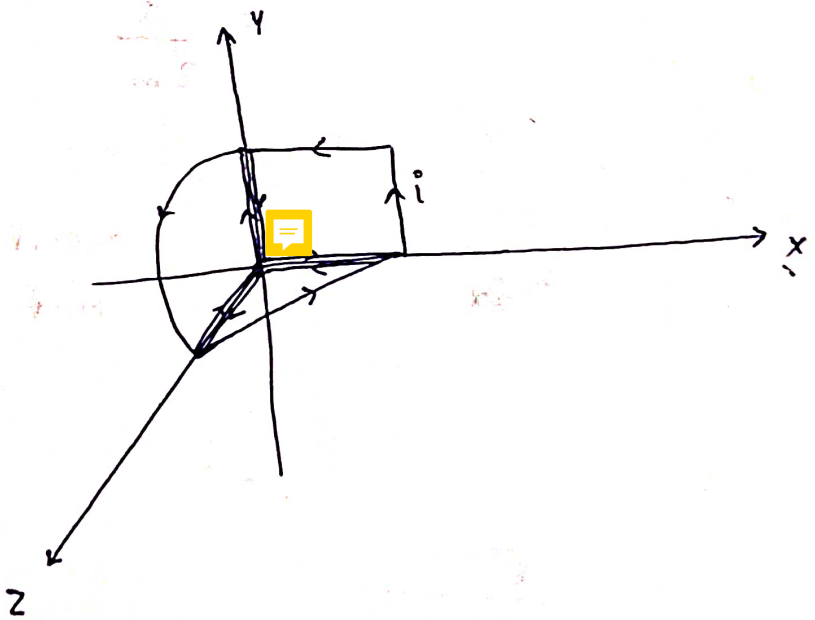
$$M = i A N$$

Q/



Find magnetic dipole moment.

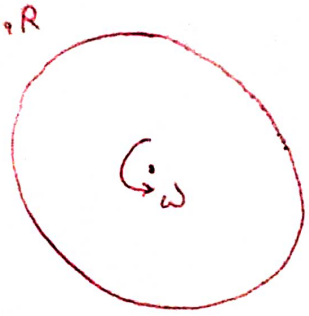
Sol/



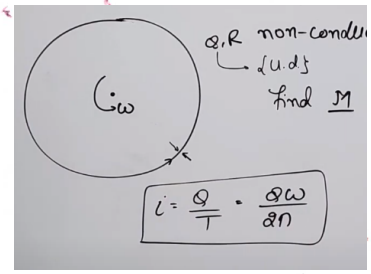
$$\vec{M} = \vec{M}_1 + \vec{M}_2 + \vec{M}_3$$

$$= iR^2 \hat{k} + i\frac{\pi R^2}{4}\hat{i} + i\frac{R^2}{2}\hat{j}$$

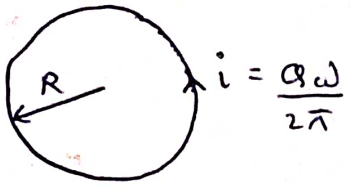
Q) m, R



Q (uniformly distributed), non-conducting ri...
Find magnetic dipole moment



thus, this can be replaced with the following current carrying loop



Sol

$$i = \frac{Q}{T} = \frac{Q\omega}{2\pi}$$

$$M = i A$$

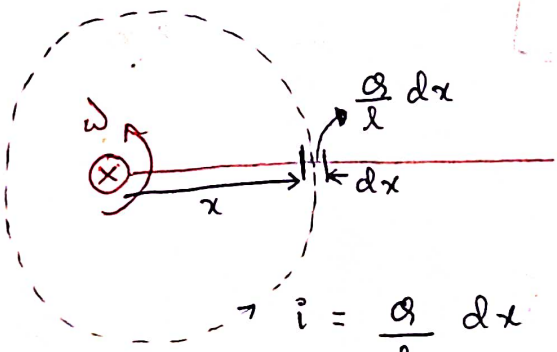
$$= \frac{Q\omega}{2\pi} \cdot \pi R^2$$

$$= \frac{Q\omega R^2}{2}$$

$$\frac{\text{Magnetic dipole moment}}{\text{Angular momentum}} = \frac{M}{L} = \frac{Q\omega R^2/2}{m R^2 \omega}$$

$$= \frac{Q}{2m}$$

Q)



Q, m, l (uniformly dist...
Find M

Sol

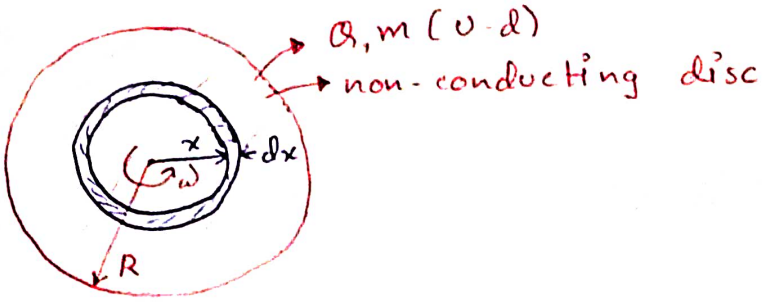
$$i = \frac{\frac{Q}{l} dx}{2\pi/\omega}$$

$$dM = \frac{Q\omega}{2\pi l} dx \, \pi x^2$$

$$M = \frac{Q\omega}{2l} \int_0^l x^2 dx = Q\omega \frac{l^2}{6}$$

$$\frac{M}{L} = \frac{Q\omega L^2/6}{\frac{ml^2}{3} \cdot \omega} = \frac{Q}{2m}$$

Q Find magnetic dipole moment



Sol

$$dq = \frac{Q}{\pi R^2} 2\pi x\, dx$$

$$i = \frac{dq}{2\pi/\omega} = \frac{2Q x\, dx\, \omega}{R^2 \cdot 2\pi}$$

$$dM = i A$$

$$= \frac{Q\omega}{\pi R^2} x\, dx \cdot \pi x^2$$

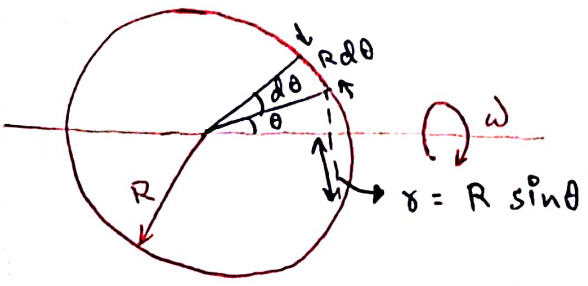
$$M = \frac{Q\omega}{R^2} \int_0^R x^3\, dx$$

$$= \frac{Q\omega}{R^2} \cdot \frac{R^4}{4}$$

$$= \frac{Q\omega R^2}{4}$$

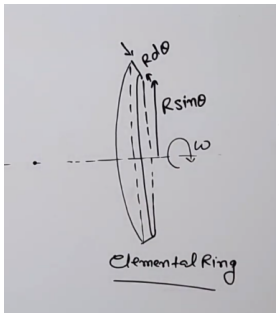
$$\frac{M}{L} = \frac{\frac{Q\omega R^2}{4}}{\frac{mR^2}{2}\omega} = \frac{Q}{2m}$$

Q Find magnetic dipole moment.



Hollow sphere.

Q, m (v·d)



Sol

$$i = \frac{Q \sin\theta \, d\theta}{2 \times 2\pi} \omega = \frac{Q}{T}$$

$$dQ = \sigma \, dA$$

$$= \frac{Q}{4\pi R^2} \cdot 2\pi R \sin\theta \, R \, d\theta$$

$$= \frac{Q}{2} \sin\theta \, d\theta$$

$$dM = \frac{Q\omega}{4\pi} \cdot \pi \sin\theta \cdot d\theta \, (R\sin\theta)^2$$

$$M = \frac{Q\omega}{4} R^2 \int_0^{\pi} (1 - \cos^2\theta) \sin\theta \, d\theta$$

$\int (1-\cos^2\theta)\sin\theta \, d\theta$

$\cos\theta = t$

$-\sin\theta \, d\theta = dt$

$= -\int (1-t^2) \, dt$

$= -\left(t - \frac{t^3}{3}\right)$

$= \left(\frac{t^3}{3} - t\right)$

$= \left(\frac{\cos^3\theta}{3} - \cos\theta\right)_0^{\pi}$

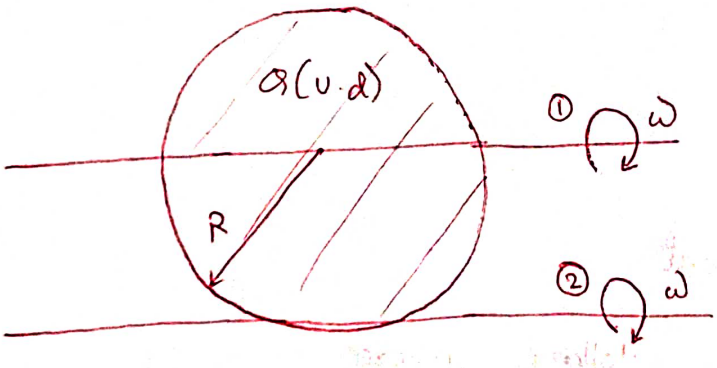
$$M = \frac{Q\omega R^2}{4} \cdot \frac{4}{3}$$

$$= \frac{Q\omega R^2}{3}$$

$$\frac{M}{L} = \frac{Q\omega R^2/3}{\frac{2}{3} m R^2 \omega} = \frac{Q}{2m}$$

Q. Find magnetic dipole moment.

Uniform non-conducting solid sphere



Sol

$$\frac{M}{L} = \frac{Q}{2m}$$

$$M = \frac{Q}{2m} L$$

① $M = \frac{Q}{2m}(I\omega)$

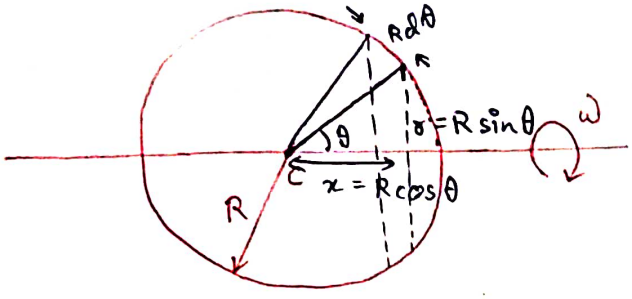
$= \frac{Q}{2m}\left(\frac{2}{5}mR^2\omega\right)$

$= \frac{Q\omega R^2}{5}$

② $M = \frac{Q}{2m} \cdot \frac{7}{5} mR^2\omega$

$= \frac{7Q\omega R^2}{10}$

Q, Find magnetic field at C.



Sol

$dQ = \sigma\, dA$

$= \frac{Q}{4\pi R^2} 2\pi R^2 \sin\theta\, d\theta$

$= \frac{Q}{2}\sin\theta\, d\theta$

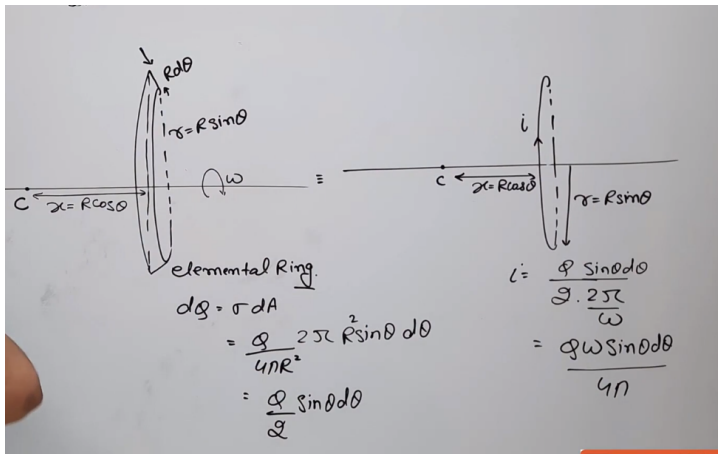
$i = \frac{Q\sin\theta\, d\theta}{2 \cdot \frac{2\pi}{\omega}}$

$= \frac{Q\omega\sin\theta\, d\theta}{4\pi}$

$dB = \frac{\mu_0 Q\omega\sin\theta\, d\theta\,(R^2\sin^2\theta)}{2\times 4\pi\left(R^2\sin^2\theta + R^2\cos^2\theta\right)^{3/2}}$

$= \frac{\mu_0 Q\omega\sin^3\theta\, d\theta}{8\pi R}$

Q, R unifor non-conducting hollow sphere



$\left(\because B = \frac{\mu_0 i R^2}{2(R^2+x^2)^{3/2}}\right)$

$$B = \frac{\mu_0 Q \omega}{8\pi R} \int_0^{\pi} (1-\cos^2\theta) \sin\theta \, d\theta$$

$$= \frac{\mu_0 Q \omega}{8\pi R} \cdot \frac{4}{3}$$

$$= \frac{\mu_0 Q \omega}{6\pi R}$$

$\int (1-\cos^2\theta)\sin\theta \, d\theta$

$\cos\theta = t \quad -\int (1-t^2)\, dt$

$\left(\frac{t^3}{3} - t\right)$

$\left(\frac{\cos^3\theta}{3} - \cos\theta\right)_0^{\pi}$

$\left(\left(-\frac{1}{3}+1\right) - \left(\frac{1}{3}-1\right)\right)$

$4/3$